# PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^K}[T]$, II: EXACT VARIANCES VIA QUADRATIC TOWERS AND SUPERSINGULAR ARTIN–SCHREIER CURVES

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Abstract. Let $q = 2^k$ and let $\mathcal{I}_f = \{f^2 + s : \deg s \le d\}$ be the Legendre interval of a monic $f \in \mathbb{F}_q[t]$ of degree $d$. Chen's Open Problem 8.2 asks why the variance of the prime-power count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ deviates by a factor as large as $q^2/3$ from the Keating–Rudnick prediction. We resolve the case $d = 4$ completely and unconditionally: using the quadratic tower $\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4} \subset \mathbb{F}_{q^8}$ we obtain a closed formula for $\Psi$ on every class of the Hayes group $\mathcal{G}_3$, from which $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$, $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$, the full structure of the quadratic-twist correlations, and an elementary proof of Carlitz's cubic evaluation over $\mathbb{F}_{2^{8k}}$ follow. For $d = 6$ we prove that the Legendre profile takes exactly three values governed by $[c_2 = 0]$ and $\mathrm{Tr}(c_4/c_2^2)$, reducing the variance to three exponential sums $A_6, X, Y$ on $U = \{\alpha \in \mathbb{F}_{q^{12}} : \mathrm{Tr}\,\alpha = \mathrm{Tr}\,\alpha^3 = \mathrm{Tr}\,\alpha^5 = 0\}$. We show that $A_6 - q^9$ and $X$ are sums of Frobenius traces of supersingular Artin–Schreier curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ and reduce them to $\mathbb{F}_2$-quadratic forms whose radicals are kernels of explicit linearised polynomials; the pure cubic and pure quintic parts are evaluated for all $k$, and for odd $k$ the mixed part is evaluated completely: it is governed by the factorisation type of the Gold-type quintic $v^5 + v + c$ over $\mathbb{F}_q$, a $\mathbb{Z}/4$-condition on the Frobenius, and the signs are fixed through the $L$-polynomials of the curves. This proves $A_6 = q^9 + (q-1)q^5$ for all odd $k$. The conjectured closed forms in the remaining cases ($A_6 = q^9 - (q-1)q^5$ for even $k$, and $X = (-1)^{k+1}q^6 + q^5$ for all $k$) are verified exactly for $q \le 128$ by an algorithm independent of the explicit formula; for $X$ we determine the structure of the corresponding order-4 character sums (whose Frobenius classes run through $S_6 \cong Sp_4(\mathbb{F}_2)$ with exact polynomial frequencies) and reduce the closed form to an explicit conjecture. The third constant $Y = q^3 m_k$ is a twisted Frobenius trace on a fixed nine-dimensional complete intersection $X = V(e_1, e_3, e_5) \subset \mathbb{A}^{12}_{\mathbb{F}_2}$ (geometrically integral, normal, with explicit two-dimensional singular locus) and is not a character sum; we discuss its weight and rank. Consequently the Legendre variance at $d = 6$ obeys $\mathrm{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$, not $q^4$. A spectral-folding analysis of the pair $\mathcal{G}^2 \subset \mathcal{G}$ identifies the amplification as the coherence of the Fourier coefficients of $\Psi$ along the fibres of $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$, which are cosets of the 2-torsion $\mathcal{G}[2]$; it is produced by the identity class and its translation orbit, whose size is decided by Lucas' theorem. Exact computations up to $d = 10$ show that the amplification is not universal: at $d = 8$ the Legendre variance is only about 2.3 times the full-group variance for $q \le 8$.

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## 1. Introduction

Let $q = 2^k$, $d \geq 2$, $n = 2d$ and $\ell = d - 1$. For a monic $f \in \mathbb{F}_q[t]$ of degree $d$ the *Legendre interval* is

$$\mathcal{I}_f = \{f^2 + s : \ s \in \mathbb{F}_q[t], \ \deg s \leq d\}, \qquad |\mathcal{I}_f| = q^{d+1},$$

and $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ counts prime powers in it with the von Mangoldt weight. In odd characteristic the sets $\mathcal{I}_f$ are ordinary short intervals and the variance of $\Psi$ is governed by the Keating–Rudnick theorem [15], which predicts $\mathrm{Var}(\Psi) \sim (d-2)q^{d+1}$ as $q \to \infty$. In characteristic 2 the squaring map on the Hayes group is a homomorphism, the Legendre intervals are the fibres over the subgroup of squares, and Chen [5] observed (Table 4 of *op. cit.*) that the variance over Legendre intervals exceeds the full-group variance by factors $4/3$, $16/3$, $64/3$ for $q = 2, 4, 8$ at $d = 4$ and asked for an explanation (Open Problem 8.2).

This paper is the second part of our study. Part I established the dictionary between Legendre intervals and quadratic characters of the Hayes group and verified, by exact computation for $q \leq 8$, the identities $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$ and $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$. Here we prove them for all $q = 2^k$, and we carry the analysis to $d = 6$, where a genuinely new phenomenon appears: two of the three constants that control the Legendre variance are elementary while the third is a Frobenius trace that is not a character sum.

**Main results.** All variances below are second moments about the global mean $q^{d+1}$ (this is the quantity tabulated in [5]); see Section 2.

- $d = 4$ **(Section 3).** Theorem 3.2 gives $\Psi$ on every class of the Hayes group $\mathcal{G}_3$ in closed form. Consequently (Theorems 3.4, 3.11, 3.12) for every $q = 2^k$

$$\frac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1), \qquad \frac{\mathrm{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}, \qquad \frac{\mathrm{Var}_{\mathcal{G}^2}}{\mathrm{Var}_{\mathcal{G}}} = \frac{q^2}{3},$$

the deviation $\Psi - q^5$ is supported on the subgroup $\{c_1 = 0\}$, the identity class carries the deficit $-(q^4 - q^3)$, and the quadratic-twist correlations $\rho(\varepsilon)$ equal 1 on the $q-1$ characters of conductor $u^2$ and $(q-3)/(3(q-1))$ on all others when $k$ is odd.

- $d = 6$, **structure (Section 4).** Theorem 4.2: on the $q^2$ Legendre classes $(0, c_2, 0, c_4, 0)$ the function $\Psi$ takes exactly three values,

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q}[c_2 = 0] + \frac{Y}{q}[c_2 \neq 0]\,\chi\Big(\frac{c_4}{c_2^2}\Big),$$

  where $A_6 = |U|$, $X = \sum_U \chi(e_2)$, $Y = \sum_U \chi(e_4)$ and $U = \{\alpha \in \mathbb{F}_{q^{12}} : \operatorname{Tr}\alpha = \operatorname{Tr}\alpha^3 = \operatorname{Tr}\alpha^5 = 0\}$. The proof uses only the $\mathbb{F}_q$-translation and scaling symmetries of $U$.
- $d = 6$, **the constants (Sections 5–6).** Theorem 5.3 reduces $A_6, X, Y$ to explicit finite sums over $\mathbb{F}_{q^6}$. Theorem 6.1 expresses $A_6 - q^9$ and $X$ through the Frobenius traces of the supersingular curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$, equivalently through $\mathbb{F}_2$-quadratic forms on $\mathbb{F}_{q^{12}}$ with explicit radicals. Lemmas 6.2 and 6.4 evaluate the pure cubic and pure quintic parts for all $k$. For odd $k$, Theorem 6.12 evaluates the mixed part: $W(\lambda_3, \lambda_5) = 2q^7$ or 0 according as the Frobenius permutes the roots of the Gold-type quintic $v^5 + v + c$, $c = (\lambda_5^3/\lambda_3^5)^{1/2}$, as a 4-cycle or not; with the exact count of 4-cycles (Proposition 6.8) this *proves* $A_6 = q^9 + (q-1)q^5$ for all odd $k$ (Corollary 6.13). What remains conjectural is Conjecture 6.5 in the other cases, $A_6 = q^9 - (q-1)q^5$ for even $k$ and $X = (-1)^{k+1}q^6 + q^5$ for all $k$, verified exactly for $q \le 128$ (Table 1); for odd $k$ the sums behind $X$ are analysed in §6.7.
- **Scaling law and $Y$ (Section 7).** Theorem 7.1 gives $\operatorname{Var}_{\mathcal{G}^2}(\Psi)/q^7$ at $d = 6$ in closed form in terms of $(q, k, m_k)$, $m_k = Y/q^3$; it implies $\operatorname{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ (Table 4) and $\limsup \operatorname{Var}_{\mathcal{G}^2}/q^9 = 1$ provided $m_k = o(q^{5/2})$. We identify $Y$ as the trace of a twisted Frobenius $\sigma F^k$ on the cohomology of a fixed nine-dimensional complete intersection $X \subset \mathbb{A}^{12}_{\mathbb{F}_2}$ with an Artin–Schreier sheaf (Propositions 7.4–7.5, Theorem 7.7) and show that the data $m_k = 13, -1, -35, 47, 733$ $(k \le 5)$ exclude a pure weight-3 object of rank $\le 5$. $Y$ is the difference of the twisted point counts of two eight-dimensional complete intersections (Proposition 7.9), and $C_6 = 1$ is equivalent to a Lefschetz-type weight bound for them.
- **Spectral folding (Section 8).** For a subgroup $H$ of a finite abelian group, the variance over $H$ is the energy of the spectrum folded along the fibres of $\widehat{G} \to \widehat{H}$, and it exceeds the full variance by at most $[G : H]$, with equality iff the function is supported on $H$. For the Hayes group $[\mathcal{G} : \mathcal{G}^2] = |\mathcal{G}[2]| = q^{d/2}$, which is why the phenomenon is confined to characteristic 2. At $d = 4$ all fibre sums are computed exactly (Theorem 8.2); at $d = 6$ the Legendre variance is $[(A_6 - q^9)^2 + (q-1)X^2]/q^4 + q^3(q-1)m_k^2$ exactly, the coherent part being $A_6$ and $X$ and the incoherent part $Y$.
- **General even $d$ (Section 9).** The translation orbit $K_d$ of the identity class has order 1 or $q$ according as $d$ is a power of 2 or not (Lucas), and $\Psi$ is constant on it. Exact variances for $d \le 10$ show that the $q^2$-amplification is special to $d = 4, 6$: at $d = 8$ the ratio of the Legendre to the full-group variance is $\approx 2.3$ for $q = 4, 8$, refuting both the $q^{2(d/2-1)}$ law of [5] and a universal $q^2$ law.

**Methods.** Two ideas carry the paper. The first is the *quadratic tower*: for $n = 2d$ the extension $\mathbb{F}_{q^n}/\mathbb{F}_{q^{n/2}}$ is quadratic, so every $\alpha \in \mathbb{F}_{q^n}$ is a root of $t^2 + at + n$ with $(a, n) \in \mathbb{F}^2_{q^{n/2}}$, and the elementary symmetric functions of the $n$ conjugates of $\alpha$ become explicit polynomials in traces of $(a, n)$. Counting pairs $(a, n)$ with the irreducibility condition $\operatorname{Tr}_{q^{n/2}/2}(n/a^2) = 1$ turns the prime-counting problem into linear algebra over $\mathbb{F}_q$ and $\mathbb{F}_2$. The second is the use of the *symmetries of* $U_d = \{p_j = 0,\ j \text{ odd} \le \ell\}$: $U_d$ is stable under $\alpha \mapsto \alpha + c$ $(c \in \mathbb{F}_q)$ and $\alpha \mapsto s\alpha$ $(s \in \mathbb{F}_q^\times)$, and the elementary symmetric functions transform by binomial coefficients mod 2; this alone forces the shape of the Legendre profile.

**Reproducibility.** All statements labelled "verified" were checked with exact integer arithmetic. The scripts that produce every number in the paper (brute force, the $d = 4$ formula, the Hayes FFT driver, the tower identities, the quadratic-form algorithm and its fine-structure variants,

the $L$-polynomial check, the figures) are in the `code/` directory of the repository and are named at the places where they are used. The Hayes-group FFT of [5] was used as an independent check for $q \le 32$ at $d = 6$ and $q \le 128$ at $d = 4$.

## 2. Preliminaries

2.1. **Hayes group, fibres, Legendre intervals.** Let $\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1 u + \cdots + c_\ell u^\ell\} \subset (\mathbb{F}_q[u]/u^{\ell+1})^\times$, $|\mathcal{G}| = q^\ell$. A monic $F = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ maps to $F^*(u) = u^n F(1/u) \bmod u^{\ell+1}$, i.e. to its top $\ell$ coefficients $(c_1, \dots, c_\ell) = (a_{n-1}, \dots, a_{n-\ell})$; the map is multiplicative and each fibre has $q^{n-\ell} = q^{d+1}$ elements.

**Lemma 2.1.** *The Legendre interval $\mathcal{I}_f$ is the fibre over the class of $f^2$, and the classes of squares form the subgroup $\mathcal{G}^2 = \{g^2\}$, of index $q^m$ with $m = \#\{j \le \ell : \ j \text{ odd}\} = \lfloor d/2 \rfloor$. Hence $\Psi(f) = \Psi(g)$ depends only on $g = [f^2] \in \mathcal{G}^2$.*

*Proof.* $f^2 + s$ has vanishing coefficients in all odd degrees $> d$, and $(1 + \sum c_j u^j)^2 = 1 + \sum c_j^2 u^{2j}$. The fibre of $[f^2]$ is $\{f^2 + s : \deg s \le d\}$ since the top $\ell = d - 1$ coefficients of $f^2 + s$ are those of $f^2$. □

2.2. **$\Psi$ as a root count.** For $\alpha \in \mathbb{F}_{q^n}$ write $\operatorname{charpoly}(\alpha) = \prod_{i<n}(t - \alpha^{q^i})$. If $F = P^j$ with $\deg P = n/j$, exactly $n/j = \Lambda(F)$ elements $\alpha$ have $\operatorname{charpoly}(\alpha) = F$. Hence

$$\Psi(g) = \sum_{F \in \text{fibre}(g)} \Lambda(F) = \#\{\alpha \in \mathbb{F}_{q^n} : \ (e_1, \dots, e_\ell)(\alpha) = g\}, \tag{1}$$

where $e_j(\alpha)$ denotes the $j$-th elementary symmetric function of the $n$ conjugates of $\alpha$. In particular $\sum_g \Psi(g) = q^n$ and the mean over $\mathcal{G}$ is $q^{d+1}$.

2.3. **Variances and quadratic twists.** Put $\varphi = \Psi - q^{d+1}$ and

$$\operatorname{Var}_{\mathcal{G}}(\Psi) = \frac{1}{|\mathcal{G}|}\sum_{g \in \mathcal{G}} \varphi(g)^2, \qquad \operatorname{Var}_{\mathcal{G}^2}(\Psi) = \frac{1}{|\mathcal{G}^2|}\sum_{g \in \mathcal{G}^2} \varphi(g)^2.$$

For a character $\chi$ of $\mathcal{G}$ let $\psi_n(\chi) = \sum_{\deg F = n} \Lambda(F)\chi(F)$. Then $\varphi(g) = |\mathcal{G}|^{-1}\sum_{\chi \ne 1} \psi_n(\chi)\bar{\chi}(g)$, and for $\varepsilon$ a character trivial on $\mathcal{G}^2$,

$$\rho(\varepsilon) := \frac{\sum_\chi \psi_n(\chi)\overline{\psi_n(\chi\varepsilon)}}{\sum_{\chi \ne 1} |\psi_n(\chi)|^2} = \frac{\sum_g \varphi(g)^2 \varepsilon(g)}{\sum_g \varphi(g)^2}, \qquad \frac{\operatorname{Var}_{\mathcal{G}^2}}{\operatorname{Var}_{\mathcal{G}}} = 1 + \sum_{\varepsilon \ne 1,\ \varepsilon|_{\mathcal{G}^2} = 1} \rho(\varepsilon). \tag{2}$$

For $\chi \ne 1$ the $L$-function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\le \ell - 1 = d - 2$ with inverse roots of absolute value $\sqrt{q}$, and $\psi_n(\chi) = -\sum_j \gamma_j^n$. Thus $\operatorname{Var}_{\mathcal{G}}(\Psi)/q^{d+1} = |\mathcal{G}|^{-1}\sum_{\chi \ne 1} |\operatorname{Tr}\Theta_\chi^n|^2$ with $\Theta_\chi \in U(d-2)$ the normalised Frobenius; the Keating–Rudnick prediction is the Haar average $d - 2$.

2.4. **Quadratic characters are Artin–Schreier characters.**

**Proposition 2.2.** *Let $p_j(g)$ be the $j$-th power sum of the "roots" of $g \in \mathcal{G}_\ell$, defined by Newton's identities (in characteristic 2: $p_j = \sum_{i<j} c_i p_{j-i} + [j \text{ odd}] c_j$). Then $p_j(gh) = p_j(g) + p_j(h)$, $p_{2j} = p_j^2$, and $g \mapsto (p_j(g))_{j \text{ odd} \le \ell}$ induces an isomorphism $\mathcal{G}/\mathcal{G}^2 \cong \mathbb{F}_q^m$. The $q^m$ quadratic characters are*

$$\varepsilon_\lambda(g) = (-1)^{\operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_2} \sum_{j \text{ odd}} \lambda_j p_j(g)}, \qquad \lambda \in \mathbb{F}_q^m,$$

*and on a root $\alpha$ of a polynomial in the class $g$, $\varepsilon_\lambda(\alpha) = \psi_0(\sum_j \lambda_j \alpha^j)$ with $\psi_0$ the canonical additive character of $\mathbb{F}_{q^n}$. In particular $\varepsilon_\lambda$ is primitive (does not factor through $\mathcal{G}_{\ell-1}$) if and only if $\lambda_\ell \ne 0$, which requires $\ell$ odd, i.e. $d$ even.*

*Proof.* Power sums are additive on products because roots are concatenated; $p_j(g^2) = 2p_j(g) = 0$ for odd $j$ and $p_{2j} = p_j^2$ in characteristic 2. The map is onto $\mathbb{F}_q^m$ because $p_j = c_j + (\text{terms in } c_{<j})$ lets one solve for the odd $c_j$; its kernel has order $q^{\ell-m} = |\mathcal{G}^2|$ and contains $\mathcal{G}^2$. Finally $p_j(\operatorname{charpoly}\alpha) = \operatorname{Tr}_{q^n/q}(\alpha^j)$ and $\operatorname{Tr}_{q/2} \circ \operatorname{Tr}_{q^n/q} = \operatorname{Tr}_{q^n/2}$. □

We write $\chi(x) = (-1)^{\mathrm{Tr}_{q/2}(x)}$ for $x \in \mathbb{F}_q$ and $W = \{w \in \mathbb{F}_q : \mathrm{Tr}_{q/2}\, w = 1\}$. Throughout, $\varepsilon_0 := (-1)^{k+1}$.

## 3. The case $d = 4$: unconditional theorems

Here $n = 8$, $\ell = 3$, $g = (c_1, c_2, c_3) = (a_7, a_6, a_5)$, $\mathcal{G}^2 = \{(0, c, 0)\}$ has order $q$, and $\mathcal{G}_0 := \{c_1 = 0\} \cong (\mathbb{F}_q^2, +)$ is a subgroup of order $q^2$ containing $\mathcal{G}^2$.

### 3.1. The quadratic tower.

**Lemma 3.1.** *For $\alpha \in \mathbb{F}_{q^8}$ put $a = \alpha + \alpha^{q^4}$, $n = \alpha^{1+q^4} \in \mathbb{F}_{q^4}$ and $\mathrm{Tr} = \mathrm{Tr}_{q^4/q}$. Then*

$$e_1(\alpha) = \mathrm{Tr}\, a, \qquad e_2(\alpha) = e_2(a) + \mathrm{Tr}\, n, \qquad e_3(\alpha) = e_3(a) + \mathrm{Tr}\, a\ \mathrm{Tr}\, n + \mathrm{Tr}(an),$$

*where $e_j(a)$ are the elementary symmetric functions of the four conjugates of $a$. A pair $(a, n)$ has exactly two preimages $\alpha$ if $t^2 + at + n$ is irreducible over $\mathbb{F}_{q^4}$ (equivalently $a \neq 0$ and $\mathrm{Tr}_{q^4/2}(n/a^2) = 1$), one preimage if $a = 0$ ($\alpha \in \mathbb{F}_{q^4}$, $n = \alpha^2$), and none otherwise.*

*Proof.* The eight conjugates of $\alpha$ are the roots of $\prod_{i<4}(t^2 + a^{q^i} t + n^{q^i})$; expand. If $t^2 + at + n$ is reducible with $a \neq 0$ its roots $\beta \in \mathbb{F}_{q^4}$ satisfy $\beta + \beta^{q^4} = 0 \neq a$. □

Two facts are used repeatedly: (i) $\mathrm{Tr}_{q^8/2}$ vanishes on $\mathbb{F}_{q^4}$, $\mathrm{Tr}_{q^4/q}$ vanishes on $\mathbb{F}_{q^2}$ and on $\mathbb{F}_q$; (ii) for $a \in \mathbb{F}_{q^4}$ the $\mathbb{F}_q$-linear forms $\mathrm{Tr}(n)$, $\mathrm{Tr}(an)$ are independent iff $a \notin \mathbb{F}_q$, and the $\mathbb{F}_2$-form $\mathrm{Tr}_{q^4/2}(a^{-2}n)$ lies in the $\mathbb{F}_2$-span of $\{\mathrm{Tr}_{q^4/2}(cn) : c \in \mathbb{F}_q + \mathbb{F}_q a\}$ iff $a \in \mathbb{F}_{q^2}$.

### 3.2. The closed formula.

**Theorem 3.2.** *For every $q = 2^k$,*

$$\Psi(c_1, c_2, c_3) = q^5 \qquad (c_1 \neq 0),$$
$$\Psi(0, c_2, c_3) = q^5 + q^3[c_3 = 0] + 2q^3\, I(c_2, c_3) - 2q^2\, K(c_2, c_3),$$

*where $I(c_2, c_3) = [\, c_2 c_3 \neq 0$ and $\mathrm{Tr}_{q/2}(c_2^3/c_3^2) = 1\,]$ and*

$$K(c_2, c_3) = \sum_{w \in W} \sum_{v \in \mathbb{F}_q} \chi\Big(\frac{c_2 v^2}{w} + \frac{\sqrt{c_2}\, v}{w} + \frac{c_3 v^3}{w^2}\Big).$$

*Proof.* We count pairs $(a, n)$ as in Lemma 3.1 with $(e_1, e_2, e_3) = (c_1, c_2, c_3)$, according to the degree of $a$ over $\mathbb{F}_q$.

*Case I: $a \notin \mathbb{F}_{q^2}$.* We need $\mathrm{Tr}\, a = c_1$ and then two independent $\mathbb{F}_q$-linear conditions on $n$, $\mathrm{Tr}\, n = c_2 + e_2(a)$ and $\mathrm{Tr}(an) = c_3 + e_3(a) + c_1(c_2 + e_2(a))$, cutting out an affine subspace of size $q^2$; the irreducibility condition is a non-constant affine $\mathbb{F}_2$-functional on it by fact (ii), so exactly $q^2/2$ values of $n$ qualify, each with two roots. The number of $a$ is $q^3 - q^2[c_1 = 0]$ (all $a \in \mathbb{F}_{q^2}$ have $\mathrm{Tr}\, a = 0$). Contribution: $q^5 - q^4[c_1 = 0]$.

*Case II: $a \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$.* Then $\mathrm{Tr}\, a = 0$, so $c_1 = 0$. Let $T = a + a^q \in \mathbb{F}_q^\times$, $N = a^{1+q}$; the conjugates are $a, a^q, a, a^q$, so $e_2(a) = T^2$ and $e_3(a) = 0$. The conditions are $\mathrm{Tr}\, n = c_2 + T^2$, $\mathrm{Tr}(an) = c_3$ ($q^2$ solutions). Writing $a^{-2} = x + ya$ with $x = (N + T^2)/N^2$, $y = T/N^2$ (from $a^2 = Ta + N$), the irreducibility functional equals $\mathrm{Tr}_{q/2}(x(c_2 + T^2) + yc_3)$ on the solution set, and $\mathrm{Tr}_{q/2}(xT^2) = \mathrm{Tr}_{q/2}(z + z^2) = 0$ with $z = T^2/N$. Hence the contribution is $2q^2$ times the number of $a$ with

$$\tau(a) := \mathrm{Tr}_{q/2}\Big(\frac{(N + T^2)c_2 + Tc_3}{N^2}\Big) = 1.$$

Such $a$ correspond two-to-one to irreducible $t^2 + Tt + N$, i.e. to $(T, N)$ with $\mathrm{Tr}_{q/2}(N/T^2) = 1$. Substituting $N = T^2 w$ ($w \in W$), $v = 1/T$ and using $\mathrm{Tr}_{q/2}(c_2 v^2/w^2) = \mathrm{Tr}_{q/2}(\sqrt{c_2}\, v/w)$ gives contribution $4q^2 M$ with $M = \#\{(v, w) \in \mathbb{F}_q^\times \times W : \tau = 1\} = q^2/4 - K/2$, i.e. $q^4 - 2q^2 K$.

*Case III: $a \in \mathbb{F}_q^\times$.* Then $c_1 = 0$, $e_2(a) = e_3(a) = 0$, the conditions read $\mathrm{Tr}\, n = c_2$, $a\,\mathrm{Tr}\, n = c_3$, and irreducibility is $\mathrm{Tr}_{q/2}(a^{-2}c_2) = 1$. This forces $c_2 \neq 0$, $a = c_3/c_2 \neq 0$ and $\mathrm{Tr}_{q/2}(c_2^3/c_3^2) = 1$; then $n$ ranges over $q^3$ elements, each with two roots. Contribution $2q^3 I(c_2, c_3)$.

*Case IV: $a = 0$.* $\alpha \in \mathbb{F}_{q^4}$, $(e_1, e_2, e_3) = (0, (\mathrm{Tr}\, \alpha)^2, 0)$; contribution $q^3[c_1 = 0][c_3 = 0]$.

Adding the four contributions gives the theorem. □

**Corollary 3.3.** *For every $q = 2^k$:*

(a) $\varphi = \Psi - q^5$ *is supported on* $\mathcal{G}_0 = \{c_1 = 0\}$.
(b) $\Psi(0,0,0) = \#\{\alpha \in \mathbb{F}_{q^8} : e_1 = e_2 = e_3 = 0\} = q^5 - q^4 + q^3$*; equivalently there are exactly* $q^4(q-1)/8$ *irreducible octics with* $a_7 = a_6 = a_5 = 0$.
(c) *For* $c \neq 0$, $\Psi(0,c,0) = q^5 + \varepsilon_0\, q^3$, *i.e.* $q^5 + q^3$ *for* $k$ *even and* $q^5 - q^3$ *for* $k$ *odd.*
(d) $A := \sum_c \Psi(0,c,0) = \#\{\alpha : \operatorname{Tr}\alpha = \operatorname{Tr}\alpha^3 = 0\}$ *equals* $q^6$ *for* $k$ *even and* $q^6 - 2q^4 + 2q^3$ *for* $k$ *odd.*

*Proof.* (a) is the first line of the theorem. (b): $K(0,0) = q^2/2$, $I(0,0) = 0$. (c): with $y = \sqrt{c}\,v$ one has $\chi(cv^2/w + \sqrt{c}\,v/w) = \chi((y^2+y)/w)$; since $y \mapsto y^2 + y$ is two-to-one onto $\{\operatorname{Tr} z = 0\}$ and $z \mapsto \operatorname{Tr}(z/w)$ is nontrivial there unless $w = 1$, $K(c,0) = q[1 \in W] = q[k \text{ odd}]$. (d) follows from (b),(c). The prime-power part of $\Psi(0,c,0)$ equals $q^3$ for every $c$ (Part I), which gives the count of irreducibles in (b). □

**Theorem 3.4.** *For every* $q = 2^k$, $\dfrac{\operatorname{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1)$.

*Proof.* By Corollary 3.3(b),(c), $\sum_{g\in\mathcal{G}^2} \varphi(g)^2 = (q^4-q^3)^2 + (q-1)q^6 = q^7(q-1)$, and $|\mathcal{G}^2| = q$. □

**Remark 3.5.** The factor $q(q-1) = (q-1)^2 + (q-1)$ splits as the contribution of the identity class (the interval around $t^8$, whose prime deficit is $q^4 - q^3$, relative size $1/q$) plus that of the other $q-1$ Legendre classes ($\pm q^3$ each). The sign in (c) depends on the parity of $k$ but the variance does not.

3.3. **Cubic Artin–Schreier sums over $\mathbb{F}_{q^8}$.** Let $S(b,a) = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(ax^3 + bx)$.

**Lemma 3.6.** *For* $a \in \mathbb{F}_q^\times$ *and* $b \in \mathbb{F}_q$, $S(b,a) = S(0,a)$.

*Proof.* For any $c \in \mathbb{F}_{q^8}$ the substitution $x \mapsto x + c$ gives $S(b,a) = \psi_0(ac^3 + bc)\, S(b + T_a(c), a)$ with $T_a(c) = ac^2 + \sqrt{ac}$, since $\operatorname{Tr}(acx^2) = \operatorname{Tr}(\sqrt{ac}\,x)$. If $c \in \mathbb{F}_{q^4}$ then $\operatorname{Tr}_{q^8/2}(ac^3 + bc) = 0$. It remains to find $c \in \mathbb{F}_{q^4}$ with $T_a(c) = b$. The map $T_a$ preserves $\mathbb{F}_{q^4}$ and is self-adjoint for $\langle u, v\rangle = \operatorname{Tr}_{q^4/2}(uv)$, so $\operatorname{im} T_a = (\ker T_a)^\perp$ with $\ker T_a = \{0\} \cup \{u \in \mathbb{F}_{q^4} : u^3 = a^{-1}\}$. For such $u \neq 0$, $\zeta = u^{q-1}$ is a cube root of unity. If $\zeta = 1$, $u \in \mathbb{F}_q$ and $\operatorname{Tr}_{q^4/2}(bu) = \operatorname{Tr}_{q/2}(4bu) = 0$. If $\zeta \neq 1$ and $k$ is even, $\zeta^q = \zeta$ forces $u^{q^4} = \zeta^4 u \neq u$, impossible; if $k$ is odd, $\zeta^q = \zeta^2$ gives $u^{q^2} = u$, so $u \in \mathbb{F}_{q^2}$ and $\operatorname{Tr}_{q^4/2}(bu) = \operatorname{Tr}_{q^2/2}(2bu) = 0$. Hence $b \perp \ker T_a$. □

**Proposition 3.7** (Carlitz's evaluation, elementary proof)**.** *For* $a \in \mathbb{F}_q^\times$ *let* $\nu = a$. *Then*

$$S(0,a) = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(ax^3) = \begin{cases} -2q^4, & a \text{ is a cube in } \mathbb{F}_{q^8}, \\ +q^4, & \text{otherwise,} \end{cases}$$

*and* $a \in \mathbb{F}_q^\times$ *is a cube in* $\mathbb{F}_{q^8}$ *iff* $k$ *is odd or* $a$ *is a cube in* $\mathbb{F}_q$.

*Proof.* By Lemma 3.6, $S(0,a) = q^{-1}\sum_{b\in\mathbb{F}_q} S(b,a) = \sum_{\alpha:\operatorname{Tr}\alpha=0} \psi_0(a\alpha^3) = \widehat{\Psi}(0,a)$ in the notation of (3) below, whose value is computed in Theorem 3.11. The cube criterion: $(q^8-1)/(q-1) \equiv 8 \equiv 2 \pmod 3$ when $q \equiv 1 \pmod 3$, while $3 \mid (q^8-1)/(q-1)$ when $q \equiv 2 \pmod 3$. □

**Remark 3.8.** The classical route [4] uses the cubic Gauss sum $g_4 = \sum_{y\in\mathbb{F}_4^\times} \chi_3(y)(-1)^{\operatorname{Tr} y} = 2$ and the Hasse–Davenport relation [13] $g_{4^m} = (-1)^{m-1}2^m$, giving $\sum_{x\in\mathbb{F}_{2^{2m}}} \psi_0(ax^3) = (-1)^{m+1}2^{m+1}$ for cubes and $(-1)^m 2^m$ otherwise; with $m = 4k$ this agrees with Proposition 3.7.

3.4. **The full-group variance.** Identify $\mathcal{G}_0 = \{(0,x,y)\}$ with $(\mathbb{F}_q^2, +)$ and set

$$\widehat{\Psi}(\mu,\nu) = \sum_{x,y\in\mathbb{F}_q} \Psi(0,x,y)\chi(\mu x + \nu y). \tag{3}$$

For $j \in \mathbb{F}_q^\times$ let $C_j(x) = x^3 + (1+j)x^2 + x + 1$, $R_j = \#\{x \in \mathbb{F}_q : C_j(x) = 0\}$, $L_j = \#\{x : C_j(x) = 0,\ \operatorname{Tr}_{q/2} x = 1\}$ and $L_j' = R_j - L_j$.

**Lemma 3.9.** *For $(\mu,\nu)\neq(0,0)$:*

$$\widehat{\Psi}(\mu,0)=-q^4\quad(\mu\neq0),\qquad \widehat{\Psi}(0,\nu)=q^4\big(1-R_0(\nu)-2[k\ \text{odd}]\big)\quad(\nu\neq0),$$
$$\widehat{\Psi}(\mu,\nu)=q^4\big(1-3L_j-L_j'\big)\qquad(\mu\nu\neq0,\ j=\mu^3/\nu^2),$$

*where $R_0(\nu)=\#\{t:t^3=\nu\}$ equals 1 for $k$ odd and $3[\nu$ is a cube$]$ for $k$ even. In particular $\widehat{\Psi}(0,\nu)=-2q^4$ for $k$ odd, and for $k$ even $-2q^4$ or $+q^4$ according as $\nu$ is a cube in $\mathbb{F}_q$ or not.*

*Proof.* Transform the three terms of Theorem 3.2. The term $q^3[y=0]$ gives $q^4[\mu=0]$. For $K$, the identity $\sum_x\chi(x\alpha+\sqrt{x}\,\beta)=q[\sqrt{\alpha}=\beta]$ yields $\widehat{K}(\mu,\nu)=q^2\#\{(v,w)\in\mathbb{F}_q\times W:\ v^3=\nu w^2,\ v^2(w+1)=\mu w^2\}$. For $I=[x,y\neq0]\frac{1-\chi(x^3/y^2)}{2}$, the substitution $x=yt$ gives $\sum_{x,y\neq0}\chi(x^3/y^2+\mu x+\nu y)=q\,R^*(\mu,\nu)-(q-1)$ with $R^*=\#\{t\neq0:t^3+\mu t+\nu=0\}$. The three cases follow; for $\mu\nu\neq0$, dividing the two conditions in $\widehat{K}$ gives $v=\nu(w+1)/\mu$ and $\nu^2(w+1)^3=\mu^3w^2$, and $t=(\nu/\mu)s$, $s=1+1/w$ transforms $t^3+\mu t+\nu=0$ into $C_j(w)=0$, so $R^*=R_j$ and $\widehat{K}=q^2L_j$. □

**Lemma 3.10.** $\Sigma:=\displaystyle\sum_{j\in\mathbb{F}_q^\times}(1-3L_j-L_j')^2=3q-4+(-1)^k$.

*Proof.* Put $c(x)=2-\chi(x)\in\{1,3\}$, so $3L_j+L_j'=\sum_{C_j(x)=0}c(x)$. Every $x\in\mathbb{F}_q\setminus\{0,1\}$ is a root of exactly one $C_j$ ($j=(x+1)^3/x^2$), and $0,1$ are roots of none. Expanding, $\Sigma=(q-1)-2\sum_{x\neq0,1}c(x)+\sum_{x\neq0,1}c(x)^2+\mathrm{OFF}$, where OFF is the sum of $c(x)c(x')$ over ordered pairs of *distinct* roots of a common $C_j$. Using $\sum_{x\neq0,1}\chi(x)=-1-(-1)^k$ one finds $\Sigma=2q-1+2(-1)^k+\mathrm{OFF}$. Distinct roots $x\neq x'$ of a common $C_j$ are exactly the pairs with $x,x'\notin\{0,1\}$ satisfying

$$x^2x'^2+xx'+x+x'=0 \tag{4}$$

(the third root is $1/(xx')$ and $e_2=1$; conversely (4) with $x,x'\notin\{0,1\}$ produces a cubic $C_j$ with $j\neq0$). On (4) one has $x+x'=p^2+p$ with $p=xx'$, hence $\chi(x)\chi(x')=1$, and by symmetry $\mathrm{OFF}=5P-4\sum_{\mathrm{pts}}\chi(x)$ where $P$ is the number of points. For fixed $x\notin\{0,1\}$, (4) is a quadratic in $x'$ with $1+\chi(x^3/(x+1)^2)$ roots, all $\notin\{0,1\}$. With $y=x+1$: $x^3/(x+1)^2=y+1+1/y+1/y^2$, whose $\chi$-value is $(-1)^k\chi(y)$, and $x+x^3/(x+1)^2=1/y+1/y^2$, whose $\chi$-value is 1. Therefore $P=\sum_{\mathrm{pts}}\chi(x)=q-3-(-1)^k$, $\mathrm{OFF}=q-3-(-1)^k$ and $\Sigma=3q-4+(-1)^k$. □

**Theorem 3.11.** *For every $q=2^k$, $\dfrac{\mathrm{Var}_{\mathcal{G}}(\Psi)}{q^5}=3-\dfrac{3}{q}$ and $\dfrac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{\mathrm{Var}_{\mathcal{G}}(\Psi)}=\dfrac{q^2}{3}$.*

*Proof.* By Corollary 3.3(a) and Parseval on $\mathcal{G}_0\cong\mathbb{F}_q^2$ (note $\sum_{\mathcal{G}_0}\varphi=0$),

$$\sum_{\mathcal{G}}\varphi^2=\frac{1}{q^2}\sum_{(\mu,\nu)\neq(0,0)}|\widehat{\Psi}(\mu,\nu)|^2.$$

Lemma 3.9 gives $(q-1)q^8$ from $\nu=0$, $\kappa_k(q-1)q^8$ from $\mu=0$ with $\kappa_k=4$ ($k$ odd) and $\kappa_k=\frac{1}{3}\cdot4+\frac{2}{3}\cdot1=2$ ($k$ even), and $q^8(q-1)\Sigma$ from $\mu\nu\neq0$ (each $j$ has exactly $q-1$ preimages $(\mu,\nu)$). By Lemma 3.10 the bracket $1+\kappa_k+\Sigma$ equals $3q$ in both parities, so $\sum_{\mathcal{G}}\varphi^2=3q^7(q-1)$. □

**Theorem 3.12.** *Let $\varepsilon_{\lambda_1,\lambda_3}$ be the quadratic characters of $\mathcal{G}_3$ (Proposition 2.2). Then*

(i) *$\rho(\varepsilon_{\lambda_1,0})=1$ for all $\lambda_1\neq0$ ($q-1$ characters of conductor $u^2$);*
(ii) *$\rho(\varepsilon_{\lambda_1,\lambda_3})=r(\lambda_3)$ is independent of $\lambda_1$ and constant on cosets of the cubes in $\mathbb{F}_q^\times$;*
(iii) *if $k$ is odd, $r\equiv\dfrac{q-3}{3(q-1)}$;*
(iv) $\rho_{\mathrm{avg}}=\dfrac{q^2/3-1}{q^2-1}\to\dfrac{1}{3}$.

*Proof.* (i),(ii): $\varphi$ is supported on $\mathcal{G}_0$ where $p_1=0$ and $p_3=c_3$, so $\varepsilon_{\lambda_1,\lambda_3}|_{\mathcal{G}_0}=\chi(\lambda_3c_3)$; the scaling automorphism $(c_1,c_2,c_3)\mapsto(sc_1,s^2c_2,s^3c_3)$ preserves $\Psi$. (iii): for $k$ odd cubing is onto,

so $\Phi(y)=\sum_x \varphi(0,x,y)^2$ is constant for $y\neq 0$, equal to $(3q^7(q-1)-q^7(q-1))/(q-1)=2q^7$ by Theorems 3.4–3.11; then $r=(q^7(q-1)-2q^7)/(3q^7(q-1))$. (iv) is immediate. □

**Remark 3.13.** For $q=8$ this gives seven characters with $\rho=1$ and 56 with $\rho=5/21=0.2381$, matching [5, Table 4] exactly; the "order-two coordinates" there contribute $7\cdot 5/21=5/3$. The excess over Keating–Rudnick comes from the primitive quadratic characters, whose $\rho$ does not decay in $q$; $\sum_{\chi\neq 1}\psi_8(\chi)=-q^7+q^6$ shows that $\mathbb{E}_\chi \operatorname{Tr}\Theta_\chi^8\to -1$, so the Frobenius classes are not Haar-distributed in $U(2)$ when $p=2$, $\ell=3$.

## 4. The case $d=6$: the three-value structure theorem

Now $n=12$, $\ell=5$, $\mathcal{G}^2=\{(0,c_2,0,c_4,0)\}$ has order $q^2$ and

$$U:=\{\alpha\in\mathbb{F}_{q^{12}}:\ \operatorname{Tr}\alpha=\operatorname{Tr}\alpha^3=\operatorname{Tr}\alpha^5=0\},\qquad \operatorname{Tr}=\operatorname{Tr}_{q^{12}/q}.$$

Since $p_3=c_3+c_1c_2+c_1^3$ and $p_5\equiv c_5$ modulo terms involving $c_1,c_3$, the Legendre classes are exactly the classes with $p_1=p_3=p_5=0$, and $\Psi(0,c_2,0,c_4,0)=\#\{\alpha\in U: e_2(\alpha)=c_2,\ e_4(\alpha)=c_4\}$. Define

$$A_6=|U|,\qquad X=\sum_{\alpha\in U}\chi(e_2(\alpha)),\qquad Y=\sum_{\alpha\in U}\chi(e_4(\alpha)).$$

**Lemma 4.1.** *$U$ is stable under $\alpha\mapsto\alpha+c$ ($c\in\mathbb{F}_q$) and $\alpha\mapsto s\alpha$ ($s\in\mathbb{F}_q^\times$). On $U$,*

$$e_2(\alpha+c)=e_2(\alpha),\qquad e_4(\alpha+c)=e_4(\alpha)+c^2e_2(\alpha)+c^4,\qquad (e_2,e_4)(s\alpha)=(s^2e_2,s^4e_4)(\alpha).$$

*Proof.* $\operatorname{Tr}(\alpha+c)^3=\operatorname{Tr}\alpha^3+c(\operatorname{Tr}\alpha)^2$ and $\operatorname{Tr}(\alpha+c)^5=\operatorname{Tr}\alpha^5+c(\operatorname{Tr}\alpha)^4$ since $12c=0$. For the elementary symmetric functions, $e_j(\alpha+c)=\sum_i\binom{12-i}{j-i}c^{j-i}e_i(\alpha)$; modulo 2, $\binom{12}{2}=66$, $\binom{11}{1}=11$, $\binom{12}{4}=495$, $\binom{11}{3}=165$, $\binom{10}{2}=45$, $\binom{9}{1}=9$, and on $U$ one has $e_1=e_3=0$. □

**Theorem 4.2** (three-value profile)**.** *For all $q=2^k$ and all $(c_2,c_4)\in\mathbb{F}_q^2$,*

$$\Psi(0,c_2,0,c_4,0)=q^7+\frac{A_6-q^9-X}{q^2}+\frac{X}{q}\,[c_2=0]+\frac{Y}{q}\,[c_2\neq 0]\,\chi\Big(\frac{c_4}{c_2^2}\Big).$$

*Equivalently, the Fourier transform $\widehat{\Psi}_H(\mu,\nu)=\sum_{\alpha\in U}\chi(\mu e_2+\nu e_4)$ equals $A_6$, $X$, $Y\chi(\mu/\sqrt{\nu})$ for $(\mu,\nu)=(0,0)$, $(\mu\neq 0,0)$, $\nu\neq 0$ respectively.*

*Proof.* Let $F(\lambda)=\sum_U\chi(\lambda e_2+e_4)$. By Lemma 4.1, $F(\lambda)=\sum_U\chi(\lambda e_2+e_4+c^2e_2+c^4)=\chi(c)F(\lambda+c^2)$, hence $F(\lambda)=\chi(\sqrt{\lambda})F(0)=\chi(\lambda)Y$. For $\nu\neq 0$ choose $s=\nu^{-1/4}$; scaling gives $\widehat{\Psi}_H(\mu,\nu)=F(\mu\nu^{-1/2})=\chi(\mu/\sqrt{\nu})Y$, and for $\mu\neq 0$ scaling gives $\widehat{\Psi}_H(\mu,0)=X$. Fourier inversion on $\mathbb{F}_q^2$ yields the displayed formula. □

**Corollary 4.3.** *Put $B=(A_6-q^9-X)/q^2$. Then $\varphi=\Psi-q^7$ equals $\varphi_0=B+X/q$ on the $q$ classes with $c_2=0$, $\varphi_A=B+Y/q$ on the $q(q-1)/2$ classes with $c_2\neq 0$, $\operatorname{Tr}_{q/2}(c_4/c_2^2)=0$, and $\varphi_B=B-Y/q$ on the remaining $q(q-1)/2$ classes, and*

$$\frac{\operatorname{Var}_{\mathcal{G}^2}(\Psi)}{q^7}=\frac{q\varphi_0^2+q(q-1)\big(B^2+Y^2/q^2\big)}{q^9}. \tag{5}$$

**Remark 4.4.** The same argument applies to every $d$: $U_d=\{p_j=0,\ j\text{ odd}\le\ell\}$ is stable under $\mathbb{F}_q$-translations and scalings, and the transformation law of $e_{2j}$ is governed by $\binom{2d-i}{2j-i}\bmod 2$ (Lucas' theorem). For $d=4$ one recovers Corollary 3.3(c).

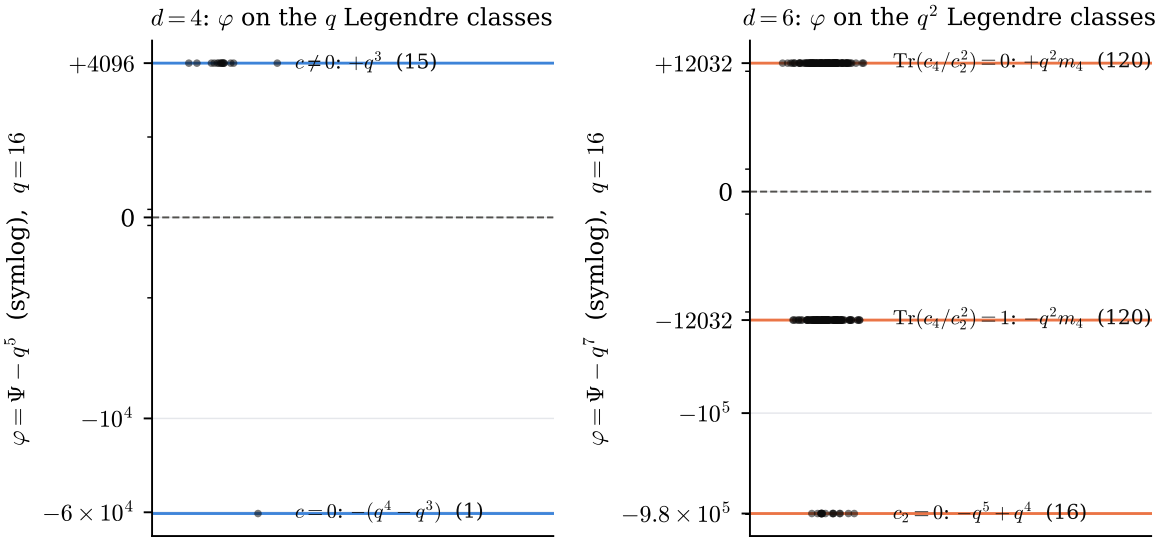



FIGURE 1. The Legendre deviations $\varphi=\Psi-q^{d+1}$ for $q=16$. Left: $d=4$, the $q$ classes $(0,c,0)$ take the two values of Corollary 3.3; the identity class carries $-(q^4-q^3)$. Right: $d=6$, the $q^2$ classes $(0,c_2,0,c_4,0)$ take exactly the three values of Theorem 4.2 ($m_4=47$). Each dot is one class (horizontal jitter only); the deviations are not continuous noise but are pinned to discrete levels by the translation symmetry of $U$.

## 5. The quadratic tower $\mathbb{F}_{q^{12}}\supset\mathbb{F}_{q^6}$

**Lemma 5.1.** *For $\alpha\in\mathbb{F}_{q^{12}}$ let $a=\alpha+\alpha^{q^6}$, $n=\alpha^{1+q^6}\in\mathbb{F}_{q^6}$, $\mathrm{Tr}=\mathrm{Tr}_{q^6/q}$, and $e_j(a)$ the elementary symmetric functions of the six conjugates of $a$. Then*

$$\begin{aligned}
e_1&=\mathrm{Tr}\,a,\qquad e_2=e_2(a)+\mathrm{Tr}\,n,\qquad e_3=e_3(a)+\mathrm{Tr}\,a\,\mathrm{Tr}\,n+\mathrm{Tr}(an),\\
e_4&=e_4(a)+e_2(a)\,\mathrm{Tr}\,n+e_1(a)\,\mathrm{Tr}(an)+\mathrm{Tr}(a^2n)+e_2(n),\\
e_5&=e_5(a)+e_3(a)\,\mathrm{Tr}\,n+e_2(a)\,\mathrm{Tr}(an)+e_1(a)\,\mathrm{Tr}(a^2n)+\mathrm{Tr}(a^3n)\\
&\qquad+e_1(a)e_2(n)+\mathrm{Tr}(an)\,\mathrm{Tr}\,n+\mathrm{Tr}(an^2),\\
p_3&=\mathrm{Tr}(a^3+an),\qquad p_5=\mathrm{Tr}(a^5+a^3n+an^2),
\end{aligned}$$

*and $e_2(\beta+\beta^q)=\mathrm{Tr}_{q^{12}/q}(\beta^{q+1})+(\mathrm{Tr}_{q^{12}/q}\,\beta)^2$ for all $\beta\in\mathbb{F}_{q^{12}}$. The multiplicity statement of Lemma 3.1 holds verbatim with $\mathbb{F}_{q^4}$ replaced by $\mathbb{F}_{q^6}$; in particular $\mathbb{F}_{q^6}\subset U$, with $e_2|_{\mathbb{F}_{q^6}}=(\mathrm{Tr}\,\alpha)^2$ and $e_4|_{\mathbb{F}_{q^6}}=e_2(\alpha)^2$.*

*Proof.* Expand $\prod_{i<6}(t^2+a_it+n_i)$ using $e_2(a\backslash j)=e_2(a)+a_je_1(a)+a_j^2$, $e_3(a\backslash j)=e_3(a)+a_je_2(a)+a_j^2e_1(a)+a_j^3$ and $e_1(a\setminus j,k)=e_1(a)+a_j+a_k$. The power sums follow from $s_k=as_{k-1}+ns_{k-2}$ for $s_k=\alpha^k+\alpha^{q^6k}$. All identities were also checked by exhaustive computation over $\mathbb{F}_{2^{12}}$. □

**Lemma 5.2** (Gauss sum of $e_2$)**.** *On $\mathbb{F}_{q^6}$ the quadratic form $e_2$ has polar form $B(x,y)=\mathrm{Tr}\,x\,\mathrm{Tr}\,y+\mathrm{Tr}(xy)$, which is nondegenerate. For every $\beta\in\mathbb{F}_{q^6}$, $\sum_{n\in\mathbb{F}_{q^6}}\chi(e_2(n)+\mathrm{Tr}(\beta n))=\chi(e_2(\beta))\,G$ with $G=\sum_n\chi(e_2(n))$.*

*Proof.* $n_0=\beta+\mathrm{Tr}\,\beta$ satisfies $B(n_0,\cdot)=\mathrm{Tr}(\beta\,\cdot)$, and $e_2(\beta+c)=e_2(\beta)+c\,\mathrm{Tr}\,\beta+c^2$ for $c\in\mathbb{F}_q$ gives $e_2(n_0)=e_2(\beta)$. □

**Theorem 5.3** (tower reduction)**.** *Let $a$ range over $\mathbb{F}_{q^6}\backslash\{0\}$ with $\mathrm{Tr}\,a=0$, $(\lambda,\mu)\in\mathbb{F}_q^2$, $\varepsilon\in\{0,1\}$, and put*

$$\beta'=\lambda a+\sqrt{\mu}\sqrt{a}+\mu a^3+\varepsilon a^{-2},\qquad \beta=e_2(a)+a^2+\beta',\qquad w=\chi(\lambda\,\mathrm{Tr}\,a^3+\mu\,\mathrm{Tr}\,a^5).$$

*Then*

$$A_6 = q^6 + q^4 \sum_a \sum_{\beta'=0} (-1)^{\varepsilon} w, \qquad X = q^4 \sum_a \chi(e_2(a)) \sum_{\beta'=1} (-1)^{\varepsilon} w,$$
$$Y = G\Big[1 + q^{-2} \sum_a \sum_{\lambda,\mu,\varepsilon} (-1)^{\varepsilon} \chi\big(e_4(a) + \lambda \operatorname{Tr} a^3 + \mu \operatorname{Tr} a^5 + e_2(\beta)\big)\Big].$$

*Proof.* For $a \neq 0$ with $\operatorname{Tr} a = 0$, the conditions $p_3 = p_5 = 0$ read $\operatorname{Tr}(an) = \operatorname{Tr} a^3$ and $\operatorname{Tr}(\sqrt{a}\, n)^2 + \operatorname{Tr}(a^3 n) = \operatorname{Tr} a^5$ (using $\operatorname{Tr}(an^2) = \operatorname{Tr}(\sqrt{a}\, n)^2$); expand their indicators and the irreducibility indicator $\frac{1}{2}\sum_{\varepsilon}(-1)^{\varepsilon}\chi(\varepsilon \operatorname{Tr}(a^{-2}n))$ by additive characters, using $\chi(\mu t^2) = \chi(\sqrt{\mu}\, t)$. The $n$-sums are $q^6[\gamma = 0]$ for $A_6, X$ (no $e_2(n)$ term) and $\chi(e_2(\beta))G$ for $Y$ by Lemma 5.2. The contribution of $\alpha \in \mathbb{F}_{q^6}$ is $q^6$, $0$, $G$ respectively. □

**Remark 5.4.** The equation $\beta' = 0$ with $\varepsilon = 0$ and $\mu \neq 0$ squares to $\mu^2 a^5 + \lambda^2 a + \mu = 0$, forcing $a \in \mathbb{F}_{q^2} \cup \mathbb{F}_{q^3}$ (a thin layer, as in Case II of Theorem 3.2). For $\varepsilon = 1$, with $u = \mu a^5 + \lambda a^3 + 1$ it becomes $u^2 + u = \lambda a^3 + 1$: the points $(a, u) \in E_\lambda(\mathbb{F}_{q^6})$ of the supersingular elliptic curve $E_\lambda : y^2 + y = \lambda x^3 + 1$ with $(u + \lambda a^3 + 1)/a^5 \in \mathbb{F}_q$. The additional term $e_2(\beta)$ in $Y$ is quadratic in $(\lambda, \sqrt{\mu})$ with cubic cross terms, which is why $Y$ does not collapse onto a thin layer.

The reductions were verified numerically for $q = 2, 4, 8$; they give $G = \varepsilon_0 q^3$ for $k \leq 3$.

## 6. Quadratic forms and supersingular Artin–Schreier curves

### 6.1. $A_6$ and $X$ as sums of Frobenius traces.

**Theorem 6.1.** *Write $\lambda = (\lambda_1, \lambda_3, \lambda_5) \in \mathbb{F}_q^3$ and $S(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$. Then*

$$A_6 - q^9 = \frac{1}{q^3} \sum_{\lambda \neq 0} S(\lambda), \tag{6}$$

*and $S(\lambda) = -\sum_j \omega_j(\lambda)^{12}$, where $\omega_j(\lambda)$ are the Frobenius eigenvalues over $\mathbb{F}_q$ of the Artin–Schreier curve $C_\lambda : y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ (genus 2, 1, 0 according as $\lambda_5 \neq 0$; $\lambda_5 = 0 \neq \lambda_3$; only $\lambda_1 \neq 0$). Moreover $\operatorname{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3) = Q_\lambda(x)$ is an $\mathbb{F}_2$-quadratic form with polar form $B_\lambda(x, y) = \operatorname{Tr}(y\, T_\lambda(x))$,*

$$T_\lambda(x) = \lambda_5 x^4 + (\lambda_5 x)^{1/4} + \lambda_3 x^2 + (\lambda_3 x)^{1/2}, \qquad R_\lambda = \ker T_\lambda = \{x : \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_3^2 x^2 + \lambda_5 x = 0\},$$

*so that $S(\lambda) \in \{0, \pm 2^{(12k + r_\lambda)/2}\}$ with $r_\lambda = \dim_{\mathbb{F}_2} R_\lambda \leq 4$ (for the Gold and Kasami–Welch monomials this is the setting of [16]), $S(\lambda) = 0$ iff $\operatorname{Tr}(\lambda_1 x) \neq Q_\lambda(x)$ for some $x \in R_\lambda$, and the sign otherwise is the Arf invariant of $Q_\lambda$ on $\mathbb{F}_{q^{12}}/R_\lambda$ (shifted by $Q_\lambda(x_0)$, $B_\lambda(x_0, \cdot) = \operatorname{Tr}(\lambda_1 \cdot)$). In particular all $C_\lambda$ are supersingular: $\omega_j(\lambda) = \sqrt{q}\, \zeta_j$ with roots of unity $\zeta_j$.*

*Proof.* $A_6 = \sum_{g \in \mathcal{G}^2} \Psi(g) = \frac{|\mathcal{G}^2|}{|\mathcal{G}|} \sum_{\varepsilon^2 = 1} \psi_{12}(\varepsilon)$ and $\psi_{12}(\varepsilon_\lambda) = S(\lambda)$ by Proposition 2.2; $S(0) = q^{12}$. The $L$-function interpretation is standard. $Q_\lambda(x) = \operatorname{Tr}(x(\lambda_5 x^4 + \lambda_3 x^2))$ is of the form $\operatorname{Tr}(xL(x))$ with $L$ linearised, whence the quadratic-form statements; the kernel of $T_\lambda$ equals that of $T_\lambda^4$. Since the same structure persists over every $\mathbb{F}_{q^m}$, $\sum_{x \in \mathbb{F}_{q^m}} (-1)^{\operatorname{Tr} f_\lambda(x)} \in \{0, \pm 2^{(mk+r)/2}\}$ for all $m$, which forces the normalised eigenvalues to be roots of unity [11]. □

Summing over $\lambda_1$ and using Hilbert 90 ($\alpha = \beta + \beta^q$, $\theta = \sigma + \sigma^{-1}$, $\operatorname{Tr} \alpha^3 = \operatorname{Tr}(\beta^2 \theta \beta)$, $\operatorname{Tr} \alpha^5 = \operatorname{Tr}(\beta^4 \theta \beta)$) one obtains the equivalent forms

$$A_6 = q^9 + \frac{1}{q^3} \sum_{(\lambda_3, \lambda_5) \neq 0} W(\lambda_3, \lambda_5), \qquad X = \frac{1}{q^3} \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5), \tag{7}$$

with $W = \sum_\beta (-1)^{\operatorname{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta)}$ and $W' = \sum_\beta (-1)^{\operatorname{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta + \beta^{q+1} + \beta)}$ (the second uses Lemma 5.1). Their polar forms are $\operatorname{Tr}(\gamma M(\theta\beta))$ with $M(y) = \lambda_3 y^2 + \sqrt{\lambda_3 y} + \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ ($+y$ for $W'$), so the radicals are $\theta^{-1}(\ker M \cap \operatorname{im} \theta)$, of dimension $2k + \dim(\ker M \cap \operatorname{im} \theta)$, where $\ker \theta = \mathbb{F}_{q^2}$ and $\operatorname{im} \theta = \ker \operatorname{Tr}_{q^{12}/q^2}$.

6.2. **The pure cubic part.**

**Lemma 6.2.** *For all $k$ and all $\lambda_3 \in \mathbb{F}_q^\times$: $W(\lambda_3, 0) = 0$ if $k$ is odd and $W(\lambda_3, 0) = -2q^7$ if $k$ is even.*

*Proof.* $\ker M = \{0\} \cup \{y : y^3 = \lambda_3^{-1}\}$, and all three cube roots lie in $\mathbb{F}_{q^{12}}$. With $\zeta = y^{q-1} \in \mu_3$: $\zeta = 1$ gives $y \in \mathbb{F}_q$; for $\zeta \neq 1$, $k$ odd gives $y \in \mathbb{F}_{q^2}$ and $k$ even gives $y \in \mathbb{F}_{q^3}$. In all cases $\mathrm{Tr}_{q^{12}/q^2} y = 0$, so $\ker M \subset \mathrm{im}\,\theta$ and $\dim R = 2k + 2$, $|W| \in \{0, 2^{7k+1}\}$. Now $Q(\beta) = \mathrm{Tr}_{q^{12}/2}(\lambda_3 \beta^2 \theta\beta)$ vanishes on $\ker\theta$, and for $\theta\beta = y$ it is independent of the choice of $\beta$ (the ambiguity $c \in \mathbb{F}_{q^2}$ contributes $\mathrm{Tr}(\lambda_3 c^2 y)$ with $c^2 y$ in a subfield of even index). If $y \in \mathbb{F}_{q^2}$, $\zeta \neq 1$: on $\mathbb{F}_{q^4}$ one has $\theta\beta = \sigma(\mathrm{Tr}_{q^4/q^2}\,\beta)$, so pick $\beta_y \in \mathbb{F}_{q^4}$ with $\mathrm{Tr}_{q^4/q^2}\,\beta_y = y^q$; then $Q(\beta_y) = \mathrm{Tr}_{q^4/2}(\lambda_3 y \beta_y^2) = \mathrm{Tr}_{q^2/2}(\lambda_3 y\,(\mathrm{Tr}_{q^4/q^2}\,\beta_y)^2) = \mathrm{Tr}_{q^2/2}(\lambda_3 y^{2q+1}) = \mathrm{Tr}_{q^2/2}(\zeta^2) = k \pmod 2$, which is 1 for $k$ odd: $Q|_R \neq 0$ and $W = 0$. If $y \in \mathbb{F}_q$ the same computation gives $\mathrm{Tr}_{q^2/2}(1) = 0$; if $y \in \mathbb{F}_{q^3}$ ($k$ even) then $\mathrm{Tr}_{q^3/q}\, y = 0$, $\theta$ is invertible on the trace-zero part of $\mathbb{F}_{q^3}$, $\beta_y \in \mathbb{F}_{q^3}$ and $Q(\beta_y) = 0$. So for $k$ even $|W| = 2q^7$. Finally $W = \sum_{\lambda_1 \in \mathbb{F}_q} S(\lambda_1, \lambda_3, 0)$ with $|S| = 2q^6$ for each term ($r = 2$), and $|W| = q \cdot 2q^6$ forces all $q$ signs to agree with that of $S(0, \lambda_3, 0) = -2q^6$ (Hasse–Davenport, $m = 6k$). □

**Remark 6.3.** For $k$ odd the analogue of Lemma 3.6 fails over $\mathbb{F}_{q^{12}}$: the cube roots lie in $\mathbb{F}_{q^2}$ and $[\mathbb{F}_{q^6} : \mathbb{F}_{q^2}] = 3$ is odd, so $\mathbb{F}_q \not\subset T(\mathbb{F}_{q^6})$. The vanishing of $W(\lambda_3, 0)$ is exactly this failure.

6.3. **The pure quintic part.** The quintic analogue uses a pure Gauss sum of order 5. Since $2^2 \equiv -1 \pmod 5$, Stickelberger's theorem on pure Gauss sums (see [2, Thm. 11.6.3]) gives, for every character $\chi$ of order 5 of $\mathbb{F}_{2^{4s}}^\times$, $G(\chi) = \sum_y \chi(y)(-1)^{\mathrm{Tr}\, y} = (-1)^{s-1} 2^{2s}$. With $4s = 12k$, i.e. $s = 3k$, and $a \in \mathbb{F}_q^\times$,

$$S(0, 0, a) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(ax^5) = \sum_{j=1}^{4} \chi^{-j}(a) G(\chi^j) = \varepsilon_0\, q^6 \big(5\,[a \in (\mathbb{F}_{q^{12}}^\times)^5] - 1\big), \tag{8}$$

and $a \in \mathbb{F}_q^\times$ is a fifth power in $\mathbb{F}_{q^{12}}$ iff $4 \nmid k$ or $a \in (\mathbb{F}_q^\times)^5$ (because $(q^{12} - 1)/(q - 1) \equiv 0 \pmod 5$ iff $q \not\equiv 1 \pmod 5$ iff $4 \nmid k$).

**Lemma 6.4** (pure quintic part)**.** *For all $k$ and all $\lambda_5 \in \mathbb{F}_q^\times$:*

$$W(0, \lambda_5) = \begin{cases} 2q^7, & k \text{ odd}, \\ 0, & k \equiv 2 \pmod 4, \\ -4q^7, & k \equiv 0 \pmod 4 \text{ and } \lambda_5 \in (\mathbb{F}_q^\times)^5, \\ +q^7, & k \equiv 0 \pmod 4 \text{ and } \lambda_5 \notin (\mathbb{F}_q^\times)^5. \end{cases}$$

*Proof.* Here $M(y) = \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ and $\ker M = \{0\} \cup \{y : y^{15} = \lambda_5^{-3}\}$. The equation $y^{15} = \lambda_5^{-3}$ is solvable in $\mathbb{F}_{q^{12}}$ iff $\lambda_5$ is a fifth power there (if $y^{15} = w^3$ then $y^5 = w\omega$ with $\omega^3 = 1$, and $\omega \in (\mathbb{F}_{q^{12}}^\times)^5$), and then, with $y_0^5 = \lambda_5^{-1}$, $\ker M = y_0 \mathbb{F}_{16}$ since $\mu_{15} \cup \{0\} = \mathbb{F}_{16} \subset \mathbb{F}_{q^{12}}$.

*Case $4 \mid k$, $\lambda_5 \notin (\mathbb{F}_q^\times)^5$.* $\ker M = 0$, the radical of the $\beta$-form is $\ker\theta = \mathbb{F}_{q^2}$ of dimension $2k$, $Q$ vanishes on it, so $|W| = 2^{(12k+2k)/2} = q^7$. Each $S(\lambda_1, 0, \lambda_5)$ has trivial radical, hence $|S| = q^6$ for all $q$ values of $\lambda_1$; $|W| = q \cdot q^6$ forces all signs to agree with that of $S(0, 0, \lambda_5) = -\varepsilon_0 q^6 = +q^6$ by (8). So $W = q^7$.

*Case $\lambda_5 \in (\mathbb{F}_{q^{12}}^\times)^5$, $k$ even.* If $4 \mid k$ choose $y_0 \in \mathbb{F}_q$; if $k \equiv 2 \pmod 4$ then $5 \mid q + 1$, so every element of $\mathbb{F}_q$ is a fifth power in $\mathbb{F}_{q^2}$ and we choose $y_0 \in \mathbb{F}_{q^2}$. Also $\mathbb{F}_{16} \subset \mathbb{F}_{q^2}$. Hence $\ker M = y_0 \mathbb{F}_{16} \subset \mathbb{F}_{q^2} \subset \mathrm{im}\,\theta$ and the radical $R = \theta^{-1}(y_0 \mathbb{F}_{16})$ has dimension $2k+4$. For $y \in y_0 \mathbb{F}_{16}$ pick $\beta_y \in \mathbb{F}_{q^4}$ with $\mathrm{Tr}_{q^4/q^2}\,\beta_y = y^q$ (as in Lemma 6.2); then $Q(\beta_y) = \mathrm{Tr}_{q^2/2}(\lambda_5 y^{4q+1})$. Write $y = y_0 c$, $c \in \mathbb{F}_{16}$, and $\rho = y_0^{q-1} \in \mu_5$ ($\rho = 1$ if $4 \mid k$). Then $\lambda_5 y^{4q+1} = \rho^4 c^{4q+1}$, and $c^{4q+1} = c^5$ if $q \equiv 1 \pmod{15}$ ($4 \mid k$), $= c^2$ if $q \equiv 4 \pmod{15}$ ($k \equiv 2 \pmod 4$). Since $c \in \mathbb{F}_{16} \subset \mathbb{F}_{q^2}$, $\mathrm{Tr}_{q^2/2}(x) = \frac{k}{2}\,\mathrm{Tr}_{16/2}(x)$ for $x \in \mathbb{F}_{16}$. If $4 \mid k$ this is 0: $Q|_R = 0$, $|W| = 2^{(14k+4)/2} = 4q^7$; on the $x$-side the radical $y_0 \mathbb{F}_{16} \subset \mathbb{F}_q$ carries $Q_x(y_0 c) = \mathrm{Tr}_{q^{12}/2}(c^5) = 3k\,\mathrm{Tr}_{16/2}(c^5) = 0$ ($c^5 \in \mathbb{F}_4$)

and the linear forms $\mathrm{Tr}(\lambda_1 y_0 c) = \mathrm{Tr}_{q/2}(12\lambda_1 y_0 c) = 0$, so all $q$ sums $S(\lambda_1, 0, \lambda_5)$ have modulus $2^{(12k+4)/2} = 4q^6$ and $|W| = q \cdot 4q^6$ forces equal signs, that of $S(0,0,\lambda_5) = 4\varepsilon_0 q^6 = -4q^6$: $W = -4q^7$. If $k \equiv 2 \pmod 4$ then $Q(\beta_{y_0 c}) = \mathrm{Tr}_{16/2}(\rho^4 c^2) = \mathrm{Tr}_{16/2}(\rho^2 c)$, which is nonzero for suitable $c$; so $Q|_R \neq 0$ and $W = 0$.

*Case $k$ odd.* Fifth powers are bijective on $\mathbb{F}_q^\times$, so $y_0 \in \mathbb{F}_q$. Now $\mathbb{F}_{16} \not\subset \mathbb{F}_{q^2}$ and $q^2 \equiv 4 \pmod{15}$, so for $c \in \mathbb{F}_{16}$, $\mathrm{Tr}_{q^{12}/q^2}(y_0 c) = y_0 \sum_{i<6} c^{4^i} = y_0(c + c^4)$, which vanishes iff $c \in \mathbb{F}_4$. Thus $\ker M \cap \mathrm{im}\,\theta = y_0 \mathbb{F}_4$ and $\dim R = 2k+2$. For $y = y_0 c$, $c \in \mathbb{F}_4^\times$: $Q(\beta_y) = \mathrm{Tr}_{q^2/2}(\lambda_5 y_0^5 c^{4q+1}) = \mathrm{Tr}_{q^2/2}(1) = 0$ because $4q+1 \equiv q+1 \equiv 0 \pmod 3$. Hence $|W| = 2^{(14k+2)/2} = 2q^7$. On the $x$-side, $Q_x(y_0 c) = \mathrm{Tr}_{q^{12}/2}(c^5) = 0$ as before, while $\mathrm{Tr}_{q^{12}/2}(\lambda_1 y_0 c) = \mathrm{Tr}_{q/2}(\lambda_1 y_0)\,\mathrm{Tr}_{16/2}(c)$ (the twelve conjugates of $c$ under $q$-Frobenius run three times through $c, c^2, c^4, c^8$); so exactly the $q/2$ values of $\lambda_1$ with $\mathrm{Tr}_{q/2}(\lambda_1 y_0) = 0$ give $|S| = 4q^6$, the others give 0, and $|W| = \frac{q}{2} \cdot 4q^6$ forces equal signs, that of $S(0,0,\lambda_5) = 4\varepsilon_0 q^6 = +4q^6$. So $W = 2q^7$. □

Lemmas 6.2 and 6.4 give, for the pure terms, $\sum_{\lambda_3 \neq 0} W(\lambda_3, 0) + \sum_{\lambda_5 \neq 0} W(0, \lambda_5) = 2(q-1)q^7$ for $k$ odd and $= -2(q-1)q^7$ for $k$ even (for $4 \mid k$ the quintic terms contribute $\frac{q-1}{5}(-4q^7) + \frac{4(q-1)}{5} q^7 = 0$). Hence Conjecture 6.5 for $A_6$ is equivalent to

$$\sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = \begin{cases} (q-1)(q-2)\, q^7, & k \text{ odd}, \\ -(q-1)(q-2)\, q^7, & k \text{ even}. \end{cases} \tag{9}$$

6.4. **Exact evaluation and the conjecture.** Every $W, W'$ is the exponential sum of an $\mathbb{F}_2$-quadratic form (plus a linear form) on $\mathbb{F}_2^{12k}$, hence computable exactly by linear algebra: radical, test of $Q + L$ on it, a symplectic basis of a complement and its Arf invariant. We implemented this (`legendre_d6_qf.py`) with $\mathbb{F}_{q^{12}} = \mathbb{F}_q[z]/(m)$ and the maps $\beta \mapsto \beta^2, \beta^{1/2}, \sigma, \theta$ as $12k \times 12k$ matrices over $\mathbb{F}_2$.

**Conjecture 6.5.** *For all $q = 2^k$: $A_6 = q^9 + \varepsilon_0(q-1)q^5$ and $X = \varepsilon_0\, q^6 + q^5$, i.e.*

$$\sum_{\lambda_5 \neq 0} W(0, \lambda_5) + \sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = (q-1)q^7(\varepsilon_0 q + 2), \qquad \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5) = q^3(\varepsilon_0 q^6 + q^5).$$

The statement for $A_6$ with $k$ odd is proved below (Corollary 6.13); the statement for $X$ (analysed in §6.7), and that for $A_6$ with $k$ even, remain conjectural.

TABLE 1. Exact values of $A_6$ and $X$ by the quadratic-form algorithm ($N = 12k$ bits). For $k \leq 5$ the values coincide with the Hayes-group FFT of [5]; $k = 6, 7$ are beyond its reach at $d = 6$. In all cases $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$.

| $k$ | $q$ | $N$ | $A_6$ | $X$ | time |
|---|---|---|---|---|---|
| 1 | 2 | 12 | 544 | 96 | $< 1$ s |
| 2 | 4 | 24 | 259 072 | −3 072 | $< 1$ s |
| 3 | 8 | 36 | 134 447 104 | 294 912 | $< 1$ s |
| 4 | 16 | 48 | 68 703 748 096 | −15 728 640 | 2 s |
| 5 | 32 | 60 | 35 185 412 276 224 | 1 107 296 256 | 9 s |
| 6 | 64 | 72 | 18 014 330 863 747 072 | −67 645 734 912 | 48 s |
| 7 | 128 | 84 | 9 223 376 400 541 548 544 | 4 432 406 249 472 | 225 s |

**Proposition 6.6** (structure of the data)**.** *For $k \leq 7$ the following hold.*

(a) *$k$ odd: $W(0, \lambda_5) = 2q^7$ for all $\lambda_5 \neq 0$; the mixed $W$ depends only on $j = \lambda_3^5/\lambda_5^3 \in \mathbb{F}_q^\times$ and equals $2q^7$ for exactly $(q-2)/2$ values of $j$ and 0 otherwise. Hence $\sum W = (q-1)\,2q^7 + \frac{(q-1)(q-2)}{2}\,2q^7 = (q-1)q^8$.*

TABLE 2. Distribution of $W(\lambda_3,\lambda_5)/q^7$ over the $q^2-1$ nontrivial pairs (radical dimension in parentheses); "pure cubic" $\lambda_5=0$, "pure quintic" $\lambda_3=0$, "mixed" $\lambda_3\lambda_5\neq 0$.

| $k$ | pure cubic $(q-1)$ | pure quintic $(q-1)$ | mixed $((q-1)^2)$ |
|---|---|---|---|
| 1 | 0 (4) | 2 (4) | 0 (6) ×1 |
| 3 | 0 (8) | 2 (8) | 2 (8) ×21; 0 (10) ×28 |
| 5 | 0 (12) | 2 (12) | 2 (12) ×465; 0 (14) ×496 |
| 7 | 0 (16) | 2 (16) | 2 (16) ×8001; 0 (18) ×8128 |
| 2 | −2 (6) | 0 (8) | −4 (8) ×3; 1 (4) ×6 |
| 4 | −2 (10) | −4 (12) ×3; 1 (8) ×12 | −4 (12) ×75; 1 (8) ×90; 0 (12) ×60 |
| 6 | −2 (14) | 0 (16) | −4 (16) ×1386; 1 (12) ×1638; 0 (16) ×945 |

(b) *$k$ even: $W(0,\lambda_5)=0$ if $k\equiv 2 \pmod 4$, while for $k\equiv 0 \pmod 4$ (so $5\mid q-1$) it equals $-4q^7$ for the $(q-1)/5$ fifth powers and $+q^7$ otherwise; the mixed $W$ take values $-4q^7$, $q^7$, $0$ with multiplicities $(q-1)(n_-,n_+,n_0)$, $(n_-,n_+,n_0)=(1,2,0),(5,6,4),(22,26,15)$ for $k=2,4,6$, satisfying $n_+-4n_-=2-q$.*

*The set of $j$ in (a) is not of the form $\{\mathrm{Tr}_{q/2}(j^m)=c\}$ for any $m$ when $k=5,7$ (it is $\{\mathrm{Tr}\, j=0\}$ when $k=3$).*

6.5. **The mixed part for odd $k$: a Gold-type quintic and a $\mathbb{Z}/4$ condition.** For $k$ odd we describe the mixed terms completely in terms of the factorisation of a single quintic over $\mathbb{F}_q$; the evaluation is completed in Theorem 6.12 below.

**Proposition 6.7.** *Let $k$ be odd, $\lambda_3\lambda_5\neq 0$, $j=\lambda_3^5/\lambda_5^3$, $t=j^{1/5}\in\mathbb{F}_q$ and $c=j^{-1/2}\in\mathbb{F}_q^\times$ (fifth and square roots are unique in $\mathbb{F}_q$).*

(a) *$W(\lambda_3,\lambda_5)$ depends only on $j$ (the orbits of $(\lambda_3,\lambda_5)\mapsto(s^3\lambda_3,s^5\lambda_5)$, $s\in\mathbb{F}_q^\times$, are the fibres of $j$).*

(b) *With $y=\lambda_5^{-1/5}u$, the kernel equation $\lambda_5^4y^{15}+\lambda_3^4y^7+\lambda_3^2y+\lambda_5=0$ becomes $P_j(u):=u^{15}+t^4u^7+t^2u+1=0$, and*

$$P_j(u)=(u^5+t^2u+1)(u^{10}+t^2u^6+u^5+1),\qquad u=t^{1/2}v\implies u^5+t^2u+1=t^{5/2}(v^5+v+c).$$

*The five roots $u_0,\dots,u_4$ of $u^5+t^2u+1$ have sum $0$, are $\mathbb{F}_2$-linearly independent apart from this relation, and $\ker L_\lambda=\{0\}\cup\{u_i\}\cup\{u_i+u_{i'}\}$ (the ten roots of the second factor are the pairwise sums). The $q$-Frobenius acts on $\ker L_\lambda$ through its permutation action on the roots $v_i$ of $v^5+v+c$, i.e. through the standard embedding $S_5\to GL_4(\mathbb{F}_2)$ on the sum-zero module.*

(c) *Let $g\in S_5$ be the Frobenius permutation of the roots of $v^5+v+c$ (its cycle type is the factorisation type of $v^5+v+c$ over $\mathbb{F}_q$). Then $\ker L_\lambda\subset\mathbb{F}_{q^{12}}$ iff $g$ has no 5-cycle, $\dim_{\mathbb{F}_2}(\ker L_\lambda\cap\operatorname{im}\theta)=2$ if $g$ is a 4-cycle (type $(1,4)$) and $=4$ for types $(1,1,1,2)$, $(2,3)$, $(1,1,1,1,1)$, $(1,2,2)$, $(1,1,3)$, and the image of $\mathrm{Tr}_{q^{12}/q}$ on $\ker L_\lambda$ is one-dimensional over $\mathbb{F}_2$ if $g$ is a 4-cycle and zero otherwise. Consequently, for a 4-cycle, $|W|\in\{0,2q^7\}$ and exactly $q/2$ of the sums $S(\lambda_1,\lambda_3,\lambda_5)$ are nonzero (with modulus $4q^6$); for the other types $|W|\in\{0,4q^7\}$ and either all or none of the $S(\lambda_1,\cdot)$ vanish.*

*Proof.* (a) For $s\in\mathbb{F}_q^\times$, $\beta\mapsto s\beta$ sends $\mathrm{Tr}(\lambda_3\beta^2\theta\beta+\lambda_5\beta^4\theta\beta)$ to the form with $(s^3\lambda_3,s^5\lambda_5)$; since $3,5\nmid q-1$ the map $s\mapsto(s^3,s^5)$ is injective, so the orbits have size $q-1$ and are the fibres of $j$. (b) Substitute $y=\lambda_5^{-1/5}u$ and divide by $\lambda_5$. The factorisation is checked by expanding: $u^5P_j(u)=(u^{10}+t^2u^6+u^5)^2+(u^{10}+t^2u^6+u^5)$. The second factor equals $u^5A(u)+1$ with $A=u^5+t^2u+1$, so its roots are not roots of $A$; $A$ is separable ($A'=(u+\sqrt{t})^4$, $A(\sqrt{t})=1$) with $e_1=0$; $\ker L_\lambda$ is a 4-dimensional $\mathbb{F}_2$-space containing the $u_i$, and a relation $u_i+u_{i'}+u_{i''}=0$ would force two of the remaining roots to coincide, while $u_i+u_{i'}+u_{i''}+u_{i'''}=0$ would force the fifth root to be 0; hence the pairwise sums are 10 distinct nonzero kernel elements outside $\{u_i\}$, which

exhausts the kernel. Frobenius commutes with the identification. (c) Rationality over $\mathbb{F}_{q^{12}}$ means $g^{12}=1$. On the sum-zero module $V\cong\mathbb{F}_2[\mathbb{Z}/4]$ of a 4-cycle, $\sum_{i<12}g^i=3(1+g+g^2+g^3)=(1+g)^3$ has rank 1 and $\sum_{i<6}g^{2i}=1+g^2=(1+g)^2$ has kernel of dimension 2; for $g$ of order $1,2,3,6$ both operators vanish on $V$ (the sums have an even number of equal terms), giving trace image 0 and $\ker\cap\operatorname{im}\theta=\ker$. The last sentence follows from the radical dimensions $2k+\dim(\ker\cap\operatorname{im}\theta)$ for $W$ and $\dim(\ker L_\lambda)=4$ for the $S(\lambda_1,\cdot)$, together with the criterion $S\neq 0$ iff $\operatorname{Tr}(\lambda_1\,\cdot)=Q_x$ on $\ker L_\lambda$: for a 4-cycle the functionals $\operatorname{Tr}(\lambda_1\cdot)|_{\ker}$ form a one-dimensional space and $Q_x|_{\ker}$ lies in it (both are determined by whether $\operatorname{Tr}_{q^{12}/q}(x)$ vanishes, as $Q_x$ is additive on the radical and takes the same value on the four roots of degree 4), so exactly half of the $\lambda_1$ qualify. □

**Proposition 6.8** (the quintic $v^5+v+c$, $k$ odd)**.** *Let $k$ be odd and $c\in\mathbb{F}_q^\times$. The factorisation type of $v^5+v+c$ over $\mathbb{F}_q$ is one of $(1,4)$, $(2,3)$, $(1,1,1,2)$, and these occur for exactly*

$$N_{(1,4)}=\frac{q-2}{2},\qquad N_{(2,3)}=\frac{q+1}{3},\qquad N_{(1,1,1,2)}=\frac{q-2}{6}$$

*values of $c$. In particular the number of $\mathbb{F}_q$-roots is $1$, $0$ or $3$ with these multiplicities, in agreement with the theorem of Helleseth–Kholosha [12] on $x^{2^l+1}+x+a$, $\gcd(l,k)=1$ (here $l=2$), which rests on Bluher's classification [3] of the roots of $x^{q+1}+ax+b$ (the $q$ of Bluher's title is a power of the characteristic, here 4); and the quintic is never irreducible.*

*Proof.* Put $t=c^{-2/5}$ (a bijection of $\mathbb{F}_q^\times$) and $A(u)=u^5+t^2u+1$, so that $u=t^{1/2}v$ identifies the roots of $v^5+v+c$ with those of $A$ (Proposition 6.7(b)) and $K=\ker L_\lambda\subset\bar{\mathbb{F}}_q$, $\lambda=(\cdot,t,1)$, is the sum-zero $\mathbb{F}_2$-span of the five roots $u_0,\dots,u_4$ of $A$, on which the $q$-Frobenius $\sigma$ acts through its permutation $g\in S_5$ of the roots.

*Only three types occur.* The function $x\mapsto\operatorname{Tr}_{q/2}(x^5+tx^3)$ on $\mathbb{F}_q$ is an $\mathbb{F}_2$-quadratic form with polar form $\operatorname{Tr}_{q/2}(y\,T_\lambda(x))$, whose radical is $\{x\in\mathbb{F}_q:T_\lambda(x)=0\}=K^\sigma$, the $\sigma$-fixed subspace of $K$. Since the polar form is alternating and nondegenerate on $\mathbb{F}_q/K^\sigma$, $\dim K^\sigma\equiv k\equiv 1\pmod 2$. On the sum-zero module of $S_5$ the fixed space of $g$ has dimension $1,1,3$ for the cycle types $(1,4),(2,3),(1,1,1,2)$ and $0,2,2,4$ for $(5),(1,1,3),(1,2,2),(1,1,1,1,1)$ (count the $g$-stable subsets of $\{0,\dots,4\}$ modulo complementation). Hence only the first three types occur.

*Counting.* (i) The map $v\mapsto v^5+v$ sends $\mathbb{F}_q$ onto the values $c$, with $v^5+v=0$ iff $v\in\{0,1\}$ (as $v^4=1$ forces $v=1$); so $N_{(1,4)}+3N_{(1,1,1,2)}=q-2$. (ii) Let $v\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$, $\bar v=v^q$, $s=v+\bar v\neq 0$, $n=v\bar v$. Then $\bar v^5+\bar v+v^5+v=(v+\bar v)(s^4+ns^2+n^2+1)$, so $v^5+v\in\mathbb{F}_q$ iff $s^4+ns^2+n^2=1$. Writing $n=s^2m$, irreducibility of $v^2+sv+n$ means $\operatorname{Tr}_{q/2}(m)=1$, and the condition becomes $m^2+m=1+s^{-4}$, which is solvable iff $\operatorname{Tr}_{q/2}(1+s^{-4})=0$ iff $\operatorname{Tr}_{q/2}(s^{-1})=1$ ($k$ odd), in which case exactly one of the two solutions $m,m+1$ has $\operatorname{Tr}m=1$. Hence exactly $2\cdot\#\{s\in\mathbb{F}_q^\times:\operatorname{Tr}(1/s)=1\}=q$ elements $v\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$ satisfy $v^5+v\in\mathbb{F}_q$, necessarily in $\mathbb{F}_q^\times$. A quintic of type $(1,4)$, $(2,3)$, $(1,1,1,2)$ has $0,2,2$ roots in $\mathbb{F}_{q^2}\setminus\mathbb{F}_q$, so $2N_{(2,3)}+2N_{(1,1,1,2)}=q$. (iii) $N_{(1,4)}+N_{(2,3)}+N_{(1,1,1,2)}=q-1$. Solving (i)–(iii) gives the three counts. □

6.6. **The mixed part for odd $k$: the theorem.** Throughout this subsection $k$ is odd, $(\lambda_3,\lambda_5)=(t,1)$ with $t\in\mathbb{F}_q^\times$ (every class $j=t^5$ has such a representative, Proposition 6.7(a)), $A(u)=u^5+t^2u+1$ with roots $u_0,\dots,u_4$, $K=\ker L_\lambda=\ker T_\lambda\subset\mathbb{F}_{q^{12}}$ (Proposition 6.7(b)), $T=T_\lambda$, $\sigma$ is the $q$-Frobenius and $\tau=\operatorname{Tr}_{q^{12}/q}$. For $b\in\mathbb{F}_q$ put

$$S(b)=S(b,t,1)=\sum_{x\in\mathbb{F}_{q^{12}}}\psi_0(x^5+tx^3+bx),\qquad Q_b(x)=\operatorname{Tr}_{q^{12}/2}(x^5+tx^3+bx),$$

so that $W(t,1)=\sum_{b\in\mathbb{F}_q}S(b)$ by (7), and $S(b)\in\{0,\pm 4q^6\}$ since $\dim K=4$.

**Lemma 6.9** (shifts)**.** *Let $V=T^{-1}(\mathbb{F}_q)=\{c\in\mathbb{F}_{q^{12}}:T(c)\in\mathbb{F}_q\}$.*

(a) *For $c\in V$ and $b\in\mathbb{F}_q$: $S(b)=(-1)^{Q_b(c)}\,S(b+T(c))$.*
(b) *$c\in V$ iff $(\sigma-1)c\in K$, and then $\tau(c)=(1+\sigma^2+\sigma^4+\cdots+\sigma^{10})\,(\sigma-1)c$.*
(c) *If $c\in V$ and $\tau(c)=0$, then $W(t,1)=(-1)^{Q_0(c)}\,W(t,1)$; in particular $W(t,1)=0$ if $Q_0(c)=1$.*

*Proof.* (a) $Q_b(x+c) = Q_b(x) + Q_b(c) + \mathrm{Tr}(x\,T(c))$ and $\mathrm{Tr}_{q^{12}/2}(xT(c)) = \mathrm{Tr}_{q^{12}/2}(T(c)x)$ with $T(c) \in \mathbb{F}_q$; substitute $x \mapsto x+c$ in $S(b)$. (b) $L_\lambda$ has coefficients in $\mathbb{F}_q$ and commutes with $\sigma$, so $L_\lambda(c) \in \mathbb{F}_q$ iff $L_\lambda((\sigma-1)c) = 0$; and $T(c) \in \mathbb{F}_q$ iff $T(c)^4 = L_\lambda(c) \in \mathbb{F}_q$. With $c_0 = (\sigma-1)c$ one has $\sigma^i c = c + \sum_{m<i} \sigma^m c_0$, hence $\tau(c) = 12c + \sum_{m\le 10}(11-m)\sigma^m c_0 = \sum_{m \text{ even}} \sigma^m c_0$. (c) Sum (a) over $b \in \mathbb{F}_q$: $Q_b(c) = Q_0(c) + \mathrm{Tr}_{q/2}(b\,\tau(c)) = Q_0(c)$, and $b \mapsto b + T(c)$ permutes $\mathbb{F}_q$. □

**Lemma 6.10** (a quadratic factor forces $W = 0$)**.** *Suppose $A(u)$ has an irreducible quadratic factor $u^2 + \kappa u + N$ over $\mathbb{F}_q$ (types $(2,3)$ and $(1,1,1,2)$). Then $N = \kappa^{-3}$, $t = \kappa^2 + \kappa^{-1/2} + \kappa^{-3}$ and $\mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Let $u$ be a root of the quadratic factor and $c \in \mathbb{F}_{q^4}$ with $c^q + c = u$. Then $c \in V$, $\tau(c) = 0$ and $Q_0(c) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Consequently $W(t,1) = 0$. Moreover every $S(b)$, $b \in \mathbb{F}_q$, equals $\pm 4q^6$, and $S(b) = -S(b + T(c))$: the $q$ sums cancel in pairs.*

*Proof.* Dividing, $A = (u^2 + \kappa u + N)(u^3 + \kappa u^2 + (\kappa^2 + N)u + \kappa^3)$ with $N\kappa^3 = 1$ and $t^2 = \kappa^4 + N\kappa^2 + N^2 = \kappa^4 + \kappa^{-1} + \kappa^{-6}$; the quadratic is irreducible iff $\mathrm{Tr}_{q/2}(N/\kappa^2) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Since $u \in \mathbb{F}_{q^2}$, $\mathrm{Tr}_{q^4/q}(u) = 2(u + u^q) = 0$ and $c$ exists. By Lemma 6.9(b), $(\sigma-1)c = u \in K$ gives $c \in V$, and $\tau(c) = (1 + \sigma^2 + \cdots + \sigma^{10})u = 6u = 0$. The conjugates of $c$ over $\mathbb{F}_q$ are $c$, $c+u$, $c+\kappa$, $c+u'$ with $u' = u^q = u + \kappa$. As $[\mathbb{F}_{q^{12}} : \mathbb{F}_{q^4}] = 3$ is odd, $Q_0(c) = \mathrm{Tr}_{q^4/2}(c^5 + tc^3) = \mathrm{Tr}_{q/2} \sum_i \big(c_i^5 + tc_i^3\big)$ over the four conjugates $c_i = c + a_i$, $a_i \in \{0, u, \kappa, u'\}$. In characteristic 2, $(c+a)^3 = c^3 + c^2 a + ca^2 + a^3$ and $(c+a)^5 = c^5 + c^4 a + ca^4 + a^5$, and $\sum a_i = \sum a_i^2 = \sum a_i^4 = 0$, so $\sum_i c_i^3 = \sum a_i^3 = u^3 + u'^3 + \kappa^3 = \kappa N = \kappa^{-2}$ and $\sum_i c_i^5 = u^5 + u'^5 + \kappa^5 = \kappa^3 N + \kappa N^2 = 1 + \kappa^{-5}$ (Newton's identities for $u + u' = \kappa$, $uu' = N$). Hence

$$Q_0(c) = \mathrm{Tr}_{q/2}\big(1 + \kappa^{-5} + t\kappa^{-2}\big), \qquad t\kappa^{-2} = 1 + \kappa^{-5/2} + \kappa^{-5},$$

and $\mathrm{Tr}(1) = 1$, $\mathrm{Tr}(\kappa^{-5/2}) = \mathrm{Tr}(\kappa^{-5})$ give $Q_0(c) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Lemma 6.9(c) gives $W = 0$. For the last assertion: by Proposition 6.7(c) the trace $\tau$ vanishes on $K$ in these types, so $\mathrm{Tr}_{q^{12}/2}(bx) = 0$ for $x \in K$; and $Q_0$ vanishes on $K$ because it is additive there and each root $u_i$ lies in a subfield $\mathbb{F}_q, \mathbb{F}_{q^2}$ or $\mathbb{F}_{q^3}$ of even index in $\mathbb{F}_{q^{12}}$. By Theorem 6.1, $S(b) \neq 0$ for all $b$, and Lemma 6.9(a) with $Q_b(c) = 1$ gives the pairing (note $T(c) \notin T(\mathbb{F}_q)$, since otherwise $c \in \mathbb{F}_q + K$ and $Q_0(c)$ would vanish). □

**Lemma 6.11** (the 4-cycle: vanishing over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ fixes the sign)**.** *Suppose $A$ has type $(1,4)$: $u_0 \in \mathbb{F}_q$ and $\sigma$ cycles $u_1 \to u_2 \to u_3 \to u_4 \to u_1$. Put $\kappa = 1 + \mathrm{Tr}_{q/2}(u_0^5 + tu_0^3) \in \mathbb{F}_2$ and, for $b \in \mathbb{F}_q$ and $m \ge 1$, $s_m(b) = \sum_{x \in \mathbb{F}_{q^m}} \psi_0(x^5 + tx^3 + bx)$, so that $S(b) = s_{12}(b)$.*

(a) *$S(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(bu_0) = \kappa$ iff $s_1(b) = 0$; this holds for exactly $q/2$ values of $b$.*
(b) *If $s_1(b) = 0$ then $s_2(b) = 0$.*
(c) *For these $b$ the $L$-polynomial of $C_b : y^2 + y = x^5 + tx^3 + bx$ over $\mathbb{F}_q$ is $1 + q^2T^4$, hence $S(b) = +4q^6$. Therefore $W(t,1) = \frac{q}{2} \cdot 4q^6 = 2q^7$.*

*Proof.* (a) By Theorem 6.1, $S(b) \neq 0$ iff $\mathrm{Tr}_{q^{12}/2}(bx) = Q_0(x)$ for all $x \in K$; both sides are additive on $K$. For $x = u_0 \in \mathbb{F}_q$ both sides vanish ($\mathrm{Tr}_{q^{12}/q}$ is $12 \cdot \mathrm{id}$ on $\mathbb{F}_q$). For $i \ge 1$, $u_i \in \mathbb{F}_{q^4}$ and $\tau(u_i) = 3\,\mathrm{Tr}_{q^4/q}(u_i) = u_1 + u_2 + u_3 + u_4 = u_0$, so $\mathrm{Tr}_{q^{12}/2}(bu_i) = \mathrm{Tr}_{q/2}(bu_0)$, while $Q_0(u_i) = \mathrm{Tr}_{q/2}\big(\sum_{i' \ge 1}(u_{i'}^5 + tu_{i'}^3)\big) = \mathrm{Tr}_{q/2}\big(p_5 - u_0^5 + t(p_3 - u_0^3)\big)$ with $p_m$ the power sums of all five roots of $A$; Newton's identities for $A$ ($e_1 = e_2 = e_3 = 0$, $e_4 = t^2$, $e_5 = 1$) give $p_3 = 0$, $p_5 = 5e_5 = 1$, so $Q_0(u_i) = \mathrm{Tr}_{q/2}(1 + u_0^5 + tu_0^3) = \kappa$. Thus $S(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(bu_0) = \kappa$ iff $\mathrm{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 1$. On the other hand $x \mapsto \mathrm{Tr}_{q/2}(x^5 + tx^3 + bx)$ is a quadratic form on $\mathbb{F}_q$ with polar form $\mathrm{Tr}_{q/2}(yT(x))$ and radical $K^\sigma = \{0, u_0\}$, so $s_1(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 0$.

(b) Over $\mathbb{F}_{q^2}$ the radical is $K^{\sigma^2} = \{0, u_0, w, w'\}$ with $w = u_1 + u_3$, $w' = \sigma w = u_2 + u_4 = w + u_0$, and $s_2(b) \neq 0$ iff $\mathrm{Tr}_{q^2/2}(x^5 + tx^3 + bx)$ vanishes at $x = u_0$ and $x = w$. At $u_0$ it does. Let $N_2 = ww' \in \mathbb{F}_q$ and let $\rho \in \mathbb{F}_q$ be the cube root of $u_0$ ($3 \nmid q-1$). We claim $N_2 = \rho$. Dividing $A$ by $u + u_0$ and using $u_0^5 + t^2 u_0 + 1 = 0$, the quartic with roots $u_1, \dots, u_4$ is $B(u) = u^4 + u_0 u^3 + u_0^2 u^2 + u_0^3 u + u_0^{-1}$; its resolvent cubic, with roots $u_1u_2 + u_3u_4$, $u_1u_3 + u_2u_4$, $u_1u_4 + u_2u_3$, is $\theta^3 + u_0^2\theta^2 + u_0^4\theta + u_0^6 + u_0 = (\theta + u_0^2)^3 + u_0$, whose unique root in $\mathbb{F}_q$ is $u_0^2 + \rho$; the $\sigma$-invariant root is $M = u_1u_3 + u_2u_4$, so $M = u_0^2 + \rho$ and $N_2 = (u_1+u_3)(u_2+u_4) = e_2(B) + M = u_0^2 + M = \rho$.

Now with $w+w'=u_0$, $ww'=\rho$: $w^3+w'^3=u_0^3+u_0\rho$ and $w^5+w'^5=u_0^5+u_0^3\rho+u_0\rho^2$, so

$$\begin{aligned}\mathrm{Tr}_{q^2/2}(w^5+tw^3+bw) &= \mathrm{Tr}_{q/2}(u_0^5+tu_0^3+bu_0)+\mathrm{Tr}_{q/2}\big(u_0^3\rho+u_0\rho^2+tu_0\rho\big)\\ &= 1+\mathrm{Tr}_{q/2}\big(\rho^{10}+\rho^5+t\rho^4\big)=1+\mathrm{Tr}_{q/2}(t\rho^4).\end{aligned}$$

From $u_0^5+t^2u_0+1=0$, $t^2=u_0^4+u_0^{-1}=\rho^{12}+\rho^{-3}$, so $t=\rho^6+\rho^{-3/2}$ and $t\rho^4=\rho^{10}+\rho^{5/2}$, whose trace is $\mathrm{Tr}(\rho^5)+\mathrm{Tr}(\rho^5)=0$. Hence the form takes the value 1 at $w$ and $s_2(b)=0$.

(c) With $\omega_1,\dots,\omega_4$ the Frobenius eigenvalues of $C_b$ (genus 2), $s_m(b)=-\sum_i\omega_i^m$ (Theorem 6.1), so $e_1=\sum\omega_i=-s_1=0$ and $e_2=\frac12\big(e_1^2-\sum\omega_i^2\big)=\frac12(s_1^2+s_2)=0$; the functional equation gives $e_3=qe_1=0$, $e_4=q^2$. So $\prod(1-\omega_iT)=1+q^2T^4$, $\omega_i^4=-q^2$, $\omega_i^{12}=-q^6$ and $S(b)=-\sum\omega_i^{12}=4q^6$. Summing over the $q/2$ values of $b$ in (a) gives $W(t,1)=2q^7$. □

**Theorem 6.12** (the mixed part, $k$ odd)**.** *Let $k$ be odd, $\lambda_3\lambda_5\neq0$, $j=\lambda_3^5/\lambda_5^3$ and $c=j^{-1/2}$. Then*

$$W(\lambda_3,\lambda_5)=\begin{cases}2q^7, & v^5+v+c \text{ has type } (1,4),\\ 0, & v^5+v+c \text{ has type } (2,3) \text{ or } (1,1,1,2),\end{cases}$$

*and no other type occurs. In the 4-cycle case exactly $q/2$ of the sums $S(\lambda_1,\lambda_3,\lambda_5)$, $\lambda_1\in\mathbb{F}_q$, are nonzero and each equals $+4q^6$; in the other two cases all $q$ sums equal $\pm4q^6$ and cancel in pairs. Equivalently, in the $\beta$-form of (7): $Q$ vanishes on the radical $\theta^{-1}(\ker L_\lambda\cap\mathbb{F}_{q^2})$ (of dimension $2k+2$) with Arf sign $+$ in the 4-cycle case, and $Q$ does not vanish on the radical (of dimension $2k+4$) otherwise.*

*Proof.* By Proposition 6.7(a),(b) we may take $(\lambda_3,\lambda_5)=(t,1)$, $t=j^{1/5}$, and the type of $v^5+v+c$ is the type of $A(u)=u^5+t^2u+1$. Proposition 6.8 restricts the types to the three listed (the scaling $(\lambda_1,\lambda_3,\lambda_5)\mapsto(s\lambda_1,s^3\lambda_3,s^5\lambda_5)$ preserves the number and signs of the nonzero $S$). Lemma 6.10 covers types $(2,3)$ and $(1,1,1,2)$, Lemma 6.11 the type $(1,4)$. For the $\beta$-form: by Proposition 6.7(c) the radical has dimension $2k+2$ or $2k+4$, so $|W|\in\{0,2q^7\}$ or $\{0,4q^7\}$ respectively, and $W=2q^7$ (resp. $W=0$) is equivalent to the stated behaviour of $Q$ on the radical. □

**Corollary 6.13.** *For all odd $k$: $\sum_{\lambda_3\lambda_5\neq0}W(\lambda_3,\lambda_5)=(q-1)(q-2)q^7$ and*

$$A_6=|U|=q^9+(q-1)q^5.$$

*That is, Conjecture 6.5 holds for $A_6$ whenever $k$ is odd.*

*Proof.* Each class $j\in\mathbb{F}_q^\times$ consists of $q-1$ pairs $(\lambda_3,\lambda_5)$ (Proposition 6.7(a)), and the 4-cycle classes number $(q-2)/2$ (Proposition 6.8); Theorem 6.12 gives $\sum_{\text{mixed}}W=(q-1)\frac{q-2}{2}2q^7$, which is (9) for odd $k$. Combine with Lemmas 6.2, 6.4 and (7). □

Theorem 6.12 was checked in two independent ways: for $k\le7$ by the exact $\mathbb{F}_2$-quadratic-form algorithm (`legendre_d6_qf_mixed.py`: every $j$, radical test and signs), and for $k\le9$ through the $L$-polynomials (`kasami_sign.py`): for every $j\in\mathbb{F}_q^\times$ and every $b\in\mathbb{F}_q$ the sums $s_1(b),s_2(b)$ over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ determine $\prod(1-\omega_iT)$ and hence $S(b)=-\sum\omega_i^{12}$; the resulting multisets $\{S(b)\}$ and the values of $W$ agree with the theorem in every case, and in the 4-cycle case the predicate $\mathrm{Tr}_{q/2}(bu_0)=\kappa$ of Lemma 6.11(a) matches $S(b)\neq0$ exactly. The types of Proposition 6.8 were enumerated for $k\le11$ (`quintic_check.py`).

The condition "$g$ is a 4-cycle" is a $\mathbb{Z}/4$-condition (the Frobenius must be a regular unipotent on the kernel), not an $\mathbb{F}_2$-trace condition; this is why no predicate $\mathrm{Tr}(f(j))=0$ describes the set of $j$ with $W\neq0$ (Proposition 6.6). Note also that the sign in the 4-cycle case is *not* obtained from an Arf-invariant computation on $\mathbb{F}_2^{12k}$ but from the two small sums $s_1,s_2$ over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$, which vanish for exactly the $b$ with $S(b)\neq0$ and force the Frobenius eigenvalues of $C_b$ to be $\sqrt{q}$ times primitive 8th roots of unity. For even $k$ the orbit invariant $j$ is not complete (cube and fifth roots need not exist in $\mathbb{F}_q$) and the analogous analysis has not been carried out; the closed form of $X$ (the sums $W'$) is open for all $k$.

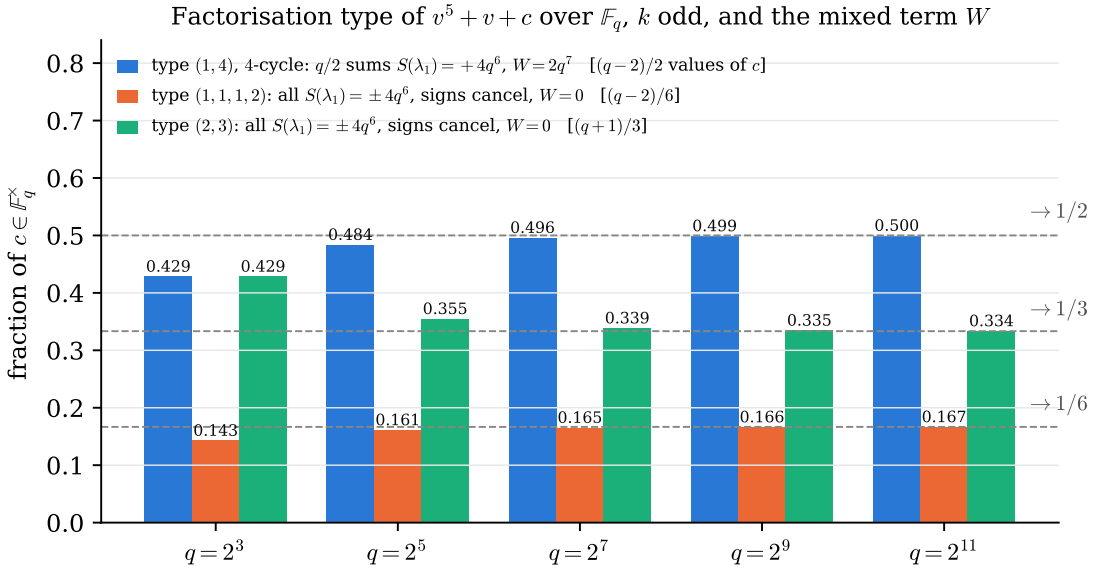



FIGURE 2. The mixed part for odd $k$ (Theorem 6.12). For each $c \in \mathbb{F}_q^\times$ the Gold-type quintic $v^5+v+c$ has one of three factorisation types over $\mathbb{F}_q$, with the exact frequencies of Proposition 6.8 (bars: $q=2^3,\dots,2^{11}$; dashed lines: the limits $\frac{1}{2},\frac{1}{3},\frac{1}{6}$). Only the 4-cycle type contributes to $A_6$: there $q/2$ of the $q$ curve sums $S(\lambda_1)$ equal $+4q^6$ and the rest vanish; for the other types all $q$ sums are $\pm 4q^6$ and cancel in pairs (Lemmas 6.10, 6.11). The "half-measure" of the 4-cycle set, $(q-2)/2$, is what produces the term $(q-1)(q-2)q^7$ of (9).

6.7. **The sums $W'$ and the constant $X$.** We now turn to $X = q^{-3}\sum_{\lambda_3,\lambda_5} W'(\lambda_3,\lambda_5)$. Throughout, $k$ is odd unless said otherwise, and we write

$$S'(\lambda_1,\lambda_3,\lambda_5) = \sum_{x\in\mathbb{F}_{q^{12}}} (-1)^{\mathrm{Tr}(\lambda_5 x^5+\lambda_3 x^3+\lambda_1 x)+\mathrm{Tr}_{q/2}\, e_2(x)},$$

$$Q'_\lambda(x) = \mathrm{Tr}(\lambda_5 x^5+\lambda_3 x^3) + \mathrm{Tr}_{q/2}\, e_2(x),$$

where $e_2(x)$ is the second elementary symmetric function of the twelve $q$-conjugates of $x$. In Witt-vector terms $S'$ is the sum of a character of order 4 of the Hayes group (the $q^3$ characters restricting to $(-1)^{\mathrm{Tr}\, c_2}$ on $\mathcal{G}^2$), and $X = q^{-3}\sum_{\lambda\in\mathbb{F}_q^3} S'(\lambda)$.

**Proposition 6.14** (structure of $W'$). *Let $H_0=\{x\in\mathbb{F}_{q^{12}} : \mathrm{Tr}_{q^{12}/q}\, x=0\}$, $L'_\lambda(x)=\lambda_5^4x^{16}+\lambda_3^4x^8+x^4+\lambda_3^2x^2+\lambda_5 x$ (so $L'_\lambda=L_\lambda+x^4$), $K'=\ker L'_\lambda\cap\mathbb{F}_{q^{12}}$, and $P=1+\sigma^2+\sigma^4+\cdots+\sigma^{10}=(\sigma+1)^2(\sigma^2+\sigma+1)^4$. Then for all $k$:*

(a) *$W'(\lambda_3,\lambda_5)=\sum_{\lambda_1\in\mathbb{F}_q} S'(\lambda) = q\sum_{x\in H_0}(-1)^{Q'_\lambda(x)}$, and $Q'_\lambda$ is invariant under $x\mapsto x+c$, $c\in\mathbb{F}_q$, on $H_0$.*

(b) *$Q'_\lambda$ is an $\mathbb{F}_2$-quadratic form on $\mathbb{F}_{q^{12}}$ with polar form $\mathrm{Tr}\big(y\,(T_\lambda(x)+x+\mathrm{Tr}_{q^{12}/q}\,x)\big)$; on $H_0$ its radical is $\{x\in H_0 : L'_\lambda(x)\in\mathbb{F}_q\}$, of dimension $k+d$ with $d=\dim_{\mathbb{F}_2}\ker(P|_{K'})$. Hence $|W'(\lambda_3,\lambda_5)|\in\{0,\,2^{d/2}q^7\}$, $d\in\{0,2,4\}$, and the radical of the $\beta$-form (7) has dimension $2k+d$.*

(c) *$d=0$ iff $K'=0$, i.e. iff no nonzero root of $L'_\lambda$ lies in $\mathbb{F}_{q^{12}}$; in that case $W'(\lambda_3,\lambda_5)=\pm q^7$. If $K'\neq 0$ then $d=\dim M_3+\dim\ker(\sigma-1)^2|_{M_1}$, where $K'=M_1\oplus M_3$ is the decomposition into the $(\sigma-1)$- and $(\sigma^2+\sigma+1)$-primary parts.*

*Proof.* (a) $\sum_{\lambda_1}(-1)^{\mathrm{Tr}(\lambda_1 x)} = q[x\in H_0]$; the $\beta$-form follows from $x=\beta+\beta^q$ ($q$-to-one onto $H_0$) and $e_2(\beta+\beta^q)=\mathrm{Tr}_{q^{12}/q}(\beta^{q+1})+(\mathrm{Tr}_{q^{12}/q}\,\beta)^2$ (Lemma 5.1). For $c\in\mathbb{F}_q$ and $x\in H_0$: $\mathrm{Tr}(x+c)^5=\mathrm{Tr}\,x^5+c(\mathrm{Tr}\,x)^4+c^4\,\mathrm{Tr}\,x$, $\mathrm{Tr}(x+c)^3=\mathrm{Tr}\,x^3+c(\mathrm{Tr}\,x)^2+c^2\,\mathrm{Tr}\,x$ and $e_2(x+$

$c) = e_2(x) + c\,\mathrm{Tr}\,x + \binom{12}{2}c^2$, all equal to their values at $x$. (b) The polar form of $\mathrm{Tr}_{q/2}\,e_2$ is $\mathrm{Tr}_{q/2}\big(\mathrm{Tr}\,x\,\mathrm{Tr}\,y + \mathrm{Tr}(xy)\big) = \mathrm{Tr}_{q^{12}/2}\big(y(x+\mathrm{Tr}\,x)\big)$, that of $\mathrm{Tr}(\lambda_5 x^5 + \lambda_3 x^3)$ is $\mathrm{Tr}(yT_\lambda(x))$. On $H_0$ the term $\mathrm{Tr}\,x$ disappears and $H_0^\perp = \mathbb{F}_q$ for the trace pairing, so the radical is $\{x \in H_0 : T_\lambda(x) + x \in \mathbb{F}_q\} = \{x \in H_0 : L'_\lambda(x) \in \mathbb{F}_q\}$ ($L'_\lambda = (T_\lambda + 1)^4$). As in Lemma 6.9(b), $L'_\lambda(x) \in \mathbb{F}_q$ iff $(\sigma-1)x \in K'$, and $\mathrm{Tr}_{q^{12}/q}\,x = P(\sigma-1)x$; since $(\sigma+1)P = 1 + \sigma + \cdots + \sigma^{11}$, $\ker P \subset H_0$, and the radical is $(\sigma-1)^{-1}(\ker P|_{K'})$, of dimension $k+d$. The exponential sum of a quadratic form on $\mathbb{F}_2^{11k}$ with radical of dimension $k+d$ has modulus 0 or $2^{(12k+d)/2}$, and $W' = q\cdot(\ldots)$. (c) $K'$ is a $\sigma$-module with $\sigma^{12} = 1$, so its minimal polynomial divides $x^{12}-1 = (x-1)^4(x^2+x+1)^4$ and has degree $\le 4$; $P$ kills $M_3$ and acts on $M_1$ as $(\sigma-1)^2$ times a unit. If $K' \neq 0$ then $K'$ contains a nonzero element killed by $\sigma - 1$ or by $\sigma^2 + \sigma + 1$, so $d > 0$. For $d = 0$ the radical on $H_0$ is $\mathbb{F}_q$, on which $Q'_\lambda$ vanishes by (a), so the sum is $\pm 2^{6k}$ and $W' = \pm q^7$. □

**Lemma 6.15** (the term $(0,0)$). *For all $k$, $W'(0,0) = \sum_{\beta\in\mathbb{F}_{q^{12}}}(-1)^{\mathrm{Tr}(\beta^{q+1}+\beta)} = \varepsilon_0\,q^7 = (-1)^{k+1}q^7$.*

*Proof.* Let $c \in \mathbb{F}_{q^{12}}$ satisfy $c^{q^2} + c = 1$ (solvable, as $\mathrm{Tr}_{q^{12}/q^2}\,1 = 6 = 0$); then $\theta c = c^q + c^{q^{-1}} = \sigma^{-1}(c^{q^2} + c) = 1$. Substituting $\beta \mapsto \beta + c$ and using $\mathrm{Tr}((\beta+c)^{q+1}) = \mathrm{Tr}(\beta^{q+1}) + \mathrm{Tr}(\beta\,\theta c) + \mathrm{Tr}(c^{q+1})$ gives $W'(0,0) = (-1)^{\mathrm{Tr}(c^{q+1}+c)}\sum_\beta(-1)^{\mathrm{Tr}(\beta^{q+1})}$. By Coulter's evaluation [8, Thm. 4.2] of $\sum_{x\in\mathbb{F}_{2^N}}(-1)^{\mathrm{Tr}(x^{2^k+1})}$ with $N = 12k$, $\gcd(k,N) = k$, $N/k = 12$ even and $a = 1$ a $(2^k+1)$-th power, the last sum equals $-(-1)^{N/2k}2^{N/2+k} = -q^7$. Finally $\mathrm{Tr}_{q^{12}/q^2}\,c = 3c + 3(c+1) = 1$ gives $\mathrm{Tr}_{q^{12}/2}\,c = \mathrm{Tr}_{q^2/2}\,1 = 0$, and $\mathrm{Tr}_{q^{12}/q^2}(c^{q+1}) = 3c\,c^q + 3(c+1)(c^q+1) = c + c^q + 1$, whence $\mathrm{Tr}_{q^{12}/2}(c^{q+1}) = \mathrm{Tr}_{q^2/2}(c + c^q) = \mathrm{Tr}_{q/2}(c + c^{q^2}) = \mathrm{Tr}_{q/2}\,1 = k \bmod 2$. So $W'(0,0) = (-1)^k(-q^7)$. □

**Lemma 6.16** (pure cubic part of $X$, $k$ odd). *Let $\lambda_3 \neq 0$. Then $K' = \{0\}\cup\{z/\lambda_3 : z^3 + z = \lambda_3^2\}$, $d = 2$, and $W'(\lambda_3, 0) \neq 0$ (hence $= \pm 2q^7$) iff the cubic $z^3 + z + \lambda_3^2$ does* not *have exactly one root in $\mathbb{F}_q$, i.e. iff it has type* (3) *or* $(1,1,1)$. *This holds for exactly $q/2$ values of $\lambda_3$.*

*Proof.* $L'_\lambda = \lambda_3^4x^8 + x^4 + \lambda_3^2x^2 = x^2(\lambda_3^2x^3 + x + \lambda_3)^2$, and $x = z/\lambda_3$ turns the cubic into $z^3 + z + \lambda_3^2$. Its roots lie in $\mathbb{F}_q$, $\mathbb{F}_{q^2}$ or $\mathbb{F}_{q^3}$, all of even index in $\mathbb{F}_{q^{12}}$, so $K' = \ker L'_\lambda$ ($\dim 2$), $P = 0$ on $K'$ and $d = 2$. In the $\beta$-form the radical is $\theta^{-1}(K')$, $Q' + \mathrm{Tr}$ is additive on it and vanishes on $\theta^{-1}(0) = \mathbb{F}_{q^2}$; for a root $y = z/\lambda_3$ let $\beta_y \in \mathbb{F}_{q^4}$ satisfy $\beta_y^{q^2} + \beta_y = y^q$ (so $\theta\beta_y = y$), and put $v(y) = \mathrm{Tr}_{q^{12}/2}(\lambda_3\beta_y^2 y + \beta_y^{q+1} + \beta_y) = \mathrm{Tr}_{q^4/2}(\lambda_3\beta_y^2 y + \beta_y^{q+1} + \beta_y)$. If $z \in \mathbb{F}_q$, the conjugates of $\beta_y$ are $\beta_y, \beta_y^q, \beta_y + y, \beta_y^q + y$ and one finds $\mathrm{Tr}_{q^4/q}\,\beta_y = 0$, $\mathrm{Tr}_{q^4/q}(\beta_y^2) = 0$, $\mathrm{Tr}_{q^4/q}(\beta_y^{q+1}) = y^2$, so $v(y) = \mathrm{Tr}_{q/2}(y) = \mathrm{Tr}_{q/2}\big(z^{1/2}/(z+1)\big)$ (from $\lambda_3^2 = z(z+1)^2$) $= \mathrm{Tr}_{q/2}\big(z/(z+1)^2\big) = \mathrm{Tr}_{q/2}\big(\frac{1}{z+1} + \frac{1}{(z+1)^2}\big) = 0$. If $z \in \mathbb{F}_{q^2}\setminus\mathbb{F}_q$ with conjugate $\bar z$, $s = z + \bar z$, $n = z\bar z$, the same computation gives $v(y) = \mathrm{Tr}_{q/2}(s/\lambda_3) + \mathrm{Tr}_{q^2/2}(\lambda_3 y\bar y^2) + \mathrm{Tr}_{q/2}(y\bar y) = \mathrm{Tr}_{q/2}\big(s/\lambda_3 + n(1+s)/\lambda_3^2\big)$; the factorisation $z^3 + z + \lambda_3^2 = (z+s)(z^2 + sz + n)$ gives $n = s^2 + 1$, $\lambda_3^2 = s(s+1)^2$, hence $v(y) = \mathrm{Tr}_{q/2}\big(s^{1/2}/(s+1)\big) + \mathrm{Tr}_{q/2}\big((s+1)/s\big) = 0 + 1 + \mathrm{Tr}_{q/2}(1/s)$, while $z^2 + sz + n$ is irreducible iff $\mathrm{Tr}_{q/2}(n/s^2) = 1 + \mathrm{Tr}_{q/2}(1/s) = 1$. So in type $(1,2)$ the form takes the value 1 on the radical and $W' = 0$; in type $(1,1,1)$ all values vanish; in type (3) the three values are equal and sum to $v(0) = 0$. The count $q/2 = \frac{q+1}{3} + \frac{q-2}{6}$ is the classical distribution of $x^3 + x + c$ for odd $k$ (the same count as in Proposition 6.8, with the roles of the types $(1,2)$ and $(1,4)$). □

**Lemma 6.17** (pure quintic part of $X$: the kernel, $k$ odd). *Let $\lambda_5 \neq 0$, $y_0 = \lambda_5^{-1/5}$ and $c = \lambda_5^2$. Then $K' = \ker L'_\lambda = y_0\{w : w^{15} + y_0^8w^3 + 1 = 0\}\cup\{0\}$, the fifteen elements $w$ are the cube roots $w^3 = \lambda_5^{-2/5}v'$ of the five roots $v'$ of $v'^5 + v' + c$, and the Frobenius orbit structure on $K'\setminus 0$ is $(1,2,4,4,4)$, $(3,6,6)$, $(1,1,1,2,2,2,2,2,2)$ according as $v^5 + v + \lambda_5^2$ has type $(1,4)$, $(2,3)$, $(1,1,1,2)$. Consequently $d = 2$ in the first case and $d = 4$ in the other two, so $|W'(0,\lambda_5)| \in \{0, 2q^7\}$ for $(q-2)/2$ values of $\lambda_5$ and $\in \{0, 4q^7\}$ for the remaining $q/2$.*

*Proof.* With $x = y_0w$ and $\lambda_5y_0^5 = 1$, $y_0^4L'_\lambda(y_0w) = w^{16} + y_0^8w^4 + w$. Put $w^3 = v$: $w^{15} + y_0^8w^3 + 1 = v^5 + y_0^8v + 1$, and $v = y_0^2v'$ gives $v'^5 + v' = y_0^{-10} = \lambda_5^2$. A root $v' \in \mathbb{F}_q$ has one cube root in $\mathbb{F}_q$ and

two in $\mathbb{F}_{q^2}$ ($3 \nmid q-1$); the roots of the irreducible quartic (type $(1,4)$) have their twelve cube roots in $\mathbb{F}_{q^4}$ forming three orbits of length 4; the three roots in $\mathbb{F}_{q^3}$ have one cube root each in $\mathbb{F}_{q^3}$ (one orbit of length 3) and six in $\mathbb{F}_{q^6}$ (one orbit); the two roots in $\mathbb{F}_{q^2}$ have six cube roots forming orbits of length 2 (type $(1,1,1,2)$) or 6 (type $(2,3)$) according to their cubicity in $\mathbb{F}_{q^2}$, which is read off from the orbit count 15. In all cases $K' \subset \mathbb{F}_{q^{12}}$, and $d$ follows from Proposition 6.14(c): the orbit structure $(1,2,4,4,4)$ is a single Jordan block of size 4 (so $\ker(\sigma-1)^2$ has dimension 2), while in the other two cases $(\sigma-1)^2$ or $\sigma^2+\sigma+1$ annihilates everything. □

**Proposition 6.18** (structure of the data for $W'$, $k$ odd)**.** *For $k = 3,5,7$ the following hold (`legendre_d6_Wp_pattern.py`; Table 3).*

(a) *Pure terms: $W'(\lambda_3,0) = 2q^7$ whenever it is nonzero (Lemma 6.16), and $W'(0,\lambda_5) = 2q^7$ or $4q^7$ according as $d = 2$ or 4 (Lemma 6.17): $\sum_{\lambda_3\neq 0} W'(\lambda_3,0) = q^8$ and $\sum_{\lambda_5\neq 0} W'(0,\lambda_5) = 3q^8 - 2q^7$.*

(b) *Mixed terms: for every $(\lambda_3,\lambda_5) \in (\mathbb{F}_q^\times)^2$ the Frobenius orbit structure on the 15 nonzero elements of $\ker L'_\lambda$ is one of the eleven patterns obtained from the action of a conjugacy class of $S_6 \cong Sp_4(\mathbb{F}_2)$ on the 15 two-element subsets of $\{1,\dots,6\}$, and the pair $(d,W')$ depends only on the class; the number of pairs in each class is the polynomial in $q$ of Table 3, exactly. The classes of order 7 and 15 of $GL_4(\mathbb{F}_2) \cong A_8$ do not occur.*

(c) *Consequently $\sum_{\lambda_3\lambda_5\neq 0} W' = q^9 - 3q^8 + q^7$ and, with Lemma 6.15 and (a), $\sum_{\lambda_3,\lambda_5} W' = q^9 + q^8$, i.e. $X = q^6 + q^5$.*

TABLE 3. The mixed sums $W'(\lambda_3,\lambda_5)$ for odd $k$, by the Frobenius class on $\ker L'_\lambda$ (cycle type in $S_6$; orbit lengths on the 15 nonzero kernel elements), with the radical excess $d$, the value, and the exact number of pairs $(\lambda_3,\lambda_5) \in (\mathbb{F}_q^\times)^2$ (verified for $k = 3,5,7$). The two classes of order 4 have the same orbit structure and opposite signs; the two classes $2^2 1^2$, $2^3$ have the same orbit structure and $W' = 0$, $4q^7$ respectively.

| class in $S_6$ | orbits on $\ker L' \setminus 0$ | $d$ | $W'/q^7$ | number of pairs |
|---|---|---|---|---|
| $1^6$ | $(1^{15})$ | 4 | 4 | $(q-2)(q-8)/720$ |
| $2\,1^4$ | $(1^7,2^4)$ | 4 | 0 | $(q-2)(q-8)/48$ |
| $2^2\,1^2$ | $(1^3,2^6)$ | 4 | 0 | $q(q-2)/16$ |
| $2^3$ | $(1^3,2^6)$ | 4 | 4 | $(q-2)(q-8)/48$ |
| $3\,1^3$ | $(1^3,3^4)$ | 4 | 4 | $(q+1)(q-2)/18$ |
| $3^2$ | $(3^5)$ | 4 | 4 | $(q+1)(q-2)/18$ |
| $3\,2\,1$ | $(1,2,3,3,6)$ | 4 | 0 | $(q+1)(q-2)/6$ |
| $4\,1^2$ | $(1,2,4,4,4)$ | 2 | 2 | $(q-2)(q-4)/8$ |
| $4\,2$ | $(1,2,4,4,4)$ | 2 | $-2$ | $q(q-2)/8$ |
| $5\,1$ | $(5,5,5)$ | 0 | $-1$ | $(q^2+1)/5$ |
| $6$ | $(3,6,6)$ | 4 | 4 | $(q+1)(q-2)/6$ |

**Conjecture 6.19** ($X$ for odd $k$)**.** *For all odd $k$: Proposition 6.18(b) holds, in particular $W'(\lambda_3,\lambda_5) = -q^7$ whenever $\ker L'_\lambda \cap \mathbb{F}_{q^{12}} = 0$; the nonzero pure sums are positive; and hence $X = q^6 + q^5$.*

Figure 3 displays the table. Several features of Table 3 are forced by Proposition 6.14: the value of $d$ in every row, the modulus $|W'| \le 2^{d/2}q^7$, and $W' = \pm q^7$ in the class $5\,1$ (where the fifteen roots lie in $\mathbb{F}_{q^5}\setminus\mathbb{F}_{q^{12}}$). The densities of the classes are those of $S_6$ (1/5 for the 5-cycles, 1/6 for the 6-cycles and for $3\,2\,1$, 1/4 for the two classes of order 4 together, …), so the geometric monodromy group of the family $\{\ker L'_\lambda\}$ over the $(\lambda_3,\lambda_5)$-plane is $Sp_4(\mathbb{F}_2) \cong S_6$ rather than $GL_4(\mathbb{F}_2) \cong A_8$; the exactness of the polynomial counts indicates that, as in Proposition 6.7(b) where $\ker L_\lambda$ is spanned by the five roots of a quintic, $\ker L'_\lambda$ should be the set of pairwise sums of the six roots of a sextic over $\mathbb{F}_q$ with vanishing $e_1$ (for $\lambda_5 \to 0$ or $\lambda_3 \to 0$ this degenerates to $u\,A_\lambda(u)$ and to the cube-root structure of Lemma 6.17). Exhibiting this sextic, and evaluating

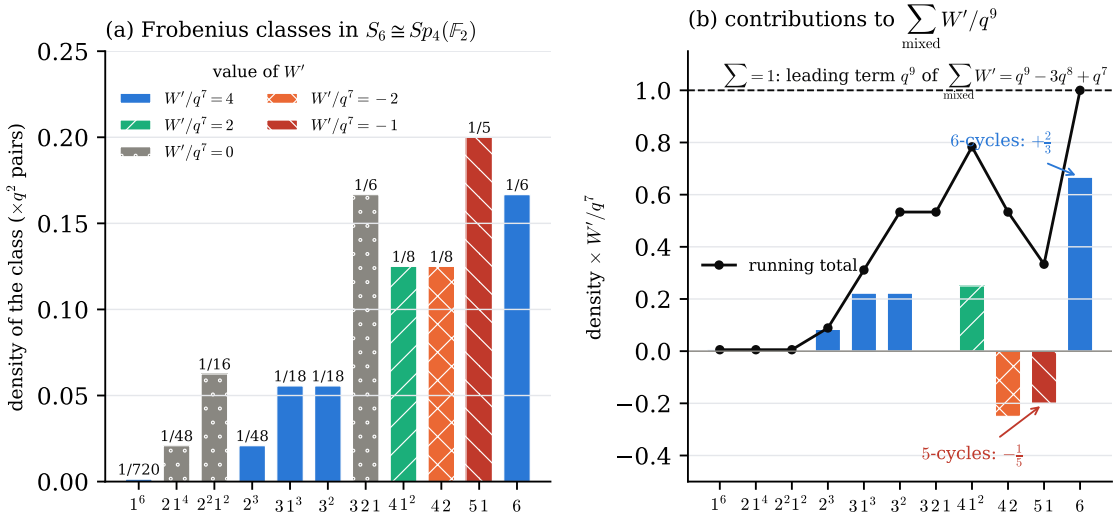



FIGURE 3. The mixed sums $W'$ for odd $k$ by Frobenius class (Table 3), the counterpart of Figure 2. (a) Asymptotic density of each of the eleven classes of $S_6 \cong Sp_4(\mathbb{F}_2)$ among the $(q-1)^2$ pairs $(\lambda_3, \lambda_5)$, coloured by the value of $W'/q^7$: the 5-cycles (1/5) carry $-q^7$; the 6-cycles (1/6), the two classes of order 3 (1/18 each), the class $2^3$ and the identity carry $+4q^7$; the two classes of order 4 (1/8 each) carry $\pm 2q^7$ and cancel to leading order; the remaining classes vanish. (b) Density times value: the running total reaches exactly 1, the leading coefficient of $\sum_{\text{mixed}} W' = q^9 - 3q^8 + q^7$ (Proposition 6.18(c)), the negative $-q^9/5$ of the 5-cycles being overcompensated by the 6-cycles ($+2q^9/3$) and the order-3 classes ($+4q^9/9$).

the form on the radical and the Arf invariant class by class as in Lemmas 6.10–6.11, is what a proof of Conjecture 6.19 requires; the sign in the class $5\,1$, which carries the leading term $-q^9/5$, and the signs of the two order-4 classes are the essential points. We have not carried this out.

## 7. The scaling law at $d = 6$ and the nature of $Y$

### 7.1. Variance formula.

**Theorem 7.1.** *Assume Conjecture 6.5 (true for $q \le 128$; for odd $k$ only its statement for $X$ is needed, by Corollary 6.13, i.e. Conjecture 6.19) and write $Y = q^3 m_k$. Then $B = -2q^3[k \text{ odd}]$,*

$$\varphi_0 = \begin{cases} q^5 + q^4 - 2q^3, & k \text{ odd}, \\ -q^5 + q^4, & k \text{ even}, \end{cases} \qquad \varphi_{A,B} = \begin{cases} -2q^3 \pm q^2 m_k, & k \text{ odd}, \\ \pm q^2 m_k, & k \text{ even}, \end{cases}$$

*and*

$$\frac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \begin{cases} \dfrac{(q-1)^2(q+2)^2}{q^2} + \dfrac{(q-1)(4q^2+m_k^2)}{q^4}, & k \text{ odd}, \\ (q-1)^2 + \dfrac{(q-1)m_k^2}{q^4}, & k \text{ even}. \end{cases}$$

*In particular* $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^7 = q^2(1 + O(1/q))$, *i.e.* $C_6 := \lim_{q\to\infty} \mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^9 = 1$, *provided* $m_k = o(q^{5/2})$.

*Proof.* Substitute into Corollary 4.3. □

**Remark 7.2.** At $d = 4$ the Legendre anomaly is a single class with relative deficit $1/q$; at $d = 6$ it is the $q$ classes with $c_2 = 0$ (the intervals around $f^2$ with $f_5 = 0$), each with relative deviation $\pm q^{-2}$ and sign $\varepsilon_0$. Both mechanisms give $\mathrm{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^2$. The full-group variance $\mathrm{Var}_{\mathcal{G}}/q^7$

TABLE 4. $d = 6$: exact variances (Hayes FFT, $q \le 32$; $q = 16$ also by CRT), the three Legendre deviations and $m_k = Y/q^3$. The last two columns show that $\mathrm{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ and not $q^4$.

| $q$ | $\mathrm{Var}_{\mathcal{G}}/q^7$ | $\mathrm{Var}_{\mathcal{G}^2}/q^7$ | ratio | $\varphi_0$ | $\varphi_A$ | $\varphi_B$ | $m_k$ | $\mathrm{Var}_{\mathcal{G}^2}/q^9$ |
|---|---|---|---|---|---|---|---|---|
| 2 | 5.7578 | 15.5625 | 2.703 | 32 | 36 | −68 | 13 | 3.891 |
| 4 | 3.7004 | 9.0117 | 2.435 | −768 | −16 | 16 | −1 | 0.563 |
| 8 | 4.7483 | 79.0935 | 16.657 | 35 840 | −3 264 | 1 216 | −35 | 1.236 |
| 16 | 4.3514 | 225.5056 | 51.824 | −983 040 | 12 032 | −12 032 | 47 | 0.881 |
| 32 | 4.2086 | 1100.8844 | 261.581 | 34 537 472 | 685 056 | −816 128 | 733 | 1.075 |

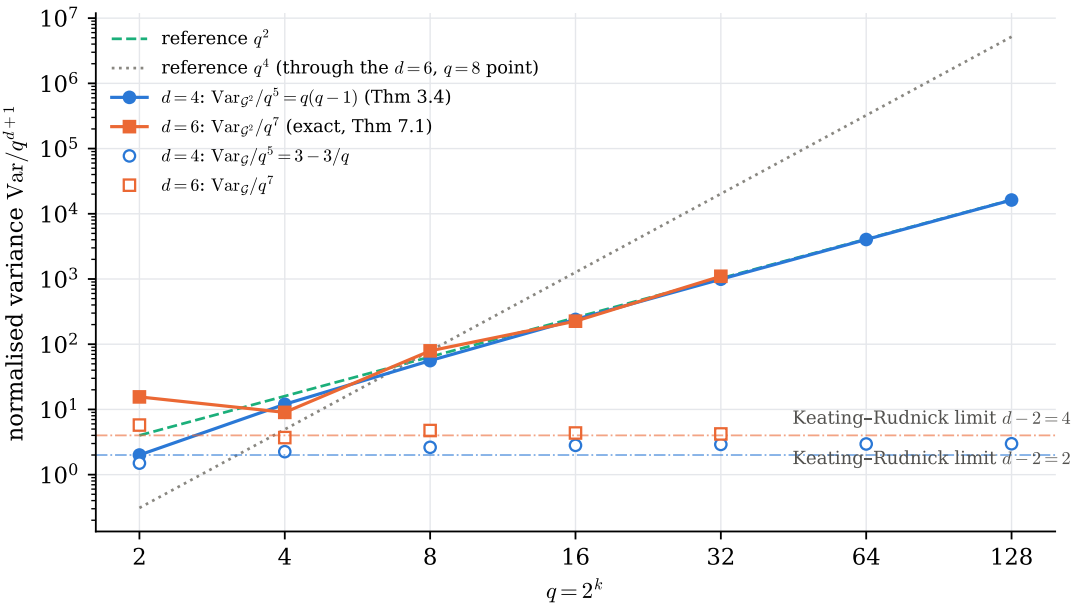



FIGURE 4. Normalised variances against $q$, both axes logarithmic. Filled markers are $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^{d+1}$, exact values ($d = 4$: $q(q-1)$ for $q \le 128$, Theorem 3.4; $d = 6$: Table 4). Open markers: the full-group variances $\mathrm{Var}_{\mathcal{G}}/q^{d+1}$, which approach 3 ($d = 4$, Theorem 3.11) and the Keating–Rudnick value 4 ($d = 6$). The dashed line is $q^2$; the dotted line is a $q^4$ law normalised through the $d = 6$, $q = 8$ point, which the data exclude. Whether $\lim \mathrm{Var}_{\mathcal{G}^2}/q^9 = 1$ at $d = 6$ depends on $m_k = o(q^{5/2})$ (Theorem 7.1).

tends to the Keating–Rudnick value 4 (4.35, 4.21 at $q = 16, 32$), unlike $d = 4$ where it tends to $3 \ne 2$. The value $q^{2(d/2-1)} = q^4$ suggested in [5] is excluded.

7.2. **The geometry behind $Y$: a fixed nine-dimensional variety and a twisted Frobenius.** We now give $Y$ an unambiguous algebro-geometric meaning. The point is that $Y_k$ is a Frobenius trace on a *fixed* variety over $\mathbb{F}_2$, but for a *twisted* Frobenius; in particular $(Y_k)_k$ is not the sequence of $\mathbb{F}_{2^k}$-point counts of any single variety.

**Definition 7.3.** Let $z = (z_0, \dots, z_{11})$ be coordinates on $\mathbb{A}^{12}_{\mathbb{F}_2}$ and $e_m(z)$ the elementary symmetric polynomials. Put

$$I = \langle e_1, e_3, e_5 \rangle \subset \mathbb{F}_2[z], \qquad X = V(I) \subset \mathbb{A}^{12}_{\mathbb{F}_2}, \qquad f = e_4|_X, \qquad \mathcal{L} = f^*\mathcal{L}_{\psi_0},$$

where $\mathcal{L}_{\psi_0}$ is the Artin–Schreier sheaf on $\mathbb{A}^1_{\mathbb{F}_2}$ attached to $\psi_0(x) = (-1)^x$, so that the trace function of $\mathcal{L} \otimes \mathbb{F}_q$ is $\psi_q \circ e_4$. Let $\sigma \in \mathrm{Aut}(\mathbb{A}^{12})$ be the cyclic shift $\sigma(z)_j = z_{j-1}$ (indices mod 12); it is defined over $\mathbb{F}_2$, preserves $I$ and $e_4$, and commutes with the absolute Frobenius $F : z \mapsto (z_j^2)$.

The $\sigma$-twisted $\mathbb{F}_q$-form $X^{(\sigma)}$ of $X$ is the $\mathbb{F}_q$-structure whose geometric Frobenius is $\sigma \circ F^k$ (the class of a 12-cycle in $H^1(\mathbb{F}_q, S_{12})$; it is the twist defining $\mathrm{Res}_{\mathbb{F}_{q^{12}}/\mathbb{F}_q}\mathbb{A}^1 \cong \mathbb{A}^{12}_{\mathbb{F}_q}$).

**Proposition 7.4.** (a) *$I = \langle p_1, p_3, p_5\rangle$ with $p_m = \sum_j z_j^m$ (Newton's identities in characteristic 2: $p_3 = e_1^3 + e_1e_2 + e_3$, $p_5 \equiv e_5 + e_2e_3 \pmod{e_1}$).*

(b) *$\alpha \mapsto (\alpha, \alpha^q, \dots, \alpha^{q^{11}})$ is a bijection $U \to X^{(\sigma)}(\mathbb{F}_q) = \{z \in X(\bar{\mathbb{F}}_2) : \sigma F^k(z) = z\}$ carrying $e_4(\alpha)$ to $e_4(z)$; hence $Y = \sum_{z\in X^{(\sigma)}(\mathbb{F}_q)} \psi_q(e_4(z))$.*

(c) *Let $\pi : \mathbb{A}^{12} \to \mathbb{A}^{12}/S_{12} = \operatorname{Spec}\mathbb{F}_2[e_1, \dots, e_{12}]$ and $\Lambda = V(e_1, e_3, e_5) \cong \mathbb{A}^9$ with coordinates $(e_2, e_4, e_6, e_7, \dots, e_{12})$. Then $X = \pi^{-1}(\Lambda)$, $\pi|_X : X \to \Lambda$ is finite flat of degree 12! and generically étale, $\Lambda(\mathbb{F}_q)$ is the set of monic $F \in \mathbb{F}_q[t]$ of degree 12 with $a_{11} = a_9 = a_7 = 0$, and $\#(\pi^{-1}(F) \cap X^{(\sigma)}(\mathbb{F}_q)) = \Lambda(F)$. Thus $Y = \sum_{F\in\Lambda(\mathbb{F}_q)} \Lambda(F)\psi_q(a_8(F))$, which is the Hayes-group form of Section 4; the projection $\Lambda \to \mathbb{A}^2$, $(e_2, e_4)$, has fibres of dimension 7.*

*Proof.* (a) is immediate from Newton's identities. (b): $\sigma F^k(z)_j = z_{j-1}^q$, so the fixed points are the tuples $z_j = z_0^{q^j}$ with $z_0^{q^{12}} = z_0$, and $e_1 = e_3 = e_5 = 0$ is $p_1 = p_3 = p_5 = 0$ by (a). (c): $\mathbb{F}_2[z]$ is free of rank 12! over $\mathbb{F}_2[e]$; the generic $F$ is separable since $F' = e_7t^4 + e_9t^2 + e_{11} \neq 0$; the $\sigma F^k$-fixed points of $\pi^{-1}(F)$ are the orderings $z_j = z_0^{q^j}$ of the roots of $F$, i.e. the $\alpha \in \mathbb{F}_{q^{12}}$ with $\mathrm{charpoly}(\alpha) = F$, of which there are $\Lambda(F)$ (cf. (1)). □

**Proposition 7.5** (geometry of $X$)**.** (a) *$X$ is a complete intersection of codimension 3 in $\mathbb{A}^{12}$ cut out by the regular sequence $(e_1, e_3, e_5)$ of degrees $1, 3, 5$; it is an affine cone of pure Krull dimension 9 and is Cohen–Macaulay. ("Nine-dimensional" refers to $\dim X$; the fibres of $X \to \Lambda$ have dimension 0 and those of $X \to \mathbb{A}^2$ dimension 7.)*

(b) *$\mathrm{Sing}(X) = X \cap \{z : \#\{z_0, \dots, z_{11}\} \le 2\}$ is the union of the 1023 planes $P_A = \{z_j = u\ (j \in A),\ z_j = v\ (j \notin A)\}$, $A \subset \{0, \dots, 11\}$ of even cardinality $0 < |A| < 12$ (A and its complement give the same plane), all containing the diagonal line; $\dim \mathrm{Sing}(X) = 2$, of codimension 7.*

(c) *$X$ is geometrically integral and normal. Consequently the geometric monodromy group of the family $\{F : e_1 = e_3 = e_5 = 0\}$, i.e. the Galois group of $t^{12} + e_2t^{10} + \cdots + e_{11}t + e_{12}$ over $\bar{\mathbb{F}}_2(e_2, e_4, e_6, \dots, e_{12})$, is the full $S_{12}$.*

*Proof.* (a) By Proposition 7.4(c), $\mathbb{F}_2[z]/I$ is a free module over the ring $\mathbb{F}_2[e]/(e_1, e_3, e_5)$, which is the polynomial ring $\mathbb{F}_2[e_2, e_4, e_6, \dots, e_{12}]$; so $(e_1, e_3, e_5)$ is a regular sequence and $X$ is finite over $\mathbb{A}^9$. (b) By the Jacobian criterion for complete intersections, $X$ is smooth at $z$ iff $de_1, de_3, de_5$ are independent there. On $X$, $\partial_j e_1 = 1$, $\partial_j e_3 = e_2(z \setminus z_j) = e_2 + z_j^2$ and $\partial_j e_5 = e_4(z \setminus z_j) = e_4 + e_2z_j^2 + z_j^4$, so the row space is that of the Vandermonde rows $(1)_j, (z_j^2)_j, (z_j^4)_j$, of rank 3 iff at least three of the $z_j$ are distinct. Conversely, if $z$ takes only the values $u, v$ with multiplicities $s, r$, $s + r = 12$, then $e_m(z) = \sum_i \binom{s}{m-i}\binom{r}{i}u^{m-i}v^i$; for $r$ even every term with $m$ odd contains a factor $\binom{\text{even}}{\text{odd}}$, which is even by Lucas' theorem, so $e_1 = e_3 = e_5 = 0$ and $P_A \subset X$; for $r$ odd, $e_1 = u + v$ forces $u = v$. (c) $X$ is generically reduced (generic fibre of $\pi$ étale) and Cohen–Macaulay, hence reduced; Cohen–Macaulay gives Serre's $S_2$ and (b) gives $R_1$, so $X$ is normal; a cone is connected, and a connected normal variety is irreducible. The same holds over $\bar{\mathbb{F}}_2$. The irreducible components of $X_{\bar{\mathbb{F}}_2}$ correspond to the cosets $S_{12}/G$ of the geometric monodromy group $G$ (the generic fibre of $\pi$ is the splitting algebra), so $G = S_{12}$. □

**Proposition 7.6** (the Artin–Schreier sheaf)**.** *The cover $Z = \{(z, y) \in X \times \mathbb{A}^1 : y^2 + y = e_4(z)\} \to X$ is étale of degree 2 and $Z$ is smooth over $X_{\mathrm{sm}}$. The function $f = e_4$ has no poles on $X$; on the projective closure $\overline{X} \subset \mathbb{P}^{12}$ its polar divisor is $4X_\infty$ with $X_\infty \cong \mathbb{P}(X) \subset \mathbb{P}^{11}$ the base of the cone. Since the pole order 4 is divisible by $p = 2$, the Swan conductor of $\mathcal{L}$ along $X_\infty$ is $< 4$. The sheaf $\mathcal{L}$ is geometrically nontrivial on $X$ (on the diagonal line $e_4 = \binom{12}{4}u^4 = u^4$, and $u^4 \notin \{g^2 + g + c\}$), so $Z$ is geometrically connected and*

$$H_c^{18}(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = H^0(X_{\bar{\mathbb{F}}_2}, \mathcal{L}^\vee)^\vee(-9) = 0.$$

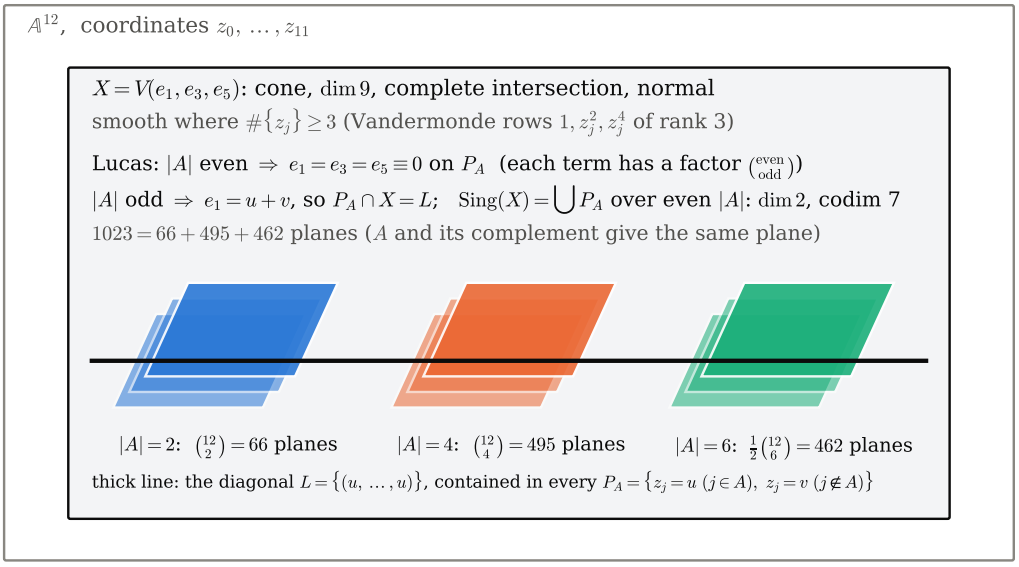



FIGURE 5. The singular locus of $X = V(e_1, e_3, e_5) \subset \mathbb{A}^{12}$ (Proposition 7.5(b)), schematically. A point with only two distinct coordinate values $u, v$ lies on the plane $P_A$ determined by the index set $A$ of the coordinates equal to $u$; by Lucas' theorem $P_A \subset X$ exactly when $|A|$ is even, and the 1023 such planes, stratified by $|A| \in \{2, 4, 6\}$ ($A$ and its complement give the same plane), all contain the diagonal line and form the two-dimensional singular locus, of codimension 7 in $X$. Outside these planes at least three coordinates are distinct and the Vandermonde rows $(1), (z_j^2), (z_j^4)$ make $X$ smooth.

**Theorem 7.7** (twisted trace formula)**.** *For every $k \geq 1$, with $F$ the geometric Frobenius of $\mathbb{F}_2$ acting on $\ell$-adic cohomology ($\ell \neq 2$) and $\sigma$ acting through its lift to $\mathcal{L}$,*

$$Y_k = \sum_{j=0}^{18} (-1)^j \operatorname{Tr}\big(\sigma \circ F^k \mid H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L})\big).$$

*Consequently $\sum_{k \geq 1} Y_k T^k = T \frac{d}{dT} \log L_\sigma(T)$ with $L_\sigma(T) = \prod_j \det(1 - \sigma F T \mid H_c^j)^{(-1)^{j+1}} \in \mathbb{Q}(T)$; $(Y_k)_k$ is a linear recurrence sequence, and $m_k = Y_k / 8^k = \sum_i \pm \zeta_i \eta_i^k$ with $\zeta_i^{12} = 1$ and $\eta_i = \lambda_i / 8$, $\lambda_i$ the eigenvalues of $F$. By Deligne [9] the $\lambda_i$ on $H_c^j$ are algebraic integers of weight $\leq j$; the normalisation $8^k = q^3$ is the Tate twist by 3, so $H_c^j(3)$ has weights $\leq j - 6$. If the cohomology were concentrated in degree $9 = \dim X$ and pure, then $m_k = \pm \sum_{i=1}^{b_9} \zeta_i \eta_i^k$ with $|\eta_i| = 2^{3/2}$ and $|m_k| \leq b_9\, q^{3/2}$.*

*Proof.* By Proposition 7.4(b), $Y_k$ is the sum of the trace function of $\mathcal{L}$ over the $\mathbb{F}_q$-points of the twisted form $X^{(\sigma)}$, whose geometric Frobenius acts on $H_c^\bullet(X_{\bar{\mathbb{F}}_2}, \mathcal{L})$ as $\sigma F^k$ ($\sigma$ has finite order and commutes with $F$); apply the Grothendieck–Lefschetz formula (SGA $4\frac{1}{2}$, Rapport; for $F \circ \sigma$ with $\sigma$ of finite order see also Deligne–Lusztig [10, proof of Thm. 3.2]). Since $\sigma$ and $F$ commute they are simultaneously triangularisable, which gives the stated shape; $L_\sigma$ has coefficients in $\mathbb{Q}(\zeta_{12})$ a priori and in $\mathbb{Q}$ because all $Y_k \in \mathbb{Z}$. The weight statements are Weil II. □

By contrast with $A_6$ and $X$, the function $(-1)^{\operatorname{Tr} e_4}$ on $\mathcal{G}^2$ is not a character: in Witt coordinates $\mathcal{G}^2 \cong W_2(\mathbb{F}_q)$, $(x_1, x_2) + (x_1', x_2') = (x_1 + x_1', x_2 + x_2' + x_1 x_1')$, and the carry destroys multiplicativity (the characters of order 4 are $i^{\operatorname{Tr} x_1}(-1)^{\operatorname{Tr} x_2 + \cdots}$). Therefore $Y$ does not decompose into Frobenius traces of individual $L$-functions of curves and no supersingularity argument applies; it is the global twisted trace on the nine-dimensional $X$.

**Proposition 7.8.** *The values $m_k = 13, -1, -35, 47, 733$ ($k = 1, \dots, 5$) satisfy $|m_k|/q^{3/2} = 4.60, 0.13, 1.55, 0.73, 4.05$. Hence, in the pure, degree-9 situation of Theorem 7.7, $b_9 \geq 5$. The elementary symmetric functions of the $\zeta_i\eta_i$ computed from $m_1, \dots, m_5$ by Newton's identities are $13, 85, 361, 1069, 2281$; if $b_9 = 5$ then $|e_5| = \prod |\eta_i| = 2^{15/2} \neq 2281$, a contradiction (the argument uses only absolute values, so the twists $\zeta_i$ do not affect it). Thus either $b_9 \geq 6$, or $H_c^\bullet$ has contributions outside the pure weight-9 part.*

Proposition 7.4(b) was checked directly: enumerating $X^{(\sigma)}(\mathbb{F}_2)$ inside $\mathbb{F}_{2^{12}}$ gives $|U| = 544$, $Y_1 = 104$, and the pushforward formula of (c) gives $(|U|, Y) = (544, 104)$ and $(259072, -64)$ for $q = 2, 4$ (script `verify_Y_direct.py`; a SageMath version is `verify_Y_sage.sage`).

**Proposition 7.9** ($Y$ as a difference of two point counts)**.** *Let $X_c = X \cap \{e_4 = c\}$ ($c \in \mathbb{F}_2$) and $N_c = \#\{\alpha \in U : e_4(\alpha) = c\} = \#X_c^{(\sigma)}(\mathbb{F}_q)$.*

(a) *$N_c = N_1$ for every $c \in \mathbb{F}_q^\times$, $Y = N_0 - N_1$ and $A_6 = N_0 + (q-1)N_1$; hence $N_1 = (A_6 - Y)/q$ and $N_0 = (A_6 + (q-1)Y)/q$. For odd $k$, by Corollary 6.13,*

$$N_1 = \#X_1^{(\sigma)}(\mathbb{F}_q) = q^8 + q^5 - q^4 - q^2 m_k,$$

*so $m_k = O(q^{3/2})$ iff $N_1 = q^8 + q^5 - q^4 + O(q^{7/2})$, and $m_k = o(q^{5/2})$ (the hypothesis of Theorem 7.1 for $C_6 = 1$) iff $N_1 = q^8 + q^5 - q^4 + o(q^{9/2})$.*

(b) *$X_0$ and $X_1$ are complete intersections of dimension 8 in $\mathbb{A}^{12}_{\mathbb{F}_2}$, stable under $\sigma$. The morphism $\varphi : \mathbb{G}_m \times X_1 \to X \setminus X_0$, $(s, z') \mapsto sz'$, satisfies $e_4 \circ \varphi = s^4$ and is finite, surjective and radicial ($s^4 = e_4(z)$ is integral and $s \mapsto s^4$ is injective in characteristic 2), hence a universal homeomorphism; since $\mathcal{L}_{\psi_0}(s^4) \cong \mathcal{L}_{\psi_0}(s)$ and $H_c^\bullet(\mathbb{G}_m, \mathcal{L}_{\psi_0}) = H_c^1$ is one-dimensional of weight 0, there is a long exact sequence of $\sigma$- and $F$-modules*

$$\cdots \to H_c^{j-1}(X_{1,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) \to H_c^j(X_{0,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to H_c^j(X_{1,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to \cdots,$$

*whose trace formula is (a). In particular the weights of $H_c^j(X, \mathcal{L})$ are bounded by those of $H_c^{j-1}(X_1)$ and $H_c^j(X_0)$.*

(c) *(lower vanishing) $H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = 0$ for $j < 9$ and $H_c^j(X_{c,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) = 0$ for $j < 8$; dually $H^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = 0$ for $j > 9$. Hence in Theorem 7.7 only $9 \leq j \leq 17$ contribute.*

*Proof.* (a) $U$ is stable under $\alpha \mapsto s\alpha$ and $e_4(s\alpha) = s^4 e_4(\alpha)$, and $s \mapsto s^4$ is a bijection of $\mathbb{F}_q^\times$; so $Y = \sum_c \chi(c) N_c = N_0 - N_1$ and $A_6 = \sum_c N_c$. (b) $e_4 \notin I$ (on the diagonal line $e_4 = u^4$), and $X$ is integral and Cohen–Macaulay, so $e_1, e_3, e_5, e_4 - c$ is a regular sequence and $X_c$ is a complete intersection of dimension 8. The coordinate ring of $\mathbb{G}_m \times X_1$ is generated over that of $X \setminus X_0$ by $s$, with $s^4 = e_4$ and $z_i' = z_i s^3/e_4$; $\varphi$ is bijective on geometric points because $s^4 = e_4(z)$ determines $s$. A universal homeomorphism induces an equivalence of étale sites, and $s^4 - s = (s^2+s)^2 + (s^2+s)$ gives $\mathcal{L}_{\psi_0}(s^4) \cong \mathcal{L}_{\psi_0}(s)$; Künneth and excision for $X_0 \subset X$ give the sequence. (c) $X$ and $X_c$ are local complete intersections of pure dimensions 9 and 8, so the shifted constant sheaves $\mathbb{Q}_\ell[9]$, $\mathbb{Q}_\ell[8]$ and the lisse rank-one twist $\mathcal{L}[9]$ are perverse; Artin's vanishing theorem for affine varieties [1, 4.1.1] gives $H_c^i(X, \mathcal{L}[9]) = 0$ for $i < 0$ and $H^i(X, \mathcal{L}[9]) = 0$ for $i > 0$. □

**Proposition 7.10** (what five values do and do not say)**.** (a) *The minimal linear recurrence over $\mathbb{Q}$ satisfied by $m_1, \dots, m_5$ (Berlekamp–Massey) has order 3, the generic value for five numbers, and is not unique; the recurrences of order 1 and 2 fitted to the first two, resp. four, values predict $m_3 = 1/13$, resp. $m_5 = 727/19$; the characteristic roots of the order-2 fit have modulus $2^{7.48}$, which is not of the form $2^{w/2}$. Five values therefore carry no information about $L_\sigma(T)$ beyond Theorem 7.7.*

(b) *If every eigenvalue of $F$ on $H_c^\bullet(X_{\bar{\mathbb{F}}_2}, \mathcal{L})$ had the form $2^{w/2}\zeta$ with $\zeta$ a root of unity (the shape of the eigenvalues behind $A_6$ and $X$), then $Y_k = \sum_w 2^{wk/2} A_w^{(k)}$ with algebraic integers $A_w^{(k)}$, and $Y_2 = -2^6$ forces: no single weight $w \geq 7$ can carry $Y$ (so the pure weight-9 case of Proposition 7.8 is impossible for every rank), and a nonzero part of weight $\leq 6$ must occur; indeed $v_2(Y_k) = 3k$ exactly for $k \leq 5$ ($m_k$ is odd).*

(c) *For odd $k$, Corollary 6.13 reads* $\mathrm{Tr}\big(\sigma F^k \mid H_c^\bullet(X_{\bar{\mathbb{F}}_2},\mathbb{Q}_\ell)\big) = 2^{9k}+2^{6k}-2^{5k}$, *and Conjecture 6.5 for $A_6$ (all $k$) is equivalent to* $\mathrm{Tr}(\sigma F^k \mid H_c^\bullet(X,\mathbb{Q}_\ell)) = 2^{9k}-(-64)^k+(-32)^k$ *for all $k$: the untwisted cohomology of $X$ is Tate-like, with $\sigma$ acting by $-1$ on the classes of weights* 12 *and* 10.

*Proof.* (a) is a computation (`L_function_spectral.py`; see Figure 6). (b) $A_w^{(k)}$ is a sum of roots of unity with signs; if $Y_2 = 2^w A$ with $A$ an algebraic integer then $2^{w-6} \mid 1$ in $\bar{\mathbb{Z}}$, so $w \le 6$; and if all weights were $\ge 7$ then $v_2(Y_2) \ge 7 > 6$. (c) is Theorem 7.7 with $\mathcal{L}$ replaced by $\mathbb{Q}_\ell$ and $A_6 = \#X^{(\sigma)}(\mathbb{F}_q)$. □

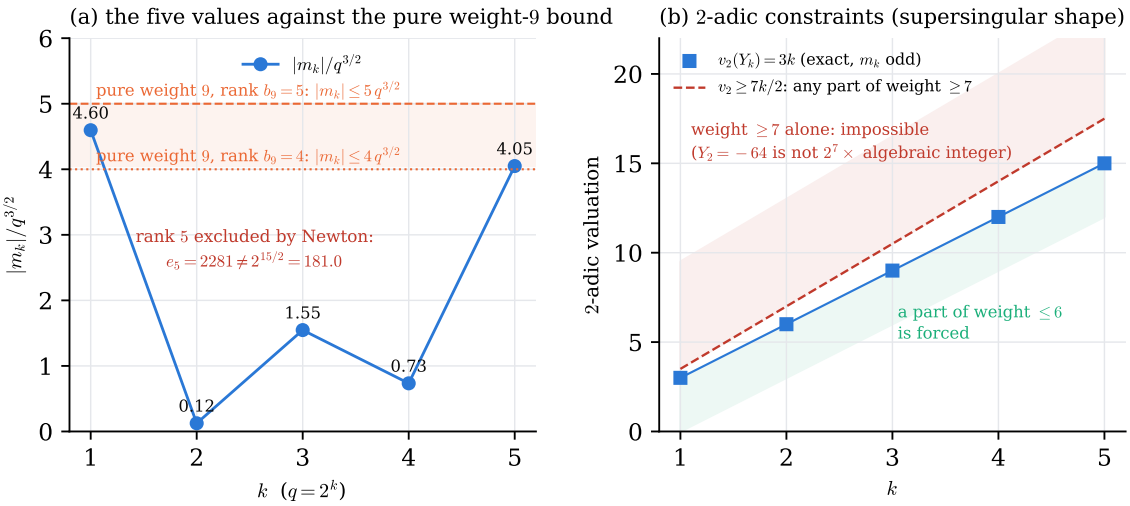



FIGURE 6. What the five values $m_k$ say (Propositions 7.8 and 7.10). (a) $|m_k|/q^{3/2}$ for $k \le 5$ against the bound $b_9$ of a pure weight-9 object of rank $b_9$: the value 4.60 at $k=1$ forces $b_9 \ge 5$, and $b_9 = 5$ is excluded because the fifth elementary symmetric function of the normalised eigenvalues computed from $m_1,\dots,m_5$ is $2281 \neq 2^{15/2}$. (b) Under the supersingular shape of the eigenvalues, a part of weight $\ge 7$ has 2-adic valuation $\ge 7k/2$, while $v_2(Y_k) = 3k$ exactly ($m_k$ odd); hence no single weight $\ge 7$ can carry $Y$ ($Y_2 = -64$) and a part of weight $\le 6$ is forced.

**Remark 7.11** (what is and is not established, and a roadmap)**.** Theorem 7.7 and Propositions 7.4–7.6, 7.9–7.10 are unconditional. By Proposition 7.9, $m_k = O(q^{3/2})$ (hence $C_6 = 1$) would follow from: *the eight-dimensional affine complete intersections $X_0 = \{e_1 = e_3 = e_4 = e_5 = 0\}$ and $X_1 = \{e_1 = e_3 = e_5 = 0,\ e_4 = 1\}$ have $H_c^j(X_{c,\bar{\mathbb{F}}_2},\mathbb{Q}_\ell)$ of weight $\le 9$ for $9 \le j \le 15$, and their top classes $H_c^{16}$ cancel in the sequence of* (b). For a smooth affine complete intersection in general position at infinity one expects even more (the cohomology of a bouquet of 8-spheres plus the top class), which would give $Y = O(q^4)$ and $m_k = O(q)$; the data $|m_k|/q \le 23$ do not exclude this for rank $\ge 23$. Two obstacles stand in the way: the singular loci (of dimension $\le 4$, where fewer than five coordinates are distinct), and the behaviour at infinity, where the projective closures of $X_0$ and $X_1$ meet the hyperplane at infinity in the same variety, the base $\mathbb{P}(X_0)$ of the cone $X_0$. We have *not* shown that $H_c^j(X_{\bar{\mathbb{F}}_2},\mathcal{L})$ vanishes for $9 < j < 18$ (Proposition 7.6 gives only $j = 18$ and Proposition 7.9(c) only $j < 9$), nor computed $b_9$: Katz's theorem on the concentration of $\sum \psi(f)$ in middle degree requires $p \nmid \deg f$ and a smooth ambient variety, whereas here $\deg e_4 = 4$ and $X$ is singular along the 1023 planes of Proposition 7.5(b); a proof would need the Swan conductor of $\mathcal{L}$ along $X_\infty$ and local computations at $\mathrm{Sing}(X)$. Hence $m_k = O(q^{3/2})$, and with it $C_6 = 1$, is supported by the data but not proved. Two further cautions. First, the variety $V = \{p_1 = p_3 = p_5 = 0\} \subset \mathbb{A}^{12}_{\mathbb{F}_q}$ obtained by choosing an $\mathbb{F}_q$-basis of $\mathbb{F}_{q^{12}}$ is exactly

$X^{(\sigma)}$, a nontrivial twist of $X \otimes \mathbb{F}_q$; so $(Y_k)_k$ is not a point-count sequence of a fixed variety, and the genuine "closed form" of $Y$ is the twisted $L$-function $L_\sigma(T)$, which Theorem 7.7 shows is a rational function determined by finitely many $Y_k$ (Berlekamp–Massey on $Y_1, Y_2, \dots$ would identify it; $k \le 5$ is not enough and $k = 6$ means $\mathbb{F}_{2^{72}}$). Second, the untwisted counts $\#X(\mathbb{F}_q)$ for $q = 2, 4, 8$ (2048, 2 099 200, 1 081 081 856) far exceed $q^9$; this is not a sign of several components but of $q < 12$, where no polynomial over $\mathbb{F}_q$ has twelve distinct roots. The tower reduction of Theorem 5.3 exhibits $Y/G - 1$ as a sum over $a \in \mathbb{F}_{q^6}$ of cubic Artin–Schreier sums over $\mathbb{F}_q$, i.e. as the trace of a relative $H^1$ of an elliptic Artin–Schreier family over a five-dimensional base; this is the natural starting point for computing $b_9$.

## 8. Spectral folding: why the restriction to squares amplifies the variance

At $d = 4$ and $d = 6$ the restriction of $\Psi$ to the subgroup of squares multiplies the normalised variance by a factor $\asymp q^2$ ($q^2/3$ exactly at $d = 4$, $\approx q^2/4$ at $d = 6$), while the full-group variances stay bounded. This section isolates the mechanism in the harmonic analysis of the pair $\mathcal{G}^2 \subset \mathcal{G}$: the variance over a subgroup is the energy of the *folded* spectrum, and folding can only increase the energy through phase coherence inside the fibres of the restriction map $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$.

8.1. **Folding.** Let $G$ be a finite abelian group, $H \le G$, $H^\perp = \{\varepsilon \in \widehat{G} : \varepsilon|_H = 1\} \cong \widehat{G/H}$, and $\varphi : G \to \mathbb{C}$ with Fourier expansion $\varphi = \sum_{\chi \in \widehat{G}} c_\chi \chi$, $c_\chi = |G|^{-1} \sum_g \varphi(g)\bar{\chi}(g)$. Write $\mathrm{Var}_G(\varphi) = |G|^{-1} \sum_{g \in G} |\varphi(g)|^2$ and $\mathrm{Var}_H(\varphi) = |H|^{-1} \sum_{h \in H} |\varphi(h)|^2$ (second moments, not centred on $H$).

**Proposition 8.1** (folding). (a) *The restriction* $\mathrm{Res} : \widehat{G} \to \widehat{H}$ *is surjective and its fibres are the cosets* $\chi H^\perp$, *each of cardinality* $|H^\perp| = [G : H]$.

(b) *(Parseval on $G$ and on $H$) With* $f_\psi = \sum_{\chi \in \mathrm{Res}^{-1}(\psi)} c_\chi$ *and* $E_\psi = \sum_{\chi \in \mathrm{Res}^{-1}(\psi)} |c_\chi|^2$,

$$\mathrm{Var}_G(\varphi) = \sum_{\psi \in \widehat{H}} E_\psi, \qquad \mathrm{Var}_H(\varphi) = \sum_{\psi \in \widehat{H}} |f_\psi|^2.$$

*Moreover, with* $F_{gH}(\psi) = |H|^{-1} \sum_{h \in H} \varphi(gh)\bar{\psi}(h)$ *the $\psi$-coefficient of $\varphi$ on the coset $gH$,*

$$f_\psi = F_H(\psi) = |H|^{-1} \sum_{h \in H} \varphi(h)\bar{\psi}(h), \qquad E_\psi = [G : H]^{-1} \sum_{gH \in G/H} |F_{gH}(\psi)|^2.$$

(c) *(Gram matrix) The matrix* $M_{\chi_1,\chi_2} = |H|^{-1} \sum_{h \in H} \chi_1(h)\overline{\chi_2(h)} = [\chi_1\bar{\chi}_2 \in H^\perp]$ *is block diagonal with all-ones blocks indexed by the fibres; its eigenvalues are* $[G : H]$ *(on vectors constant on fibres) and* $0$, *and* $\mathrm{Var}_H(\varphi) = c^* M c$, $\mathrm{Var}_G(\varphi) = c^* c$. *Writing* $M = \sum_{\varepsilon \in H^\perp} P_\varepsilon$ *with* $P_\varepsilon$ *the translation* $\chi \mapsto \chi\varepsilon$, *the Rayleigh quotient is*

$$\frac{\mathrm{Var}_H(\varphi)}{\mathrm{Var}_G(\varphi)} = 1 + \sum_{\varepsilon \in H^\perp \setminus 1} \rho(\varepsilon), \qquad \rho(\varepsilon) = \frac{\langle c, P_\varepsilon c\rangle}{\langle c, c\rangle},$$

*which is (2).*

(d) *(coherence bound)* $|f_\psi|^2 \le [G : H]\, E_\psi$ *for every $\psi$, hence* $\mathrm{Var}_H(\varphi) \le [G : H] \mathrm{Var}_G(\varphi)$, *with equality iff $c_\chi$ is constant on every fibre, i.e. iff $\varphi$ is supported on $H$. We call* $\kappa_\psi = |f_\psi|^2/([G : H]E_\psi) \in [0, 1]$ *the coherence of the fibre over $\psi$.*

*For the Hayes group $G = \mathcal{G}_\ell$ and $H = \mathcal{G}^2$ one has $[G : H] = |\mathcal{G}[2]| = q^{\lfloor d/2 \rfloor}$ (Lemma 2.1). In odd characteristic $\mathcal{G}_\ell$ has odd order, $\mathcal{G}[2] = 1$, $\mathcal{G}^2 = \mathcal{G}$, and there is nothing to fold: the Legendre intervals are then all the fibres of $\mathcal{G}$, and their variance is the full-group variance of Keating–Rudnick. The amplification is thus a consequence of the 2-torsion of the Hayes group, which exists only in characteristic 2 and has order $q^{d/2}$; the actual factor, $\asymp q^2$ rather than the maximal $q^{d/2}$, measures how much of the spectral energy is coherent along the fibres.*

*Proof.* (a) Characters of $H$ extend to $G$ and two extensions differ by an element of $H^\perp$; $|H^\perp| = |\widehat{G/H}| = [G : H]$. (b) $\varphi|_H = \sum_\psi f_\psi \psi$ and Parseval on $H$. The projection of $\varphi$ onto the fibre

$\chi_0 H^\perp$ is

$$\varphi_\psi(g) = \sum_{\chi\in\chi_0 H^\perp} c_\chi \chi(g) = \chi_0(g)\,|H|^{-1} \sum_{h\in H} \varphi(gh)\bar\chi_0(h),$$

so $|\varphi_\psi(g)| = |F_{gH}(\psi)|$ and $E_\psi = |G|^{-1}\sum_g |\varphi_\psi(g)|^2$. (c) Orthogonality of characters on $H$; $c^*Mc = \sum_\psi |\sum_{\text{fibre}} c_\chi|^2$. (d) Cauchy–Schwarz on each fibre; equality iff $c_{\chi\varepsilon} = c_\chi$ for all $\varepsilon \in H^\perp$ iff $\varphi(g)\varepsilon(g) = \varphi(g)$ for all $g$ and $\varepsilon \in H^\perp$ iff $\varphi$ vanishes off $H$. □

8.2. **The case $d = 4$: a single class folded $q^2$ times.**

**Theorem 8.2.** *Let $d = 4$, $\varphi = \Psi - q^5$ on $\mathcal{G}_3$, $H = \mathcal{G}^2 = \{(0,c,0)\} \cong \mathbb{F}_q$ with characters $\psi_\mu(c) = \chi(\mu c)$. Then the fibre sums are*

$$f_{\psi_0} = \big((-1)^k - 1\big)(q^3 - q^2), \qquad f_{\psi_\mu} = -q^3 + q^2 - (-1)^k q^2 \quad (\mu \neq 0),$$

*so that $\mathrm{Var}_{\mathcal{G}^2} = \sum_\mu |f_{\psi_\mu}|^2 = q^7 - q^6$, while $\mathrm{Var}_{\mathcal{G}} = \sum_\mu E_{\psi_\mu} = 3q^5 - 3q^4$, of which the identity class alone contributes $(q^4 - q^3)^2/q^3 = q^5 - 2q^4 + q^3$. Hence*

$$\frac{\mathrm{Var}_{\mathcal{G}^2}}{\mathrm{Var}_{\mathcal{G}}} = \frac{q^2}{3}, \qquad \frac{\mathrm{Var}_{\mathcal{G}}}{q^5} = \underbrace{1 - \frac{2}{q} + \frac{1}{q^2}}_{\text{identity class}} + \underbrace{2 - \frac{1}{q} - \frac{1}{q^2}}_{\text{all other classes}},$$

*the second bracket tending to the Keating–Rudnick value $d - 2 = 2$. The fibre over $\psi_\mu$ ($\mu \neq 0$) consists of the $q^2$ characters $\chi$ of $\mathcal{G}_3$ with $\chi^2 = \varepsilon_{(\sqrt{\mu},0)}$ (order 4), the fibre over $\psi_0$ of the $q^2$ quadratic characters $\varepsilon_\lambda$ with $c_{\varepsilon_\lambda} = S_\lambda/q^3$, $S_\lambda = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(\lambda_3 x^3 + \lambda_1 x)$.*

*Proof.* By Theorem 3.2, $\varphi(0,0,0) = -q^4 + q^3$ and $\varphi(0,c,0) = (-1)^k q^3$ for $c \neq 0$; so $f_{\psi_\mu} = q^{-1}\sum_c \varphi(0,c,0)\chi(\mu c) = (-1)^k q^3[\mu = 0] + (-q^4 + q^3 - (-1)^k q^3)/q$. The sum of squares is $q^6(q-1)$ in both parities, and Theorem 3.11 gives $\mathrm{Var}_{\mathcal{G}}$. For $g \in \mathcal{G}_3$, $\chi(g^2) = \chi(g)^2$ and $c_2(g^2) = c_1(g)^2$, so $\chi|_{\mathcal{G}^2} = \psi_\mu$ iff $\chi^2 = \varepsilon_{(\sqrt{\mu},0)}$; for $\mu = 0$ these are the characters trivial on $\mathcal{G}^2$, i.e. the $\varepsilon_\lambda$ of Proposition 2.2, whose coefficients are the Artin–Schreier sums of Section 3. □

Thus at $d = 4$ the whole anomaly is the identity class: its deficit $-(q^4 - q^3)$ (the missing $\alpha \in \mathbb{F}_{q^8}$ with $e_1 = e_2 = e_3 = 0$) has a flat Fourier transform on $\widehat{\mathcal{G}^2}$, $f_{\psi_\mu} \approx -q^3$ for every $\mu$, so every fibre is coherent (Table 5: $\kappa \approx 1/3$); on the full group the same class is diluted by the factor $[\mathcal{G} : \mathcal{G}^2] = q^2$ and contributes only the "1" in $\mathrm{Var}_{\mathcal{G}}/q^5 \to 3 = 2 + 1$. Theorem 3.12 is the same statement in the language of $\rho$: $\rho(\varepsilon) = 1$ exactly for the $\varepsilon$ trivial on $\{c_1 = 0\} \supset \operatorname{supp}\varphi$ (translation by such $\varepsilon$ fixes the coefficient vector), and the remaining $\rho(\varepsilon) = (q-3)/(3(q-1)) \to 1/3$ is the coherence along the $\lambda_3$-direction.

8.3. **The case $d = 6$: $q$ coherent fibres out of $q^2$.**

**Proposition 8.3.** *Let $d = 6$, $H = \mathcal{G}^2 = \{(c_2, c_4)\} \cong W_2(\mathbb{F}_q)$ with the Witt law $(c_2,c_4)(c_2',c_4') = (c_2 + c_2', c_4 + c_4' + c_2c_2')$, whose $q^2$ characters are $\psi_{(a,b)}(x) = i^{\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}((a,b)\cdot x)}$: $\psi_{(0,b)} = \chi(\sqrt{b}\,c_2)$ has order $\le 2$ ($q$ characters) and $\psi_{(a,b)}$, $a \neq 0$, has order 4 ($q^2 - q$ characters). Then, for all $k$,*

$$f_{\psi_{(0,0)}} = \frac{A_6 - q^9}{q^2}, \qquad f_{\psi_{(0,b)}} = \frac{X}{q^2}\ (b \neq 0), \qquad \sum_{a\neq 0,\, b} |f_{\psi_{(a,b)}}|^2 = q^3(q-1)m_k^2,$$

*and consequently*

$$\mathrm{Var}_{\mathcal{G}^2}(\Psi) = \frac{(A_6 - q^9)^2 + (q-1)X^2}{q^4} + q^3(q-1)\,m_k^2. \tag{10}$$

*The fibre over $\psi_{(0,0)}$ consists of the $q^3$ quadratic characters $\varepsilon_\lambda$ (coefficients $S(\lambda)/q^5$, Theorem 6.1), the fibre over $\psi_{(0,b)}$ of the $q^3$ characters of order 4 with $\chi^2 = \varepsilon_{(b^{1/4},0,0)}$ (coefficients $S'(\lambda)/q^5$ up to the scaling $b$, §6.7), and the fibres over the order-4 characters of $\mathcal{G}^2$ of characters of order 8 of $\mathcal{G}_5$, whose coefficients are Witt-vector sums of length 3 and are not quadratic-form sums.*

*Proof.* $f_\psi = q^{-2}\sum_{g\in\mathcal{G}^2}\varphi(g)\bar\psi(g)$. For $\psi=\psi_{(0,b)}$ this is $q^{-2}\big(\sum_{\mathcal{G}^2}\Psi(g)\chi(\sqrt{b}\,c_2) - q^7\cdot q^2[b=0]\big)$, i.e. $(A_6-q^9)/q^2$ or $X_{\sqrt{b}}/q^2 = X/q^2$ (scaling $\alpha\mapsto s\alpha$, $e_2\mapsto s^2e_2$). By Theorem 4.2, $\varphi|_{\mathcal{G}^2} = \varphi_0[c_2=0] + (B + q^2 m_k\,\chi(c_4/c_2^2))[c_2\neq 0]$; the first two terms are functions of $c_2$ alone, hence combinations of the $\psi_{(0,b)}$, while $\sum_{c_4}\chi(c_4/c_2^2)\bar\psi_{(0,b)}(c_2,c_4)=0$ for $c_2\neq 0$, so the last term is supported on the order-4 characters, where Parseval gives $q^{-2}\sum_{c_2\neq0,c_4}(q^2m_k)^2 = q^3(q-1)m_k^2$. The description of the fibres follows from $\chi(g^2)=\chi(g)^2$ and $c_2(g^2)=c_1(g)^2$ as at $d=4$ (characters with $\chi^2=\varepsilon$ of order 2 have order 4; those restricting to an order-4 character of $\mathcal{G}^2$ have order 8). □

Formula (10) separates the variance into a *coherent* part carried by the $q$ fibres over the characters of order $\le 2$, which is determined by the two constants $A_6$ and $X$, and an *incoherent* part $q^3(q-1)m_k^2$ carried by the $q^2-q$ fibres over the order-4 characters, which is $Y$. With $A_6 = q^9+(q-1)q^5$ (Corollary 6.13) and $X=q^6+q^5$ (Conjecture 6.19) the coherent part is $(q-1)(q+3)q^7$, and the three-value variance of Theorem 7.1 is recovered. The coherent fibres have sums $\approx q^4$, which is $q^3$ coefficients of typical size $q$ added with the same sign: the order-4 sums $S'(\lambda)$ of §6.7 take the values $0,\pm q^6,\pm 2q^6,\pm 4q^6$ with the strong positive bias displayed in Table 3, and the quadratic sums $S(\lambda)$ are $+4q^6$ on the 4-cycle classes of Theorem 6.12. This Arf-invariant bias of the supersingular quadratic forms is the arithmetic source of the coherence; the incoherent part, by contrast, involves no quadratic form and stays of the size $q^3(q-1)m_k^2$, negligible against $q^9$ as long as $m_k = o(q^{5/2})$. Since $\mathrm{Var}_{\mathcal{G}}/q^7\to 4$ (Table 4), the amplification is $\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}\approx q^2/4$: $q$ coherent fibres with coherence $\kappa\approx 1/10$ out of $|\mathcal{G}[2]| = q^3$ possible, carrying about $1/7$ of the total energy at $q=16$ (Table 5 and Figure 7).

TABLE 5. Spectral folding data (`fourier_spectrum_folding.py`): the fibres of $\widehat{\mathcal{G}}\to\widehat{\mathcal{G}^2}$ grouped by the order of the character of $\mathcal{G}^2$; "share" is the fraction of $\mathrm{Var}_{\mathcal{G}^2}$, resp. $\mathrm{Var}_{\mathcal{G}}$, carried by the group of fibres, and $\kappa$ the coherence of Proposition 8.1(d). At $d=4$ all classes come from Theorem 3.2; at $d=6$ from the Hayes FFT.

| $d$ | $q$ | $\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}$ | $[\mathcal{G}:\mathcal{G}^2]$ | order of $\psi$ | #fibres | share of $\mathrm{Var}_{\mathcal{G}^2}$ / $\mathrm{Var}_{\mathcal{G}}$ | $\kappa$ |
|---|---|---|---|---|---|---|---|
| 4 | 8 | $21.33 = q^2/3$ | 64 | 1 | 1 | 0.438 / 0.167 | 0.875 |
| | | | | 2 | 7 | 0.563 / 0.833 | 0.225 |
| 4 | 32 | $341.33 = q^2/3$ | 1024 | 1 | 1 | 0.121 / 0.042 | 0.969 |
| | | | | 2 | 31 | 0.879 / 0.958 | 0.306 |
| 6 | 8 | $16.66 = 0.260\,q^2$ | 512 | 1 | 1 | 0.077 / 0.036 | 0.070 |
| | | | | 2 | 7 | 0.896 / 0.228 | 0.128 |
| | | | | 4 | 56 | 0.027 / 0.736 | 0.001 |
| 6 | 16 | $51.82 = 0.202\,q^2$ | 4096 | 1 | 1 | 0.062 / 0.008 | 0.095 |
| | | | | 2 | 15 | 0.935 / 0.138 | 0.086 |
| | | | | 4 | 240 | 0.002 / 0.854 | $3\cdot 10^{-5}$ |

**Remark 8.4.** The two cases are instances of one pattern. In both, the deviation $\varphi$ is large on a small subgroup $K\subset\mathcal{G}^2$ ($K=\{1\}$ at $d=4$ with $\varphi\approx -q^4$; $K=\{c_2=0\}$ of order $q$ at $d=6$ with $\varphi\approx\varepsilon_0 q^5$), and the Fourier transform of a function supported on $K$ is constant along the fibres of $\widehat{\mathcal{G}^2}\to\widehat{K}$. Folding to $\mathcal{G}^2$ therefore adds these coefficients in phase, whereas on $\mathcal{G}$ they are spread over $[\mathcal{G}:\mathcal{G}^2]$ times more classes. The size of $\varphi$ on $K$ is in turn a statement about the number of $\alpha\in\mathbb{F}_{q^n}$ with prescribed vanishing of symmetric functions ($e_1=e_2=e_3=0$ at $d=4$; $e_1=e_2=e_3=e_5=0$ at $d=6$, Proposition 7.9), i.e. about the point counts of the varieties $X$, $X_0$, $X_1$: the "$q^2$" is $|K|\varphi_K^2/|\mathcal{G}^2|$ divided by $q^{d+1}$, which equals $q^2$ in both cases ($1\cdot q^8/q$ and $q\cdot q^{10}/q^2$ against $q^5$ and $q^7$). What is specific to characteristic 2 is not the folding identity but the existence of the subgroup $\mathcal{G}^2$ of index $q^{d/2}$ and the Arf-invariant bias that makes $\varphi_K$ as large as $q^d$, resp. $q^{d-1}$.

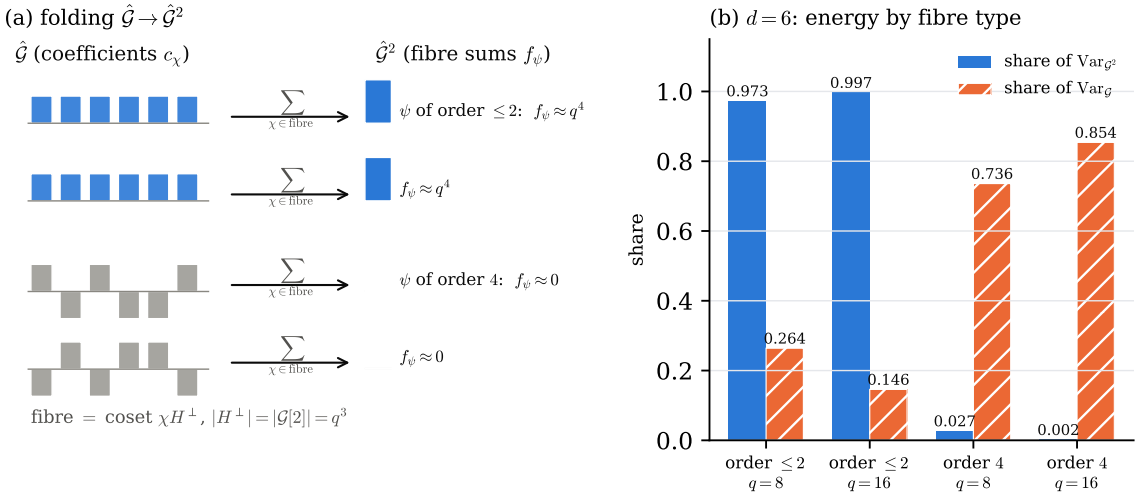



FIGURE 7. Spectral folding at $d = 6$. (a) The restriction map $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$: each fibre is a coset of $H^\perp = \widehat{\mathcal{G}/\mathcal{G}^2}$ of size $|\mathcal{G}[2]| = q^3$, and the fibre sum $f_\psi$ is the $\psi$-coefficient of $\varphi$ on $\mathcal{G}^2$. Over the $q$ characters of $\mathcal{G}^2$ of order $\le 2$ the coefficients add in phase (they are the biased supersingular sums $S$, $S'$ of Sections 6.6–6.7); over the $q^2 - q$ characters of order 4 they cancel. (b) Shares of $\mathrm{Var}_{\mathcal{G}^2}$ and of $\mathrm{Var}_{\mathcal{G}}$ carried by the two kinds of fibres for $q = 8, 16$ (Table 5): the coherent fibres hold 97–100% of the Legendre variance but only 15–26% of the full-group energy.

## 9. Discussion, open problems, and general even $d$

9.1. **Translation orbits for general even $d$.** The symmetry that produced the three-value profile at $d = 6$ exists for every even $d$, and Lucas' theorem decides how it acts on the identity class.

**Proposition 9.1.** *Let $d$ be even, $n = 2d$, $\ell = d - 1$, and $U_d = \{\alpha \in \mathbb{F}_{q^n} : \operatorname{Tr} \alpha^j = 0$ for odd $j \le \ell\}$, the set of $\alpha$ whose class lies in $\mathcal{G}^2$.*

(a) *For $c \in \mathbb{F}_q$ the translation $\alpha \mapsto \alpha + c$ preserves $U_d$ and induces a bijection $\tau_c$ of $\mathcal{G}^2$ with $\Psi \circ \tau_c = \Psi$; explicitly $e_j(\alpha + c) = \sum_{i \le j} \binom{n-i}{j-i} e_i(\alpha) c^{j-i}$, so $\tau_c$ depends only on the binomial coefficients $\binom{2d-i}{j-i} \bmod 2$, $i \le j \le d - 1$.*

(b) *The orbit of the identity class is $K_d = \{(1 + cu)^{2d} \bmod u^d : c \in \mathbb{F}_q\} = \{(\binom{2d}{j} c^j)_{j \le d-1} : c \in \mathbb{F}_q\}$, and $\Psi$ is constant on $K_d$. By Lucas' theorem $\binom{2d}{j}$ is odd iff the binary digits of $j$ are among those of $2d$; hence $|K_d| = 1$ if $d$ is a power of 2, and $|K_d| = q$ otherwise, the first nonzero coordinate of $K_d$ being $c^{2^{v+1}}$ in position $2^{v+1}$ where $2^v \parallel d$. In particular $K_4 = \{1\}$, $K_6 = \{c_2 = 0\} = \{1 + c^4 u^4\}$ (Section 4), $K_8 = \{1\}$, $K_{10} = \{1 + c^4 u^4\}$, $K_{12} = \{1 + c^8 u^8\}$, $K_{14} = \{1 + c^4 u^4 + c^8 u^8 + c^{12} u^{12}\}$.*

(c) *(lower bound) $\mathrm{Var}_{\mathcal{G}^2}(\Psi) \ge |K_d|\, \varphi(1)^2 / |\mathcal{G}^2|$ with $|\mathcal{G}^2| = q^{d/2-1}$, where $\varphi(1) = \Psi(1) - q^{d+1} = \#\{\alpha \in \mathbb{F}_{q^{2d}} : e_1(\alpha) = \cdots = e_{d-1}(\alpha) = 0\} - q^{d+1}$ is the twisted point count of the complete intersection $V_d = V(e_1, \ldots, e_{d-1}) \subset \mathbb{A}^{2d}_{\mathbb{F}_2}$ of dimension $d + 1$, minus its main term. Hence $\mathrm{Var}_{\mathcal{G}^2} / q^{d+1} \to \infty$ as soon as $|\varphi(1)| / q^{3d/4} \to \infty$.*

*Proof.* (a) $\operatorname{Tr}(\alpha + c)^j = \sum_i \binom{j}{i} c^{j-i} \operatorname{Tr} \alpha^i$ and on $U_d$ all $\operatorname{Tr} \alpha^i$ with $i < j$ vanish (odd $i$ by definition, even $i$ because $\operatorname{Tr} \alpha^{2i} = (\operatorname{Tr} \alpha^i)^2$), while $\binom{j}{0} c^j \operatorname{Tr} 1 = n c^j = 0$; the formula for $e_j(\alpha + c)$ is the expansion of $\prod_i (t - c - \alpha_i)$, and it shows that $(e_i(\alpha))_{i \le j}$ determines $(e_i(\alpha + c))_{i \le j}$, so $\tau_c$ is well defined on classes; it is bijective with inverse $\tau_{-c} = \tau_c$, and $\Psi \circ \tau_c = \Psi$ because $\alpha \mapsto \alpha + c$ is a bijection of $\mathbb{F}_{q^n}$ compatible with the classes. (b) For $\alpha$ in the identity class only the term $i = 0$ survives, giving $e_j(\alpha + c) = \binom{n}{j} c^j$, i.e. the class of $(1 + cu)^n$. Lucas' theorem gives the parity of

$\binom{2d}{j}$; the least $j \geq 1$ with $\binom{2d}{j}$ odd is the lowest set bit $2^{v+1}$ of $2d$, which is $\leq d-1$ iff $d$ is not a power of 2; if it is $\leq d-1$, then $c \mapsto c^{2^{v+1}}$ is injective and $|K_d| = q$. (c) is the contribution of $K_d$ to the second moment. □

9.2. **The variance for $d = 8$ and $d = 10$, and what is and is not universal.** Table 6 collects the exact Legendre variances for $d \leq 10$ (Hayes FFT, `general_d_scan.py`; the $d \leq 7$ values agree with [5, Table 4]). Three facts stand out (Figure 8).

TABLE 6. Legendre and full-group variances for even $d \leq 10$ (second moments about $q^{d+1}$), the identity-class deviation $\varphi(1)$, the orbit $K_d$, and the share of $\sum_{\mathcal{G}^2} \varphi^2$ carried by $K_d$.

| $d$ | $q$ | $\mathrm{Var}_{\mathcal{G}}/q^{d+1}$ | $\mathrm{Var}_{\mathcal{G}^2}/q^{d+1}$ | ratio | ratio$/q^2$ | $\varphi(1)/q^{d-1}$ | $\lvert K_d\rvert$ | share of $K_d$ |
|---|---|---|---|---|---|---|---|---|
| 4 | 4 | 2.250 | 12.00 | 5.33 | 0.333 | −3.000 | 1 | 0.750 |
| 4 | 8 | 2.625 | 56.00 | 21.33 | 0.333 | −7.000 | 1 | 0.875 |
| 4 | 32 | 2.906 | 992.00 | 341.33 | 0.333 | −31.000 | 1 | 0.969 |
| 6 | 4 | 3.700 | 9.01 | 2.44 | 0.152 | −0.750 | 4 | 0.999 |
| 6 | 8 | 4.748 | 79.09 | 16.66 | 0.260 | +1.094 | 8 | 0.968 |
| 6 | 32 | 4.209 | 1100.88 | 261.58 | 0.255 | +1.029 | 32 | 0.986 |
| 8 | 2 | 4.881 | 1.59 | 0.33 | 0.082 | −0.313 | 1 | 0.245 |
| 8 | 4 | 5.878 | 13.25 | 2.25 | 0.141 | −0.387 | 1 | 0.181 |
| 8 | 8 | 5.927 | 13.49 | 2.28 | 0.036 | −0.234 | 1 | 0.260 |
| 10 | 2 | 7.192 | 7.91 | 1.10 | 0.275 | +0.094 | 2 | 0.018 |
| 10 | 4 | 7.693 | 10.47 | 1.36 | 0.085 | −0.076 | 4 | 0.140 |

First, the full-group variance follows Keating–Rudnick at every $d$ ($\mathrm{Var}_{\mathcal{G}}/q^{d+1} \to d-2$, with the exact deviation $1 - 3/q$ at $d = 4$ coming from the identity class, Theorem 8.2). Second, the $q^2$-amplification of the Legendre variance is a feature of $d = 4$ and $d = 6$: there the identity orbit $K_d$ carries essentially all of $\sum_{\mathcal{G}^2} \varphi^2$, because $\varphi(1)$ is of order $q^d$ (exactly $-q^4 + q^3$) resp. $q^{d-1}$ (exactly $\varepsilon_0 q^5 + q^4 - 2q^3$ for odd $k$), far above the threshold $q^{3d/4}$ of Proposition 9.1(c). Third, at $d = 8$ the amplification *disappears*: $\mathrm{Var}_{\mathcal{G}^2}/q^9$ is 13.25 and 13.49 at $q = 4, 8$ and the ratio to the full-group variance is 2.25, 2.28, with the identity class ($|K_8| = 1$) carrying only a quarter of the Legendre second moment and $\varphi(1) \approx -1.9\, q^6 = O(q^{3d/4})$ exactly at the threshold; at $d = 10$ the ratio is 1.10, 1.36 for $q = 2, 4$. This refutes the law $\mathrm{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^{2(d/2-1)}$ suggested in [5] (it would give $q^6$ at $d = 8$), and it equally refutes the natural guess that the $q^2$ of $d = 4, 6$ is universal. In the language of Section 8: at $d = 8$ the index $[\mathcal{G} : \mathcal{G}^2] = q^4$ is as large as ever, but the deviation is no longer concentrated on a small subgroup, so the fibres of $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$ are incoherent and the Rayleigh quotient stays bounded.

**Conjecture 9.2** (general even $d$)**.** *Let $d \geq 4$ be even and $q = 2^k \to \infty$. Then $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^{d+1} \to d-2$, and the ratio $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/\mathrm{Var}_{\mathcal{G}}(\Psi)$ is $\asymp q^2$ for $d = 4$ and $d = 6$ and bounded for $d = 8$ and $d = 10$. More precisely, the ratio is unbounded iff the identity-class deviation satisfies $|\varphi(1)|/q^{3d/4} \to \infty$, i.e. iff the twisted point count of $V_d = V(e_1, \ldots, e_{d-1})$ has a subleading term of weight $> 3d/2$ with nonzero $\sigma$-twisted trace; this holds for $d = 4, 6$ (weights $8 = 2d$ and $10 = 2(d-1)$) and fails for $d = 8$.*

The first assertion is the Keating–Rudnick asymptotic, which in characteristic 2 rests on Katz's equidistribution theorem [14] and is not at issue here; the equivalence in the last sentence is Proposition 9.1(c) in one direction, and in the other direction it asserts that the classes outside $K_d$ never produce an amplification by themselves, which is what Table 6 shows for $d \leq 10$ and what the folding picture of Section 8 predicts (coherence requires a deviation concentrated on a subgroup, and the only subgroup singled out by the symmetries is $K_d$). We have no proof of either direction beyond $d = 6$.

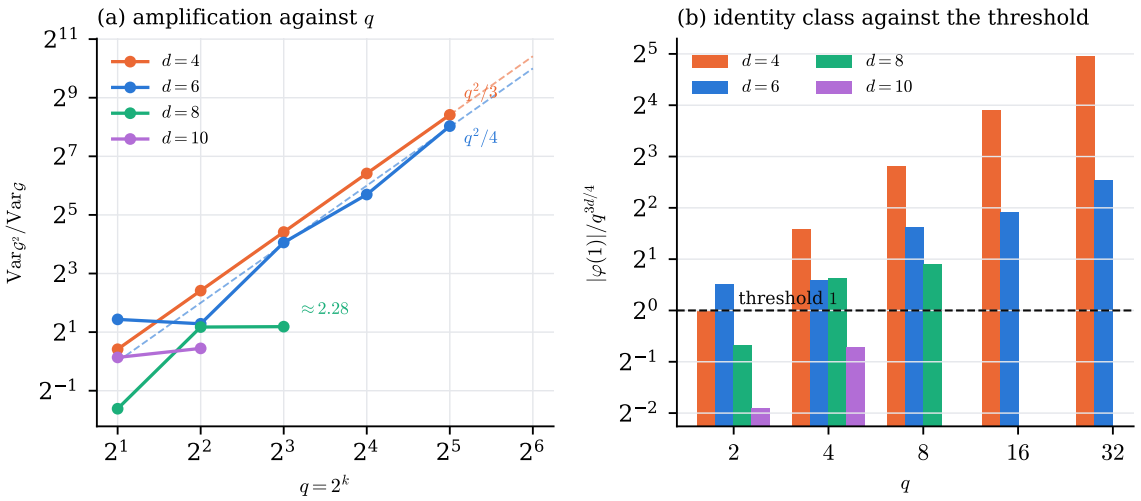



FIGURE 8. The Legendre amplification for even $d$ (data of Table 6). (a) $\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}$ against $q$ on logarithmic scales: $q^2/3$ exactly at $d=4$, about $q^2/4$ at $d=6$, and no growth at $d=8$ (2.25, 2.28 at $q=4,8$) or $d=10$. (b) The identity-class deviation $|\varphi(1)|$ relative to the threshold $q^{3d/4}$ of Proposition 9.1(c): far above it for $d=4$ (where it equals $q-1$) and $d=6$, at it for $d=8$ (1.5, 1.9 at $q=4,8$), below it for $d=10$.

### 9.3. **Open problems.**

**Open Problem 9.1.** (*the constant $X$*) Prove Conjecture 6.19: exhibit the sextic over $\mathbb{F}_q$ with $e_1=0$ whose pairwise root sums form $\ker L'_\lambda$, so that the Frobenius acts on $\ker L'_\lambda$ through $Sp_4(\mathbb{F}_2)\cong S_6$ with the class frequencies of Table 3, and evaluate the Arf invariants class by class, in particular the sign $-q^7$ on the class of 5-cycles, which carries the leading term. This would make $X=q^6+q^5$ a theorem for odd $k$ and Theorem 7.1 unconditional for odd $k$. The even-$k$ case, where the orbit invariant $j$ is not complete, is open for both $A_6$ and $X$.

**Open Problem 9.2.** (*the constant $Y$*) Prove $m_k=O(q^{3/2})$, or at least $m_k=o(q^{5/2})$, which gives $C_6=\lim \mathrm{Var}_{\mathcal{G}^2}/q^9=1$. By Proposition 7.9 this is a weight bound for $H^j_c$, $9\le j\le 15$, of the eight-dimensional complete intersections $X_0=\{e_1=e_3=e_4=e_5=0\}$ and $X_1=\{e_1=e_3=e_5=0,\ e_4=1\}$ in $\mathbb{A}^{12}$, i.e. a Lefschetz-type statement for singular affine complete intersections whose closures meet the hyperplane at infinity in the base $\mathbb{P}(X_0)$ of the cone $X_0$; the local ingredients are the Swan conductors of the Artin–Schreier sheaf $\mathcal{L}_\psi(e_4)$ along $\mathbb{P}(X_0)$ and the vanishing cycles at the singular loci (coordinates with fewer than five distinct values).

**Open Problem 9.3.** (*general $d$*) Determine the subleading $\sigma$-twisted Tate terms of the point count $\Psi(1)=\#V_d^{(\sigma)}(\mathbb{F}_q)$, where $V_d=V(e_1,\dots,e_{d-1})\subset\mathbb{A}^{2d}$: for $d=4$ they are $-q^4+q^3$, for $d=6$ and odd $k$ they are $q^5+q^4-2q^3$ (Theorem 7.1), and for $d=8$ the exact values $\varphi(1)=-40,-6336,-491008$ at $q=2,4,8$ are of size $-1.5\,q^6$ and $-1.9\,q^6$ for $q=4,8$, i.e. at the threshold $q^{3d/4}$. Decide Conjecture 9.2; in particular whether the amplification can reappear for larger $d$ when $d$ is not a power of 2 and $|K_d|=q$.

Note added. Open Problem 9.1 has since been taken up in Parts III and IV [6, 7]. The sextic asked for is $B_\lambda(u)=\lambda_5^2u^6+\lambda_3^2u^2+\lambda_5u+1$: the identity $L'_\lambda=x^4B_\lambda^2+B_\lambda+1$ shows that $\ker L'_\lambda$ is the set of pairwise sums of its six roots, which explains the action of the Frobenius through $S_6\cong Sp_4(\mathbb{F}_2)$ and identifies the class of $\lambda$ with the factorisation type of $B_\lambda$ over $\mathbb{F}_q$ (Part III). The sums $W'$ are Frobenius traces of rank-four Witt–Artin–Schreier sheaves; their values are now proved in all eleven classes for odd $k$ (the value $-q^7$ on the class of 5-cycles in Part III, the remaining signs in Part IV through a sixteen-point formula on the coset $T'^{-1}_\lambda(1)$ and a trace-parity theorem). The closed form $X=q^6+q^5$ for odd $k$ therefore depends only on

the polynomial class counts of Table 3, which remain conjectural (verified for $q \le 512$). Open Problems 9.2 and 9.3 are untouched.

9.4. **Final checklist.** Table 7 lists every statement of the paper with its status (proved, conditional, or open) and the range in which it was checked by exact computation.

TABLE 7. Final checklist: status of the main identities. "exact" means verified by exact integer computation; the method is indicated.

| Statement | Status | Range checked |
|---|---|---|
| $d = 4$: Theorem 3.2 (all classes) | proved | formula = FFT class by class, $q \le 128$ |
| $d = 4$: $\mathrm{Var}_{\mathcal{G}^2}/q^5 = q(q-1)$, $\mathrm{Var}_{\mathcal{G}}/q^5 = 3 - 3/q$ | proved | Lemma 3.10 also checked to $q = 2^{22}$ |
| $d = 4$: $\rho$ structure (Theorem 3.12) | proved | matches [5, Table 4] |
| $d = 6$: three-value profile (Theorem 4.2) | proved | all $q^2$ classes, $q \le 32$ |
| $d = 6$: tower identities and reductions | proved | $\mathbb{F}_{2^{12}}$ exhaustive; $q = 2, 4, 8$ |
| $d = 6$: pure cubic and pure quintic parts (Lemmas 6.2, 6.4) | proved, all $k$ | exact, $q \le 128$ |
| $d = 6$: mixed part, $k$ odd (Theorem 6.12); types of $v^5 + v + c$ (Proposition 6.8) | proved | quadratic forms $k \le 7$; $L$-polynomials $k \le 9$; types $k \le 11$ |
| $d = 6$: $A_6 = q^9 + (q-1)q^5$, $k$ odd (Corollary 6.13) | proved | exact, $q \le 128$ (Table 1) |
| $d = 6$: $X$, $k$ odd: structure of $W'$ (Prop. 6.14), $W'(0,0)$, pure cubic vanishing pattern, pure quintic kernel (Lemmas 6.15–6.17) | proved | exact, $q \le 128$ |
| $d = 6$: $X = q^6 + q^5$, $k$ odd (Conjecture 6.19; $S_6$-structure of Table 3) | open | exact, $q \le 128$; table $k \le 7$ |
| $d = 6$: $A_6$ and $X$ for even $k$ (Conjecture 6.5) | open | exact, $q \le 128$ (Table 1) |
| $d = 6$: $\mathrm{Var}_{\mathcal{G}^2}/q^7$ formula (Theorem 7.1) | proved given Conj. 6.5 ($X$-part only for odd $k$) | $q \le 32$ |
| $Y = N_0 - N_1$, lower vanishing $H_c^j(X, \mathcal{L}) = 0$ $(j < 9)$, spectral constraints (Props. 7.9, 7.10) | proved | $k \le 5$ |
| $C_6 = 1$ | needs $m_k = o(q^{5/2})$, Prop. 7.9(a) | $q \le 32$ |
| spectral folding: $\mathrm{Var}_{\mathcal{G}^2} = \sum_\psi \lvert f_\psi \rvert^2$, fibre sums at $d = 4$ (Theorem 8.2) and $d = 6$, (10) | proved | $d = 4$: $q \le 32$; $d = 6$: $q \le 16$ |
| general even $d$: translation orbits, $K_d$ by Lucas, lower bound (Proposition 9.1) | proved | $d \le 10$ (Table 6) |
| general even $d$: $q^2$-amplification only for $d = 4, 6$; bounded ratio at $d = 8$ (Conjecture 9.2) | open | $d = 8$: $q \le 8$; $d = 10$: $q \le 4$ |

## References


[1] A. A. Beilinson, J. Bernstein and P. Deligne, *Faisceaux pervers*, Astérisque **100** (1982).

[2] B. C. Berndt, R. J. Evans and K. S. Williams, *Gauss and Jacobi Sums*, Wiley, New York, 1998.

[3] A. W. Bluher, *On $x^{q+1}+ax+b$*, Finite Fields Appl. **10** (2004), 285–305.

[4] L. Carlitz, *Explicit evaluation of certain exponential sums*, Math. Scand. **44** (1979), 5–16.

[5] R. Chen, *Primes in Legendre intervals over $\mathbb{F}_q[t]$: an exact explicit-formula algorithm, verification for $d \leq 10$, and an anomaly in characteristic 2*, preprint (2026), version 2, DOI 10.5281/zenodo.23225896 (version 1: 10.5281/zenodo.23219609).

[6] R. Chen, *Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, III: the hexad of a quartic linearised polynomial, the symplectic group $Sp_4(\mathbb{F}_2)$, and the Witt sums behind the constant $X$*, preprint (2026), DOI 10.5281/zenodo.23290724.

[7] R. Chen, *Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, IV: point counts on hexad modular curves, Brown invariants of Witt sheaves, and the constant $X$*, preprint (2026), version 2, DOI 10.5281/zenodo.23308682.

[8] R. S. Coulter, *On the evaluation of a class of Weil sums in characteristic* 2, New Zealand J. Math. **28** (1999), 171–184.

[9] P. Deligne, *La conjecture de Weil. II*, Publ. Math. IHES **52** (1980), 137–252.

[10] P. Deligne and G. Lusztig, *Representations of reductive groups over finite fields*, Ann. of Math. (2) **103** (1976), 103–161.

[11] G. van der Geer and M. van der Vlugt, *Reed–Muller codes and supersingular curves. I*, Compositio Math. **84** (1992), 333–367.

[12] T. Helleseth and A. Kholosha, *On the equation $x^{2^l+1}+x+a=0$ over $GF(2^k)$*, Finite Fields Appl. **14** (2008), 159–176.

[13] H. Hasse and H. Davenport, *Die Nullstellen der Kongruenzzetafunktionen in gewissen zyklischen Fällen*, J. reine angew. Math. **172** (1935), 151–182.

[14] N. M. Katz, *Witt vectors and a question of Keating and Rudnick*, Int. Math. Res. Not. IMRN (2013), 3613–3638.

[15] J. P. Keating and Z. Rudnick, *The variance of the number of prime polynomials in short intervals and in residue classes*, Int. Math. Res. Not. IMRN (2014), 259–288.

[16] J. Lahtonen, G. McGuire and H. N. Ward, *Gold and Kasami–Welch functions, quadratic forms, and bent functions*, Adv. Math. Commun. **1** (2007), 243–250.



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